

Leading Sustainability Pathways

Enerqa Consultancy

A Qatari Company

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1. About Enerqa

About **ENERQA** As of 2026

Existence

- 🌐 Qatari Company
- www.enerqa.co.uk
- **Office:** in London, Doha, Muscat, Riyadh and Beijing
- **Network** in ~ 22 countries

Team

Enerqa's expanding team of over 35 professionals includes numerous PhD holders, Master's graduates, and certified experts, offering extensive technical and practical expertise in energy and environmental fields.

Roster of Experts includes: >10 PhD holders, >10 MSc holders and a lot of Professionals

Experience

The core team and associates of the company collectively bring over 325 years of combined experience in these domains.

Enerqa Consultancy is a Qatari company registered in Qatar, specializing in climate change, energy, sustainability, and environmental consultancy services. Our vision is to contribute meaningfully to the nation's ambitious climate goals and support its leadership in regional and global climate action.

The company provides strategic solutions to government entities, private sector organizations, and international institutions, helping them align with global climate commitments and national sustainability goals.

With a strong track record in energy, transport, waste management, and industrial emissions, Enerqa provides expertise in GHG inventory development, NDC tracking, climate finance, and ESG integration. The company collaborates with stakeholders to develop long-term sustainability strategies, enhance institutional frameworks, and facilitate knowledge transfer through tailored capacity-building programs.

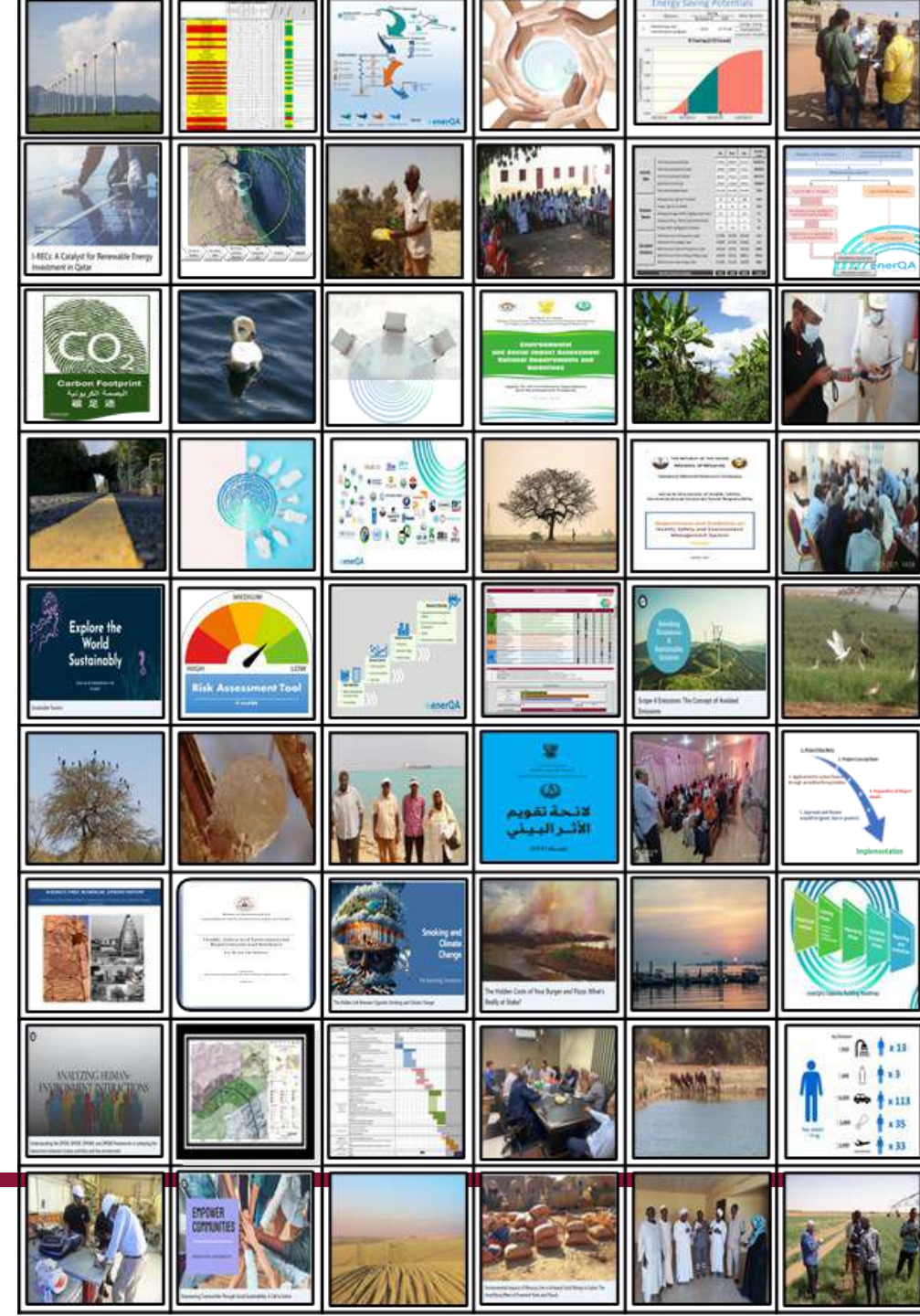
Enerqa's mission is to drive impactful climate action by bridging technical expertise with policy development. By leveraging its experience in transparency frameworks and sustainable development, Enerqa continues to support national and regional efforts to achieve low-carbon growth and resilience in alignment with Qatar National Vision 2030 and global climate commitments.

Enerqa is proud to be a part of Qatar's journey to net-zero.

Introduction to Enerqa Consultancy

With a team of highly skilled experts and a portfolio of over 50 completed projects, we support governments, international organizations, and private sector entities in aligning with global climate frameworks and sustainability goals. Enerqa plays a critical role in GHG inventory development, emissions modeling, and impact assessments, enabling stakeholders to accurately measure and track their carbon footprint. Our expertise extends to the design and implementation of MRV systems, ensuring compliance with both national and international climate regulations, such as the Paris Agreement and Nationally Determined Contributions (NDCs). By integrating robust methodologies and best practices, we help organizations establish transparent, verifiable, and science-based reporting mechanisms. Our consultancy also excels in climate risk assessment, mitigation strategies, and additionality identification, assisting policymakers and businesses in developing low-carbon pathways.

Whether through sector-specific mitigation measures, climate adaptation planning, or financial mechanisms like carbon credit markets, Enerqa ensures that climate action is both effective and economically viable. In addition to our technical services, Enerqa is actively engaged in capacity building and awareness programs aimed at empowering stakeholders with the knowledge and skills



Leading the way to decarbonization and circular economy

Vision

To be part of transforming MENA region from a resources-based economy to a knowledge-based economy and to be a leading catalyst for climate change action, environmental awareness, and sustainable development through guiding institutions, firms, entities, and startups in implementing robust environmental and social safeguards, optimizing energy use, fighting climate change, and creating lasting value for communities and the planet.

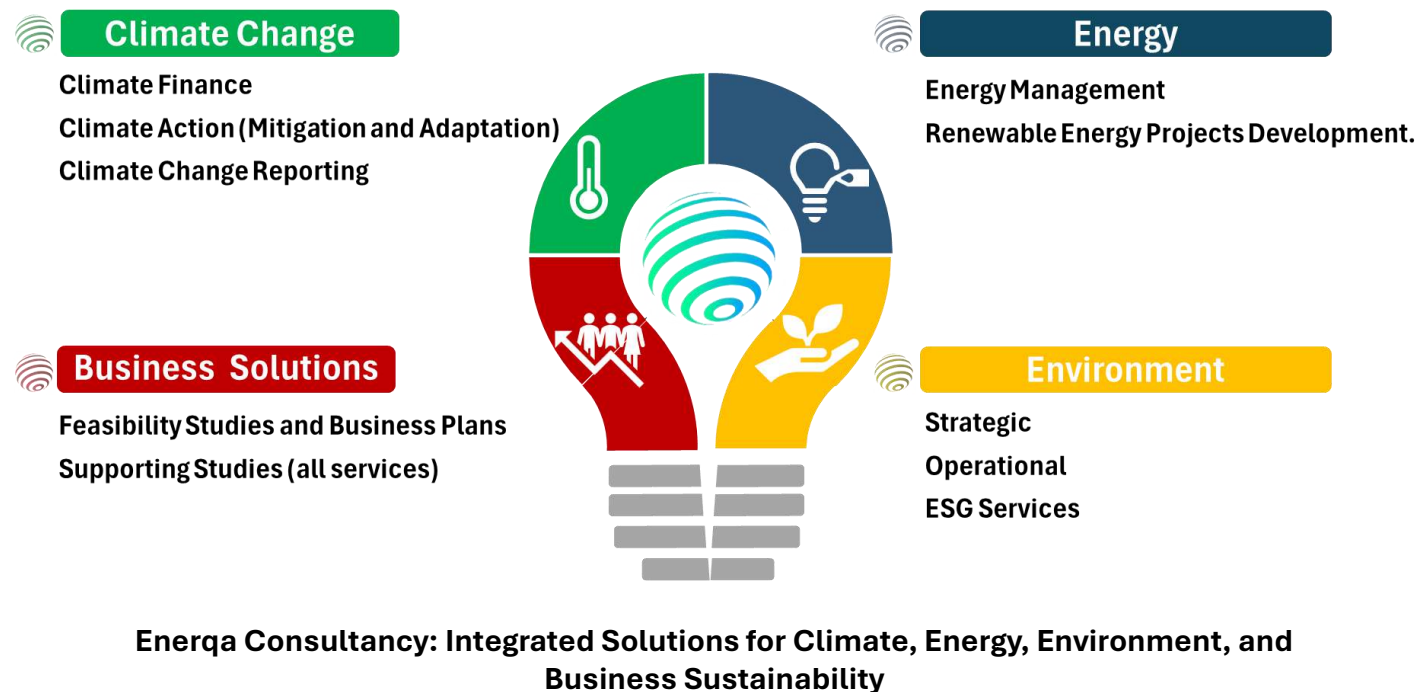
From knowledge to practice

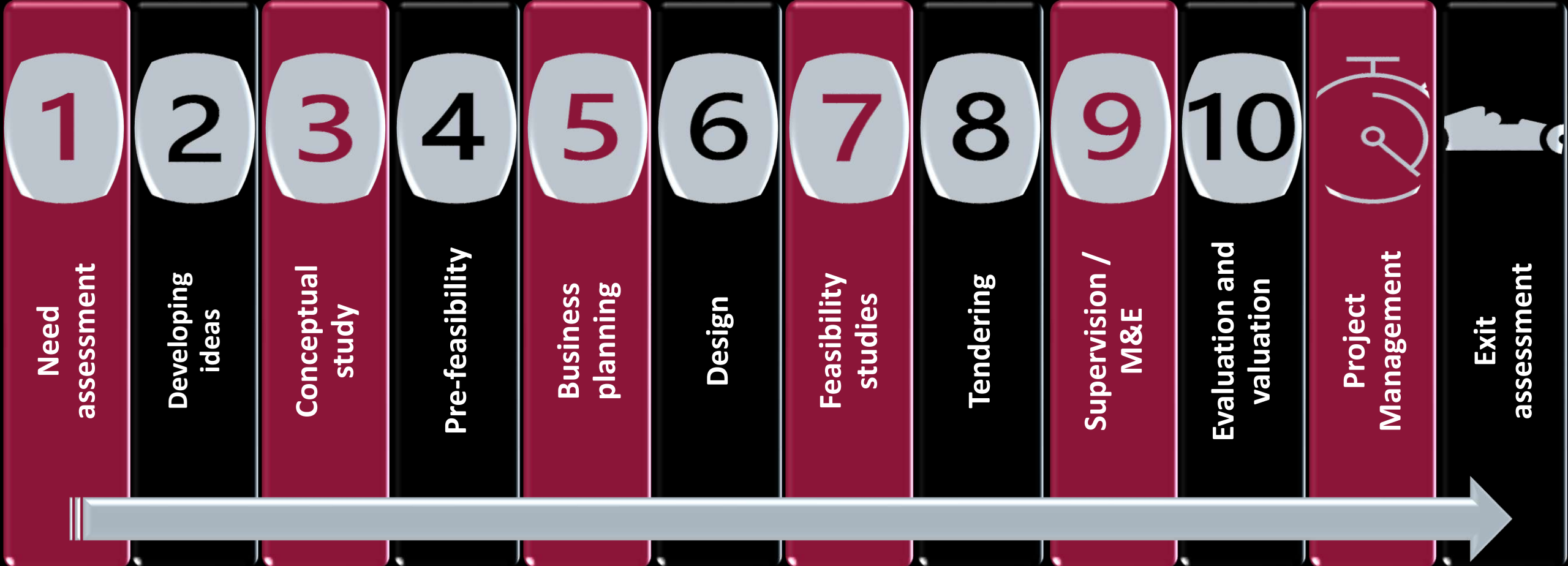
Mission

Committed to avail knowledge and good practices in the areas of energy, climate change, environmental & social safeguards and related business solutions to its clients everywhere in the world.

Enerqa Consultancy provides specialized services across four key domains: Climate Change, Energy, Environment, and Business Solutions. In the Climate Change domain, Enerqa supports governments, corporations, and institutions in addressing climate challenges through greenhouse gas (GHG) inventories, carbon footprint assessments, climate risk analysis, and the development of climate adaptation and mitigation strategies. The company helps clients align with global climate commitments, such as the Paris Agreement and Nationally Determined Contributions (NDCs), by enhancing Measurement, Reporting, and Verification (MRV) systems and integrating climate action into policy frameworks. Within the Energy sector, Enerqa provides consultancy services that optimize energy efficiency, promote renewable energy adoption, and support energy transition strategies. The company specializes in feasibility studies for solar, wind, and other renewable energy projects, as well as energy audits and grid integration solutions. Through advanced modeling and strategic planning, Enerqa assists stakeholders in reducing energy costs, enhancing sustainability, and ensuring compliance with international energy regulations.

In the Environment field, Enerqa delivers environmental impact assessments (EIA), sustainability reporting, and ecological footprint analysis. The consultancy aids organizations in improving environmental performance, adopting circular economy practices, and ensuring regulatory compliance. With expertise in waste management, water conservation, and biodiversity protection, Enerqa contributes to the development of policies and frameworks that enhance environmental sustainability at national and corporate levels. For Business Solutions, Enerqa offers strategic advisory services that help organizations integrate Environmental, Social, and Governance (ESG) principles into their operations. The company provides ESG gap assessments, corporate sustainability planning, and green finance advisory. Through tailored consultancy, Enerqa enables businesses to align with global sustainability standards, enhance resilience, and leverage climate finance opportunities for long-term growth.





At Enerqa, we guide projects from concept to completion with a structured, results-driven approach. Our consultancy services ensure that every stage of your project is backed by expert analysis, strategic planning, and sustainable execution. We start with needs assessment, identifying key challenges and opportunities. From there, we develop ideas and conduct conceptual studies to shape innovative solutions. Our pre-feasibility and feasibility studies provide critical insights, ensuring viability and sustainability. We craft business plans and designs that align with market realities and stakeholder objectives. Our expertise extends to tendering, ensuring competitive and compliant procurement, followed by supervision, monitoring, and evaluation to maintain quality and efficiency. As projects evolve, we conduct evaluation and valuation, ensuring they deliver impact and value. With our project management expertise, we oversee execution to mitigate risks and optimize performance. Finally, our exit assessment ensures a smooth transition, securing long-term success and sustainability. Enerqa is your partner in turning ideas into actionable, high-impact projects—driving innovation, efficiency, and sustainability.



ENERQA's ROSTER

Enerqa's roster of experts includes a distinguished selection of professionals specializing in energy, environment, climate change, and business modeling.

Our expertise extends beyond those listed, as we collaborate with a broader network of specialists covering a wide range of disciplines, from renewable energy, carbon management, and ESG compliance to climate risk assessment, circular economy solutions, and sustainable finance. Our team is continuously expanding, ensuring we provide comprehensive, tailored solutions to meet the evolving needs of our clients across various industries and sectors.



Eng. Ali Bakhreiba

With 28 years of cross-industry expertise in engineering, management, and innovation, in USA and MENA region. He has a proven track record of leading teams across various industries, including engineering management, mechanical design, R&D, industrial consultancy, and automotive plant establishment. He holds a Master's degree in Mechanical Engineering from Bradley University (USA) and has secured two U.S. patents for innovative engineering solutions: US D 651,212, "Touch screen panel for UPS," and US 8,127,243, "UPS with improved manual user interface." His experience spans both the USA and MENA regions, and he is fluent in English and Arabic.

Dr. Samuel Bwalya

He has an extensive background in economics and development, having served as an Economics Lecturer at the University of Zambia (UNZA) and as Director of Research and Planning at the Zambia Revenue Authority. Dr. Bwalya also held the position of Managing Director at the Development Bank of Zambia (DBZ). In addition to his roles in academia and public service, he has been involved in green economy consulting with the Government of Zambia.

Dr. Hamid Elhag

Dr. Elhag is a sustainability and ESG expert with extensive experience in environmental auditing, energy transition, and corporate sustainability strategies. He has led ESG assessments, regulatory compliance audits, and policy reforms in the energy and industrial sectors, ensuring alignment with global sustainability standards. As a Technical Consultant and Lead Auditor, he collaborates with international firms on ESG reporting, carbon footprint reduction, and sustainable resource management. His current research at Aalto University, Finland, focuses on hydrogen safety and clean energy innovations to drive a low-carbon future.

Dr. Ir. Anas M. Hassan

Dr. Ir. Hassan is a Petroleum Engineering Specialist with a PhD from Universiti Teknologi PETRONAS in collaboration with TU-Delft and is currently a Post-Doctoral Research Fellow at Khalifa University. His expertise includes Reservoir Engineering, Enhanced Oil Recovery, and Transport Phenomena, integrating RFID, IoT, AI, and Circular Economy. He has authored numerous publications and presented at SPE ATCE, ADIPEC, and OTC. Dr. Hassan is a registered professional engineer in the Netherlands, an SPE Certified Petroleum Engineer, and an associate editor for leading international journals.

Alexandre Borde, PhD

Dr. Borde is a climate finance and low-carbon transition expert with over 20 years of experience. He has served as a Senior External Carbon Finance Expert for the UNFCCC, contributing to the Registration and Issuance Team. His background includes lecturing at Université de Versailles Saint-Quentin-en-Yvelines and Sciences Po, as well as roles at Dexia Asset Management, the French Ministry of Ecology, the World Bank, and the FAO.

Ammar M. Gomaha Gaber

Mr. Gaber is climatologist and modeling expert specializing in climate prediction, meteorological modeling, and environmental simulations. He has extensive experience in weather forecasting, air quality modeling, and GIS-based climate risk assessments. His work includes developing and implementing climate adaptation models, drought monitoring systems, and predictive analytics for agriculture and water resources. Ammar has contributed to advanced climate simulations, data-driven forecasting, and scientific research on climate variability and extreme weather events.



Islam Moh. Mahmoud, PhD

A registered Technical Expert Reviewer in the UNFCCC Roster of Experts, and a certified climate and renewable energy finance expert. Dr. Islam brings specialized expertise in renewable energy finance, sustainable business models, and mainstreaming climate finance into national policies. She has contributed to key national-level initiatives, including the Capacity Building Initiative for Transparency (CBIT), and has collaborated with governments and international organizations to deliver impactful, results-driven projects.

Abdelrahman Eltahir, PhD

25+ years of experience in Environmental Science. Worked on a wide range of projects, including ESIs for mining, renewable energy, agribusiness, and infrastructure projects. He is also an expert in environmental management, climate change, and occupational health and safety.

Amr Nasreldin, CEM, MEM

Amr Nasreldin is a registered Technical Expert Reviewer in the UNFCCC Roster of Experts and a Certified Energy Manager (CEM) from the Association of Energy Engineers. His expertise encompasses technical analysis, climate change mitigation strategies, and policy development and assessment within the energy sector.

Quosay Awad Ahmed, PhD

Dr. Ahmed is an Assistant Professor of Petroleum Engineering, and an environmental and sustainability expert. He has extensive experience in greenhouse gas emissions reporting, environmental impact assessments, and sustainability initiatives. Dr. Quosay has collaborated with leading international organizations like the UNEP, UNDP, UNIDO, UNFCCC, and IPCC. He is skilled in preparing comprehensive greenhouse gas inventories, climate change mitigation strategies, and environmental assessments, aligning them with global standards. He has also number of publications in oil and gas, energy and transport sectors.

Jacob Park, PhD

Dr. Park is an Associate Professor specializing in the social and environmental dimensions of innovation, entrepreneurship, and international business, with expertise in emerging economies. He serves as Co-Executive Editor of Subsistence Marketplaces and Associate Editor for several other journals. A leading scholar, he is Thematic Chair of Business for the IUCN World Conservation Union and a Coordinating Lead Author for the UN Environment Global Environment Outlook (GEO-7) Report. His extensive experience includes service on the Renewable Energy and Adaptation to Climate Technologies Investment Committee and prior roles in investment management. He has received numerous awards, including the Responsible Research in Management Award and the U.S. EPA Environmental Merit Award.

Muzamil Hassan, PhD

Dr. Hassan is a process engineering consultant. He works also as instructor in Chemical Engineering at the University of Doha for Science and Technology. His research focuses on heterogeneous catalysis and sustainable energy conversion. He earned his Ph.D. from the University of Yamanashi, Japan in 2018, where his research on catalytic partial oxidation of methane under supercritical water conditions was funded by the Japanese government.

Mariassole Pepa, PhD

Dr. Pepa is a researcher in Human Geography and a proven track record in international cooperation and development. With 8 years of experience, Pepa has played key roles in several national and international research projects and international academic programs.

Mohamed M. Ahmed, MSc

Mohamed is a certified IBM data analyst with a strong background in engineering and business. He has experience in the energy and climate change sector, leveraging his skills in data analysis, wrangling, and modeling using tools like Excel, SQL, and Python. He is also proficient in data visualization and storytelling using Tableau and Power BI. His engineering background, particularly in oil and gas, coupled with his business acumen, makes him well-equipped to analyze complex data and provide valuable insights.

Yaseen Mohamed

A highly skilled analyst who holds substantial potential in his field of Corporate Finance, Carbon Finance and Management Accounting, and the winner of 3 awards in 3 consecutive years at Carleton University.

Sean White, PhD

With a substantial experience of 30+ years in Forestry and Environmental Impact Assessment, Mr. White has been a lead consultant in many major projects in many countries



Mr. Giovanni Fabbio

Mr. Fabbio is a seasoned financial professional with extensive experience in navigating complex financial landscapes across the Middle East, Europe, and Asia. He specializes in M&A, turnaround strategies, and financial restructuring. With a focus on sustainable growth and profitability, Giovanni has successfully led teams in various sectors, including manufacturing, recycling, IT, and construction. He is adept at identifying and addressing inefficiencies, optimizing operations, and driving financial performance.

Arig Bakhiet, PhD

20+ years of experience in Renewable Energy, Energy Management and Climate Change, which included the development of national renewable energy strategies, establishment of renewable energy business incubators and assessment of climate change mitigation potentials.

Khalid Elmeiri, PhD

12 years of academic and industrial experience in Energy and Environmental economics, and Business development. The scope of his research is the energy mix strategies of power generation for oil economies in MENA region, Europe and North America.

Eng. Yousef Ibrahim Mohammed

Eng. Mohammed is an accomplished Environmental and Sustainability Specialist with over 10 years of experience in the oil and gas, cement, and mining industries. He has expertise in Environmental Impact Assessments, greenhouse gas management, and ESG reporting, with a strong focus on advancing sustainability in industrial operations. Currently spearheading initiatives at Enerqa Consultancy, Yousef is dedicated to driving environmental stewardship and fostering sustainable development in Qatar and beyond.

Mr. Duncan Millard

Mr. Millard is an energy statistics expert with over 15 years of experience. He served as Chief Statistician for both the UK Department of Energy and Climate Change and the International Energy Agency. Currently, he advises countries, businesses, and organizations on energy data analysis for informed decision-making, particularly during energy transitions. His work includes developing energy statistics training programs in Africa and supporting countries in building energy data for Nationally Determined Contributions (NDCs), including work with Saudi Arabia, Malawi, Uganda, Nigeria, Zambia, Sudan and Mozambique. He has also worked with ASEAN on energy transition indicators.

Eng. Yaseen Hatim

Yaseen Hatim is a skilled process engineer and environmental consultant specializing in energy management, renewable energy systems, GHG emission calculations, and sustainable industrial practices. A certified ISO 50001 Lead Energy Auditor, he has led projects in environmental assessments, feasibility studies, and energy optimization. Yaseen is committed to sustainability, creative problem-solving, and advancing solutions that balance operational efficiency with environmental responsibility.

Eng. Reem Elmalik

A chemical engineer actively working to advance the renewable energy sector and reduce our reliance on fossil fuels. By educating the next generation, she aims to inspire a new wave of environmental stewards who will drive positive change for a sustainable future.

Eng. Abdelaziz Ahmed

Abdelaziz is a skilled data analyst with a strong background in engineering and sustainability. With a B.Eng. in Electrical and Electronics Engineering and a Data Analytics Professional Certificate, he has contributed to impactful sustainability projects with organizations like the UNDP and AFDP. He is passionate about leveraging data analytics to drive business growth and contribute to a sustainable future.

Ashish Shrestha, MSc

Eng. Shrestha is an electrical engineering consultant and academic with a focus on sustainable energy solutions. He has led many research initiatives aimed at rural electrification in Nepal's mountainous regions.

Mohammed Yassin, MSc

Eng. Yassin is a senior Electrical Engineer with 10+ years of experience, and a demonstrated history of working in the international trade and development industry.





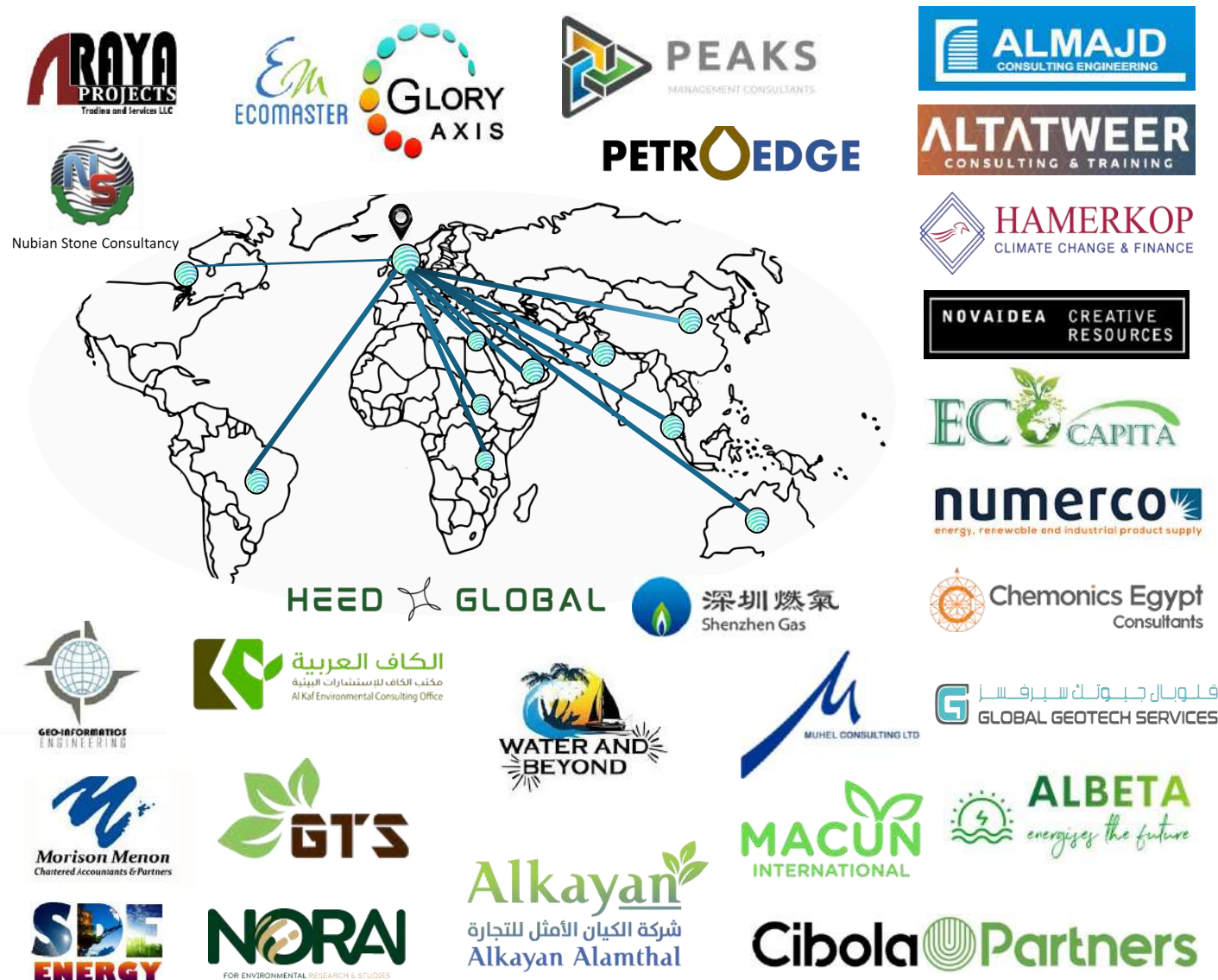
A Global Network of Experts



Enerqa brings together a diverse roster of associates, partners, and experts from over 25 countries across all regions of the world. Our specialists cover a broad spectrum of subspecialties relevant to our advisory services, ensuring comprehensive, multidisciplinary expertise for every project. Beyond our core team, Enerqa maintains a wide network of outsourced and on-call specialists, allowing us to scale and adapt to the unique needs of each engagement. Whether it's a highly specialized technical challenge or strategic project oversight, our global network ensures that the right expertise is always within reach.

At Enerqa, our strength lies in our global network of business partners—a diverse coalition of firms and experts spanning multiple industries and regions. As shown in the image, our extended network connects us with specialized consultancies, engineering firms, environmental experts, financial advisors, and technical specialists from across the world. Through these strategic partnerships, we enhance our ability to deliver comprehensive, integrated solutions that address the complex challenges our clients face. Whether it's technical advisory, feasibility studies, project management, or sustainability consulting, our partners bring local expertise, industry insights, and cutting-edge innovations to every project. By collaborating with trusted firms and specialists, we ensure that our clients receive tailored, high-quality solutions, backed by global knowledge and regional expertise—turning visions into reality with efficiency, impact, and sustainability.

EnerQA's Partners



2. Tools and Modalities



Introduction to Enerqa's tools, software and customized applications

Enerqa Consultancy leverages a suite of proprietary and globally recognized tools to deliver impactful climate change solutions. Our comprehensive approach supports clients in navigating the complexities of GHG emissions reporting, Measurement, Reporting, and Verification (MRV), energy transition modelling, and policy impact assessments. We combine cutting-edge technology with deep methodological expertise to drive effective and sustainable outcomes.

Our Arsenal for a Sustainable Future:

We utilize a powerful combination of internally developed tools and established international methodologies, ensuring accuracy, transparency, and alignment with best practices.

Enerqa's Proprietary Tools:

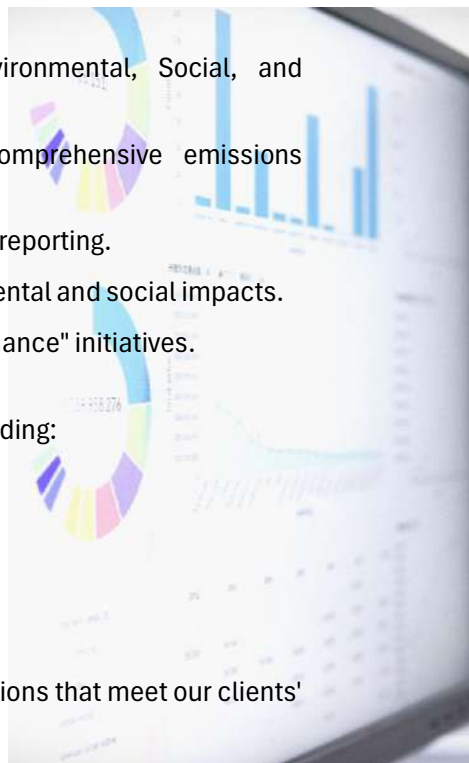
- **ESG Readiness Tool:** Assessing and enhancing your Environmental, Social, and Governance performance.
- **GHG Emissions Calculator:** Streamlining accurate and comprehensive emissions accounting.
- **MRV Tool:** Facilitating robust data management and transparent reporting.
- **ESIA Risk Assessment Tool:** Identifying and mitigating environmental and social impacts.
- **Green Project Scoring Tool:** Determining eligibility for "Green Finance" initiatives.

Key Methodologies & Frameworks:

We also leverage leading global standards and methodologies, including:

- IPCC Guidelines
- GHG Protocol
- LEAP (Long-range Energy Alternatives Planning System)
- Verra's Verified Carbon Standard (VCS)
- GORD- GCC
- UNFCCC Methodologies

This integrated approach empowers Enerqa to provide tailored solutions that meet our clients' specific needs and contribute to a cleaner, more sustainable future.



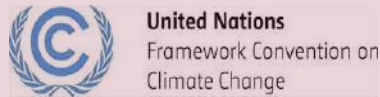
Imagine a small business striving to reduce its carbon footprint but struggling to track emissions efficiently. Or a large corporation needing to align with ESG regulations and investor expectations while managing complex supply chains. Perhaps a government entity is aiming to implement national sustainability policies, or an NGO wants to measure and report the impact of its climate projects with credibility. No matter the sector, the challenges are similar: where to start, how to measure impact, and how to ensure real progress.

At Enerqa, we understand that every organization is at a different stage in its sustainability journey—which is why our suite of customizable sustainability tools is designed to adapt to your unique needs, expertise level, and operational scale. For businesses and corporates, our ESG Readiness Tool provides a clear roadmap to ESG excellence, helping companies identify gaps, track progress, and align with global sustainability standards. If you're a government body or policymaker, our GHG Emissions Calculator offers a precise, scalable solution for national carbon accounting and reporting. NGOs and project developers can rely on our MRV Tool for transparent monitoring and verification, ensuring their sustainability efforts meet funding requirements and deliver real impact. And for investors or institutions seeking green finance, our Green Project Scoring Tool simplifies the process of assessing project eligibility, making it easier to access climate funding and drive sustainable investments.

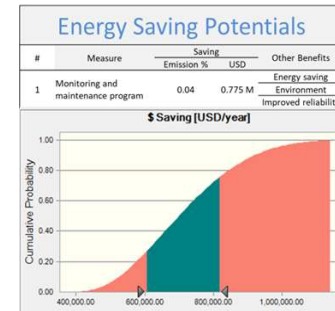
But what makes these tools truly powerful is how they work together. A corporate using our GHG Emissions Calculator to track its carbon footprint can seamlessly integrate the MRV Tool for compliance reporting. An SME looking to improve ESG performance can pair our ESG Readiness Tool with the Green Project Scoring Tool to unlock sustainability-linked financing opportunities. A government planning a climate strategy can utilize our ESIA Risk Assessment Tool to ensure environmental and social safeguards before launching a national initiative.

No matter your sector, expertise level, or sustainability goals, Enerqa's tools are designed to simplify the process, eliminate guesswork, and provide actionable insights that drive measurable results. You don't need to be a sustainability expert to get started—we provide the technology, guidance, and adaptability to make sustainability work for you.

The future of sustainability is here—customized, integrated, and built to help you succeed. Let Enerqa's smart tools power your sustainability journey.



	Min	Mean	Max	Simulation results
Activity Data				
Total Crude processed bbl/year	23130241	29964017	33313587	29383316.3
Total Crude processed Tonne/year	3209998	4286810	4783351	4204598.0
Total Crude processed CUM/year	3869553	4763122	5312012	4671775.6
Light products Tonne/year	2534543	3225658	3583158	3170188.9
Total Crude processed PJ/year	141.5339	183.3498	263.8458	179.8
Emission factors				
Refining process, [kg CH4 / PJ refined]	90	745	1400	745.0
Storage, [kg CH4 / PJ refined]	20	135	250	135.0
Refining and storage, NMVOC, [kg/Mg crude oil input]	0.67	0.2	0.61	0.3
Venting and flaring, NMVOC, [g/m3 refined feed]	1	2	6	3.5
Storage, NMVOC, [kg/Mg gasoline handled]	0.01	0.06	0.6	0.3
Calculated Emissions				
CH4 Emission from refining process, t/year	12.73805	136.5556	285.5842	133.9
CH4 Emission from storage, t/year	2.830679	24.75223	50.56146	24.3
NMVOC Emissions from refining process, t/year	230.9299	857.362	2916.624	1429.6
NMVOC Emissions from venting and flaring, t/year	3298.998	8573.62	26688.11	14716.1
NMVOC Emissions from storage, t/year	25.34543	193.5515	2149.895	966.9
Total VOC Emissions Tonne/year	3571	9786	34091	12771

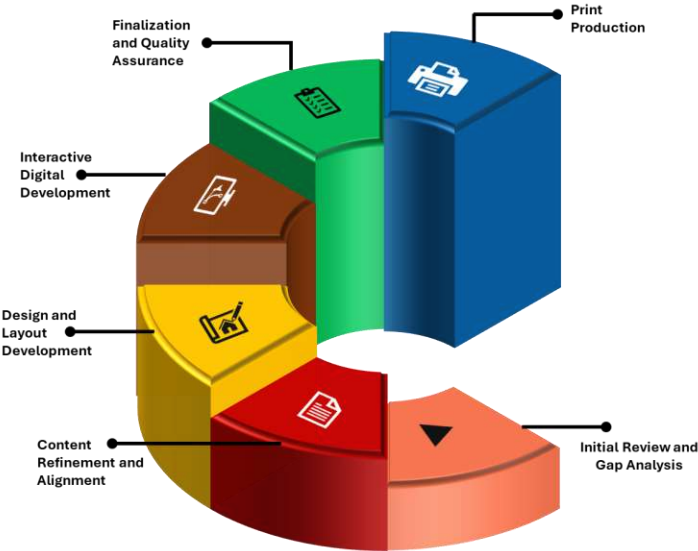


ESG Readiness Checklist									
Item	Criteria	Questions to Consider	1	2	3	4	5	6	7
Energy Management	Renewable Energy Use	What percentage of energy comes from renewable sources?							
	Energy Efficiency	What are the total greenhouse gas emissions?							
	Energy Usage	How to reduce energy consumption and improve?							
	Renewable Energy	How the organization plans to increase its renewable energy?							
Climate Change	Climate Change	How the organization plans to increase its renewable energy?							
	Climate Change	How the organization plans to increase its renewable energy?							
	Climate Change	How the organization plans to increase its renewable energy?							
	Climate Change	How the organization plans to increase its renewable energy?							
Social	Human Rights	How the organization plans to increase its renewable energy?							
	Human Rights	How the organization plans to increase its renewable energy?							
	Human Rights	How the organization plans to increase its renewable energy?							
	Human Rights	How the organization plans to increase its renewable energy?							
Governance	Board Diversity	How the organization plans to increase its renewable energy?							
	Board Diversity	How the organization plans to increase its renewable energy?							
	Board Diversity	How the organization plans to increase its renewable energy?							
	Board Diversity	How the organization plans to increase its renewable energy?							



Summary of Results		
Technology, Grid Connectivity and BESS Selection	Annual electricity consumption to be from Solar PV (MWh)	2,200 MWh
Technology selected	Monocrystalline Silicon	High
Performance factor	0.85	
System Performance Degradation (%)	0.50%	
Effective Irradiance Factor (Based on tracking technology)	1.00	
Ground Cover Ratio	0.50	
Grid connection	On-grid	
Battery Energy Storage System	Yes	

Enerqa Consultancy offers a comprehensive suite of Sustainability and ESG services, helping clients navigate the evolving landscape of environmental, social, and governance standards. Our expertise includes readiness assessments to evaluate organizational preparedness, gap assessments to identify areas for improvement, and specialized studies to provide in-depth analysis on key sustainability challenges. We support clients in ESG reporting, ensuring alignment with global frameworks, while our editing and auditing services enhance accuracy, credibility, and compliance with sustainability standards. Through tailored solutions, Enerqa empowers businesses to integrate sustainability into their operations, drive impact, and meet stakeholder expectations.



- Readiness
- Gap assessment
- Specialized studies
- Reporting
- Editing
- Auditing





EMPOWERING ORGANISATIONS TOWARDS SUSTAINABLE PRACTICES



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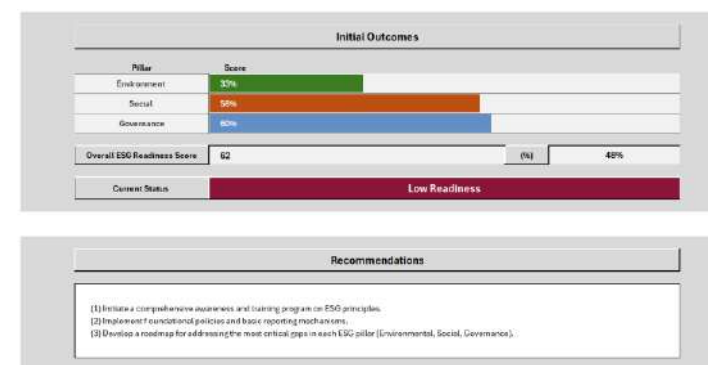
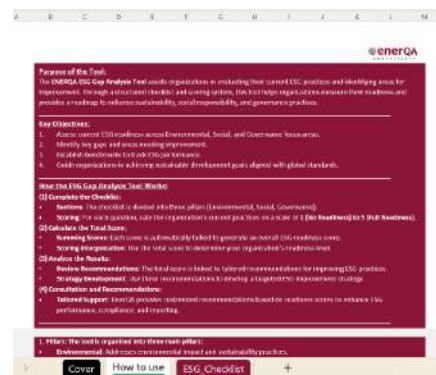
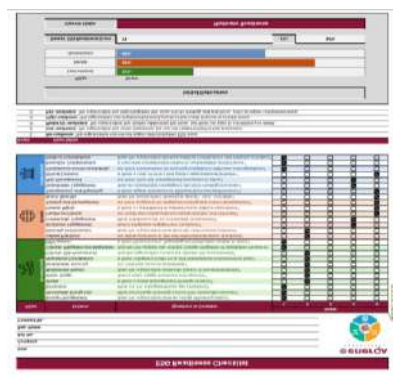
ENVIRONMENTAL, SOCIAL AND GOVERNANCE (ESG)

ESG stands for Environmental, Social and Governance. It refers to the three central factors used to measure the sustainability and societal impact of an investment in a company or business.



EnerQA has launched a free-to-download ESG Readiness Tool, designed on Microsoft Excel to provide an intuitive, user-friendly solution for assessing and improving ESG performance. The tool simplifies the evaluation of Environmental, Social, and Governance pillars through structured criteria and guiding questions, allowing users to input their practices, score their readiness, and receive tailored recommendations for improvement.

The tool's Excel format makes it accessible and customizable, enabling organizations of all sizes, especially SMEs, to adapt it to their specific needs without requiring additional software or expertise. By offering actionable insights based on scoring, it empowers users to address ESG gaps and align with sustainability and governance standards efficiently. EnerQA's tool is an ideal resource for companies aiming to enhance transparency, meet stakeholder expectations, and align with global goals. Download it now and take the first step toward measurable ESG excellence.



GreenScale Pro



	A	B	C	D	E	F	G	H	I	J	Score	
1		Project's Green Scoring Scale										5
2												4
3												3
4												2
5												1
6												
7												
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Category	Weight (%)	Scoring					Weighted Score	Remarks
		1	2	3	4	5		
Alignment with SDGs	15%				4		0.6	Targeting multiple SDGs improve
GHG Emissions Reduction	15%					5	0.75	Higher reductions result in better
Renewable Energy and energy efficiency	10%			3			0.3	Use of innovative solutions score
Water and Circular Economy	15%					5	0.75	Efficiency and recycling initiatives matter.
Pollution Prevention and Biodiversity	15%				4		0.6	Measures to improve environmental quality.
Paradigm Shift Potential	10%			3			0.3	Innovative, replicable, or scalable projects score higher.
Uniqueness and Scalability	10%			3			0.3	Innovative and scalable projects rate higher.
Alignment with Qatar Vision 2030	10%					5	0.5	Strong alignment contributes to score.
Total	100%						4.1	Overall project rating.

Fianl Project Score	4.1
Justification	The project's 4.1 score demonstrates its strong impact on sustainability and its alignment with key green project criteria. However, there is potential for further innovation and scalability, particularly in clean transportation and renewable energy integration, to enhance its overall impact.

<>

ScreeningGreen ScoringSDGsGHG reductionREWater and Circular Economy

Uniqueness and Scalability

Score	Definition	Project scale	Justification
5	Exceptional: Groundbreaking and highly scalable project with global replication potential.		The Al Asriya Factory demonstrates moderate alignment with the uniqueness and scalability criteria, justifying a score of 3. Its adoption of rotary tilting furnaces, oxy-fuel combustion systems, and advanced sorting technologies reflects a commitment to sustainable and efficient recycling practices. These technologies enhance material recovery and energy efficiency, making the factory a model for replication
4	High: Strongly innovative and scalable with		
3			
2			
1			

GHG Reduction Scale

Score	Definition	Project scale	Justification
5	Exceptional Reduction: Significant GHG reduction, exceeding 50% of baseline emissions.		The Al Asriya Factory excels in reducing GHG emissions through innovative practices and advanced technologies. By utilizing 100% aluminum scrap in Phase 1 and 70% in Phase 2, the factory achieves up to 95% energy savings compared to primary aluminum production.
4	High Reduction: GHG reduction		
3			
2			
1			

Water and Circular Economy

Score	Definition	Project scale	Justification
5	Exceptional: Comprehensive recycling and water conservation systems with a circular economy model.		The Al Asriya Factory demonstrates exemplary performance in Water and Circular Economy criteria, justifying a score of 5. Its closed-loop water cooling system reduces freshwater intake by up to 90% and minimizes wastewater discharge, showcasing a commitment to water conservation. The factory's water recycling and reuse practices ensure maximum utilization of industrial water, decreasing reliance on external water supplies and promoting resource efficiency. By adopting measures to minimize evaporation losses, the factory conserves significant water volumes, even in high-temperature environments. These practices not only align with but exceed Qatar's environmental regulations for water usage and discharge, reinforcing sustainable operations.
4	High: Significant recycling infrastructure or advanced water conservation measures.		In terms of circular economy, the factory employs 100% aluminum scrap in Phase 1 and 70% in Phase 2, significantly reducing reliance on virgin materials. Its advanced sorting technologies maximize material recovery, while closed-loop processes optimize resource efficiency and reduce operational waste. By extending the lifecycle of resources and creating a circular industrial ecosystem, the factory contributes to global circular economy goals and Qatar's environmental vision. These integrated strategies highlight the factory's leadership in sustainability and resource conservation
3	Moderate: Moderate-scale recycling programs or water efficiency improvements.		
2	Limited: Limited focus on recycling or water efficiency.		
1	Minimal: Negligible recycling or water conservation impact.		

Steps

- Step 1: Project Screening
- Step 2: Green Scoring
- Step 3: Detailed Criteria Evaluation
- Step 4: Recommendations and Justifications

Criteria

- Sustainable Development Goals.
- Renewable energy and energy efficiency measures.
- Pollution & Biodiversity
- Uniqueness & Scalability
- GHG Reduction
- Water and Circular Economy
- Paradigm Shift
- Country Long-term Vision

Enerqa’s Risk Assessment Tool: A Comprehensive Approach to Environmental and Social Risk Management

Enerqa has developed a Risk Assessment Tool that integrates qualitative and quantitative methodologies to systematically evaluate environmental and social risks across all project phases. This tool is particularly instrumental in Environmental and Social Impact Assessment (ESIA) studies and other environmental and climate-related projects. It enables project developers, regulators, and stakeholders to identify, analyze, and mitigate potential risks while ensuring compliance with sustainability standards.

Key Features of the Risk Assessment Tool

- Structured Risk Evaluation
- Multi-Dimensional Risk Scoring
- Decision-Support Framework
- Adaptability Across Project Phases
- Compliance and Sustainability Alignment

Risk Rating (RR) = Probability of Occurrence (OV) x Severity of Consequences Value (CV)																				
Probability /Likelihood	Consequence (Extent + Duration + Magnitude)																			
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
	2	2	4	6	8	10	12	14	16	18	20	22	24	26	28	30	32	34	36	40
	3	3	6	9	12	15	18	21	24	27	30	33	36	39	42	45	48	51	54	60
	4	4	8	12	16	20	24	28	32	36	40	44	48	52	56	60	64	68	72	80
5	5	10	15	20	25	30	35	40	45	50	55	60	65	70	75	80	85	90	95	100

Activities (What are the Project Actions Causing Risks?)	Receptors (Who or What is Affected?)	Impacts (What are the Potential Consequences?)
a. Construction Phase Activities Site preparation (clearing vegetation, excavation) Earthworks and grading Transportation of materials (dirt, equipment, fuels) Use of heavy machinery (diesel, oil leaks, fuel spills) Waste management (solid waste, hazardous waste disposal) Water use and management (diversion, pollution discharge)	a. Human Receptors Local communities (residents, workers) Public health (vulnerable populations, schools, hospitals) Neighborhood areas (roads, tourist areas) Cultural heritage sites b. Ecological Receptors Flora (terrestrial and aquatic vegetation) Fauna (wildlife, sensitive species, migratory birds) Protected areas (national parks, wildlife sanctuaries) Biodiversity hotspots Water bodies (rivers, lakes, wetlands)	a. Environmental Impacts Air Pollution: Emission of pollutants (PM, NOx, SOx, CO2) Water Pollution: Contamination of water sources due to spills, effluents, or leachates Soil Contamination: Chemical spills, hazardous waste, soil degradation Noise and Vibration: Impact on local communities and wildlife Reliable Destruction: Deforestation, loss of cover crops, sandbars Climate Change Contributions: Increase in greenhouse gas emissions
b. Operational Phase Activities Industrial operations (manufacturing, processing) Energy and emissions (power generation, fuel burning) Effluent discharges (wastewater treatment) Waste generation and disposal (solid, liquid, gaseous waste) Transportation and logistics (truck fleet operations)	c. Physical Environment Air quality (atmospheric, ambient air) Soil and geology (landslides, contamination potential) Surface water and groundwater Climate (temperature, precipitation patterns)	b. Human Health and Safety Impacts Occupational Health Risks: Exposure to hazardous substances, dusts, and noise Community Health Risks: Waterborne diseases, respiratory issues from air pollution Food Safety Risks: Contamination of agricultural produce or fish stocks
c. Decommissioning Phase Activities Demolition and dismantling (waste management, emissions) Site rehabilitation (land restoration, waste disposal) Handling of hazardous materials (leakages, oil residues)	d. Socioeconomic Repercussions Livelihoods (fishing, farming, tourism) Infrastructure (transportation networks, utilities) Economic sectors (agriculture, industry, tourism)	c. Socioeconomic Impacts Displacement: Relocation of communities due to project expansion Livelihood Disruptions: Loss of income for farmers, fishers, and tourism Traffic Congestion and Accidents: Increased road use leading to potential accidents Resource Conflicts: Competing demands for water, land, or energy resources



Enerqa’s Risk Assessment Tool

Enerqa’s Comprehensive Risk Assessment Tool for Environmental and Climate Projects

Enerqa has developed a state-of-the-art Risk Assessment Tool that integrates both qualitative and quantitative methodologies to evaluate environmental and social risks at every phase of a project. Designed to support Environmental and Social Impact Assessment (ESIA) and other environmental and climate-related initiatives, this tool provides a structured, data-driven approach to identifying, analyzing, and mitigating risks that could impact project sustainability and compliance.

Key Features of the Risk Assessment Tool

- Holistic Risk Evaluation – Covers environmental, social, and climate-related risks, ensuring a comprehensive assessment of potential impacts.
- Integrated Methodology – Utilizes a hybrid approach combining qualitative insights (expert judgment, stakeholder engagement) and quantitative analysis (risk probability, impact severity modeling).
- Multi-Phase Application – Adaptable across all project phases, from feasibility and planning to implementation and decommissioning.
- Decision-Support Framework – Provides actionable risk mitigation strategies, supporting project developers, policymakers, and investors in making informed, sustainable decisions.

Action Phase	Receptor (Environmental Component)	Potential Impact	Consequence	Probability of Occurrence (OV)										Severity of Consequences (CV)	Risk Rating (RR)	Mitigation Measures (Prevention/Reduction)	Consequence Severity			
				Low	Medium	High	Very High	Extremely High	Unlikely	Medium	High	Very High	Extremely High				Low	Medium	High	Very High
				1	2	3	4	5	6	7	8	9	10				1	2	3	4
Construction Phase	Noise	High noise levels from equipment	Disturbance to local communities and wildlife	2	3	4	5	6	7	8	9	10	11	Medium	12	Implement noise barriers, use sound-dampening equipment, and schedule noisy activities during off-peak hours.	2	3	4	5
	Air Quality	Dust and emissions from equipment	Respiratory irritation, reduced visibility	3	4	5	6	7	8	9	10	11	12	Medium	13	Use water sprays to suppress dust, maintain equipment, and monitor air quality.	3	4	5	6
	Water Quality	Spills of oil, fuel, or chemicals	Contamination of surface water bodies	4	5	6	7	8	9	10	11	12	13	High	14	Implement strict spill prevention protocols, use spill kits, and monitor water quality.	4	5	6	7
	Soil Quality	Disturbance to soil structure and vegetation	Loss of topsoil and nutrients, erosion	5	6	7	8	9	10	11	12	13	14	High	15	Implement erosion control measures, revegetate disturbed areas, and monitor soil health.	5	6	7	8
	Waste Management	Improper disposal of construction waste	Visual pollution, soil contamination	6	7	8	9	10	11	12	13	14	15	High	16	Implement waste management plans, segregate waste, and ensure proper disposal.	6	7	8	9
Operational Phase	Water Quality	Effluent discharges from processing	Water pollution, degradation of aquatic life	3	4	5	6	7	8	9	10	11	12	Medium	17	Implement effluent treatment, monitor discharge, and maintain water quality standards.	3	4	5	6
	Air Quality	Greenhouse gas emissions from energy use	Contribution to climate change	4	5	6	7	8	9	10	11	12	13	Medium	18	Optimize energy efficiency, use renewable energy sources, and monitor emissions.	4	5	6	7
	Soil Quality	Leakages from storage tanks or pipelines	Soil contamination, groundwater pollution	5	6	7	8	9	10	11	12	13	14	High	19	Implement leak detection systems, use secondary containment, and monitor soil and groundwater.	5	6	7	8
	Waste Management	Improper disposal of industrial waste	Visual pollution, soil and water contamination	6	7	8	9	10	11	12	13	14	15	High	20	Implement waste management plans, segregate waste, and ensure proper disposal.	6	7	8	9
	Water Quality	Extraction of water for industrial use	Depletion of local water resources	7	8	9	10	11	12	13	14	15	16	Very High	21	Implement water conservation measures, use recycled water, and monitor water levels.	7	8	9	10
Decommissioning Phase	Soil Quality	Disturbance from land clearing and construction	Loss of topsoil and nutrients, erosion	3	4	5	6	7	8	9	10	11	12	Medium	22	Implement erosion control measures, revegetate disturbed areas, and monitor soil health.	3	4	5	6
	Air Quality	Dust and emissions from equipment	Respiratory irritation, reduced visibility	4	5	6	7	8	9	10	11	12	13	Medium	23	Use water sprays to suppress dust, maintain equipment, and monitor air quality.	4	5	6	7
	Water Quality	Spills of oil, fuel, or chemicals	Contamination of surface water bodies	5	6	7	8	9	10	11	12	13	14	High	24	Implement strict spill prevention protocols, use spill kits, and monitor water quality.	5	6	7	8
	Soil Quality	Disturbance to soil structure and vegetation	Loss of topsoil and nutrients, erosion	6	7	8	9	10	11	12	13	14	15	High	25	Implement erosion control measures, revegetate disturbed areas, and monitor soil health.	6	7	8	9
	Waste Management	Improper disposal of construction waste	Visual pollution, soil and water contamination	7	8	9	10	11	12	13	14	15	16	Very High	26	Implement waste management plans, segregate waste, and ensure proper disposal.	7	8	9	10



GHG365

GHG365 is a comprehensive carbon footprint assessment tool designed to help organizations measure, manage, and mitigate their greenhouse gas (GHG) emissions. Built on international standards such as IPCC guidelines and the GHG Protocol, the tool allows users to input various activity data—including fuel consumption, electricity usage, transportation, waste, and more—to generate detailed emissions reports across Scope 1, 2, and 3 categories

Operational Boundary Questions - Emissions Sources to Include

Guidance

Use the questions below to help you determine which emissions sources should be included in the inventory.

Emissions Source Questions

A typical office-based organization will likely have the following (Scope 1 (direct) and Scope 2 (indirect)) emissions sources:

- Stationary Combustion
- Refrigeration and A/C
- Electricity

If you answer "yes" to a question below, that emissions source should be included in your inventory. For each facility within the defined organizational boundary, collect the necessary data for the selected time period. Use the corresponding Excel sheet to quantify these emissions.

Tip: you may need to ask your landlord about heating sources, steam purchased and refrigerants

Stationary Combustion	Yes or No?
Do you have facilities that burn fuels on-site (e.g., natural gas, propane, coal, fuel oil for heating, diesel fuel for backup generators, biomass fuels)?	Yes
Mobile Sources	
Do any vehicles fall within your organizational boundary? This can include cars, trucks, propane forklifts, aircraft, boats. Only vehicles owned or leased by your company should be included here.	?
Refrigeration and Air Conditioning	
Do your facilities use refrigeration or air conditioning equipment?	Yes
Purchased Gases	
Do you purchase any industrial gases for use in your business? These gases may be purchased for use in manufacturing, testing, or laboratories.	Yes
Electricity	
Does your inventory include facilities that use electricity?	Yes
Steam	
Do you purchase steam for heating or cooling in your facilities?	Yes

The questions below refer to optional emissions sources. If you answer "yes" you may choose whether or not to include these emissions sources in your inventory. Use the corresponding sheet to enter data.

Business Travel	Yes or No?
Do your employees travel for business using transportation other than company-owned or leased vehicles (e.g., commercial airline flights, rental cars, trains)?	Yes
Employee Commuting	
Do your employees commute to work in personal vehicles or use public transportation?	Yes

Carbon Footprint Tool

Introduction

This Carbon Footprint Calculator is designed by **enerQA consultancy** which is a comprehensive tool designed to assist Small and Medium Enterprises (SMEs) in estimating their greenhouse gas (GHG) emissions. Developed in alignment with globally recognized standards such as the IPCC Guidelines, GHG Protocol, and EPA standards, this calculator provides a structured framework to calculate emissions across Scope 1 (direct emissions), Scope 2 (indirect emissions from energy use), and Scope 3 (other indirect emissions). The tool simplifies the process of carbon footprint calculation, empowering organizations to understand their environmental impact and take informed steps toward sustainability.

Office 21, Al-Mana Twin Tower, Street no. 231, Zone 23, P.O.



GHG365

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2. User Guide.
3. Organization.
4. Boundary Questions.
5. Report.
6. Fuels.
7. Refrigerant.
8. Owned Vehicles.
9. Waste.
10. Flight and Accommodation.
11. Business Travel, Land and Sea.
12. Freight goods.
13. Employees Commuting.
14. Water.
15. Offsets.
16. Unit Conversion.
17. Raw Data.

Back to Report

Option 1

Material Balance Method: Enter company-wide total gases stored and transferred (by gas) in Table 1.

Choose the appropriate gas from the drop down menu.

Inventory Change = difference of gas stored in inventory from beginning to end of reporting period.

(Includes only gas stored on-site (i.e. cylinders) and not gas contained within equipment).

Transferred Amount = gas purchased minus gas sold disposed during reporting period.

Gas purchased includes: Purchased for inventory, as part of equipment servicing (not from inventory) within purchased equipment and gas returned to the site after off-site recharging.

Gas sold/disposed includes: Returns to supplier, sales or disposals (including within equipment), and gas sent off-site for recycling, reclamation or destruction.

Capacity Change = capacity of all units at beginning minus capacity of all units at end of reporting period.

(Can be assumed to be capacity of new units minus capacity of retired units).

Table 1. Company-Wide Refrigeration Gas CO₂ Equivalent Emissions - Material Balance

Gas	Gas OWP	Inventory Change (lb)	Transferred Amount (lb)	Capacity Change (lb)	CO ₂ Equivalent Emissions (lb)
C2F6	12,200	100.0	20.0	50.0	170.0
SF6	22,800	1,000.0			1,000.0
N2O	298				
C-24F8	10,300				
CO2	1				
CO2	1				
N2O	298				
HFC 152a	124				
HFC 22	14,600				
HFC 134a	4,470				
Total (tonne)					0.5

Option 2. Material Balance Method (Simplified): Enter organization-wide total gases in units (or gas) in Table 2.

Choose the appropriate gas from the drop down menu.

New units are those introduced during reporting period. Do not include any data for new units pre-charged by supplier, disposed units were disposed of during the reporting period, and existing units are all others.

Charge/Recharge = gas added to units by organization or a contractor (do not include pre-charge by manufacturer).

Capacity = sum of the full capacity for all units (do not include new units pre-charged by manufacturer).

Amount recovered = total gas recovered from all retired units.

Table 2. Organization-Wide Refrigeration Gas CO₂ Equivalent Emissions - Simplified Material Balance

Gas	Gas OWP	New Units Charge (kg)	Existing Units Recharge (kg)	Disposed Units Capacity (kg)	Recovered (kg)	CO ₂ Equivalent Emissions (kg)
CO2	1					0.0
CH4	28					0.0
CO2	1					0.0
N2O	298					0.0
CH4	28					0.0
N2O	298					0.0
HFC 22	14,600					0.0
HFC 152a	124					0.0
HFC 134a	4,470					0.0
HFC 152a	124					0.0
HFC 134a	4,470					0.0
Total (tonne)						0.0

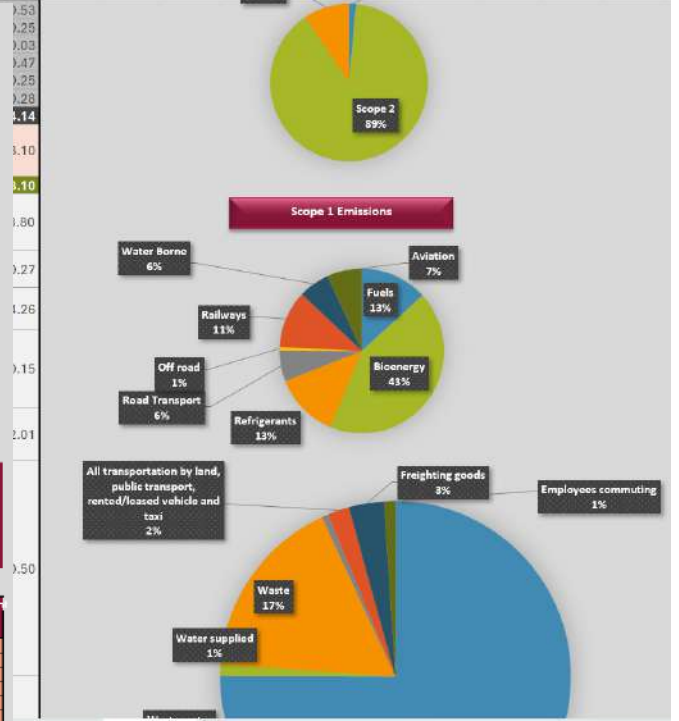
1. Enter the amount of fuel used in litres.
2. Enter the amount of the Fuel in Litres.
3. Enter the Amount of Bioenergy source (kg).

Do not include here the Combustion of fuels in transportation devices as automobiles, trucks, buses, Trains, Boats, ships, barges, vessels, etc. Please enter the amount of each applicable fuel.

Fuel/Type	English description	Amount (litres)	Density (kg/l)	kg CO ₂ e
Crude Oil		100	0.9	269.05
Crude Oil		100	1.2	267.72
Refined Gas Oil			0.841	
Refined Gas Oil	Motor Gasoline		0.7459	
Refined Gas Oil	Aviation		0.8	
Refined Gas Oil	Gasoline		0.8	
Refined Gas Oil	Jet Gasoline		0.804	
Other Kerosene			0.81	
Shale Oil			1.9	
Crude/Refined Oil			0.85	
Refined Fuel Oil			0.92	
Liquefied Petroleum Gases			0.525	
Ethane			1.2662	
Propane			0.24	
Butane			1.01	
Isobutane			0.6	
Petroleum Coke			2.25	
Refinery Feedstocks			0	
Refinery Gas 2			0	
Paraffin Waxes			0.9	
Refinery Soot and Tar			0.87	

Type	Fuel	Amount (kg)	kg CO ₂ e
Solid Biofuels	Wood/Wood Wastes	1000	1,778.9
	Sugarcane Lys		0
	Black Liquor		0
	Other Primary Solid Biofuels		0
Liquid Biofuels	Charcoal		0
	Bioethanol		0
	Biomass		0
	Other Liquid Biofuels		0
Gas Biofuels	Landfill Gas		0
	Sludge Gas		0
	Driver Biogas		0
	Municipal Waste		0
Other non fossil fuels	Municipal Waste		0
	Household Waste		0
	Household Waste		0
	Household Waste		0
Total (tonne)			1.79

A. Heaters, Incinerators, engines, flames, etc. and direct control of the reporting organization. All factors





ESRQ For Infrastructure Real-estate Developers

Enerqa Sustainability & Resilience Quantification

This tool quantifies sustainability and climate resilience performance for buildings and infrastructure using Enerqa's weighted indicator framework.

Enter project data and evidence, score each indicator, and instantly generate:

- overall ESRQ score,
- category performance,
- confidence-adjusted results,
- and a prioritized action roadmap.



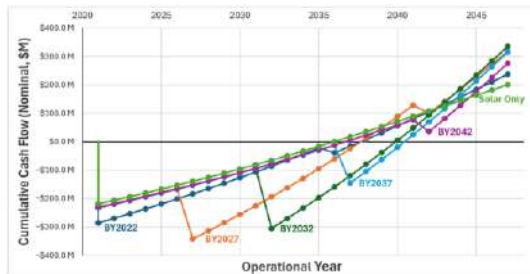
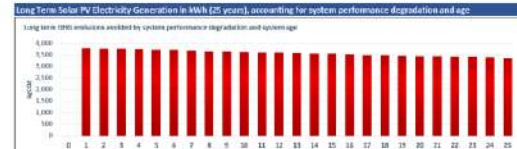
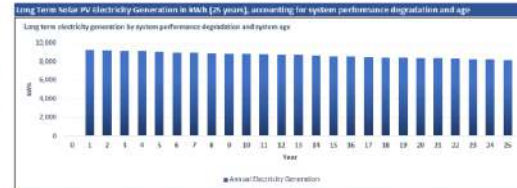
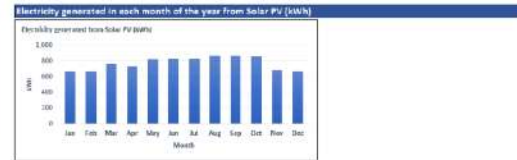
ENERQA

Sustainability & Resilience Quantification (ESRQ)

DECISION-SUPPORT TOOL FOR DEVELOPERS & FINANCIERS



Technology, Grid Connectivity and BESS Selection	
Annual electricity consumption to be from Solar PV (kWh)	7,200 kWh
Technology selected	Monocrystalline Silicon
Technology Efficiency	0.2
Performance Ratio	0.87
System Profile Factor (degradation %)	0.90%
Effective Irradiance Factor (based on tracking technology)	1.00
Ground Cover Ratio	0.50
Grid connection	On-grid
Battery Energy Storage System	Yes
Total Area Production Needed (m ² /m ²)	21.48
Solar PV Capacity and Panels	
Solar PV Capacity Required	4.24 kWp
Total surface area of panels needed	23.20 m ²
Land area needed @ ground-mounting	47.39 m ²
Inverter and BESS Technology	
Available Inverter technology	Hybrid Inverter
Available Battery technology	Lithium Iron Phosphate (LFP)
Inverter	
Hybrid Inverter	
Efficiency (%)	98.00%
Bridge Rectifier (ac-cv)	1:50
Topology	Transformerless
Output Waveform	Pure-Sine-Wave
Safety Margin	15.00%
Total DC capacity required (kWh/Day)	29.42
Required inverter size (kW)	3.5
DC input voltage (V)	300
Inverter Life (years)	15
BESS	
Lithium Iron Phosphate (LFP)	
Battery capacity (kWh)	13
Battery capacity (kWh)	65.8
Depth of Discharge	80.00%
Round-Trip Efficiency	95.00%
Temperature derating factor	0.00%



Additional Tool Characteristics

- ✓ **User-Friendly Interface:** Features drop-down menus for selections, buttons for data pulling (e.g., irradiance), and default values for quick assessments. Includes guides on Monte Carlo analysis for uncertainty (mentioned in "Explanation" sheet).
- ✓ **Outputs and Visualizations:** Summarizes results in dedicated sheets (e.g., "Summary of Results" for overview, charts for long-term energy/GHG/cash flows).
- ✓ **Limitations:** Assumes 25-year project life; does not handle real-time data or advanced simulations; regional costs are estimates and may require updates.
- ✓ **Scope:** Primarily for residential/commercial applications, with utility-scale considerations via technology choices.



easySOLAR tool's Features :

1. User Input and Configuration
2. Technical Sizing and Optimization

- Location and Irradiance Data
- Load and Consumption Profiling
- Solar PV Array Sizing
- Inverter Selection and Sizing
- BESS Sizing
- Long-Term Energy Assessment

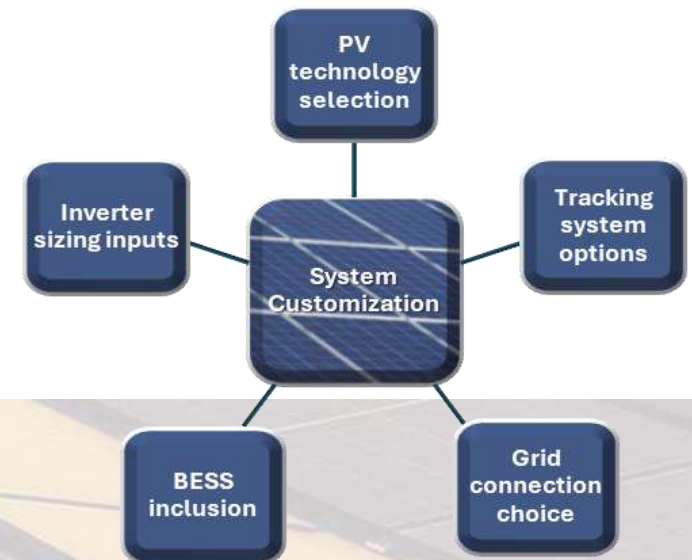
3. Financial Analysis

- Cost Estimation (CAPEX and OPE)
- Economic Metrics
- Savings Calculation

4. Environmental Impact Assessment and GHG Emissions Avoidance Calculations

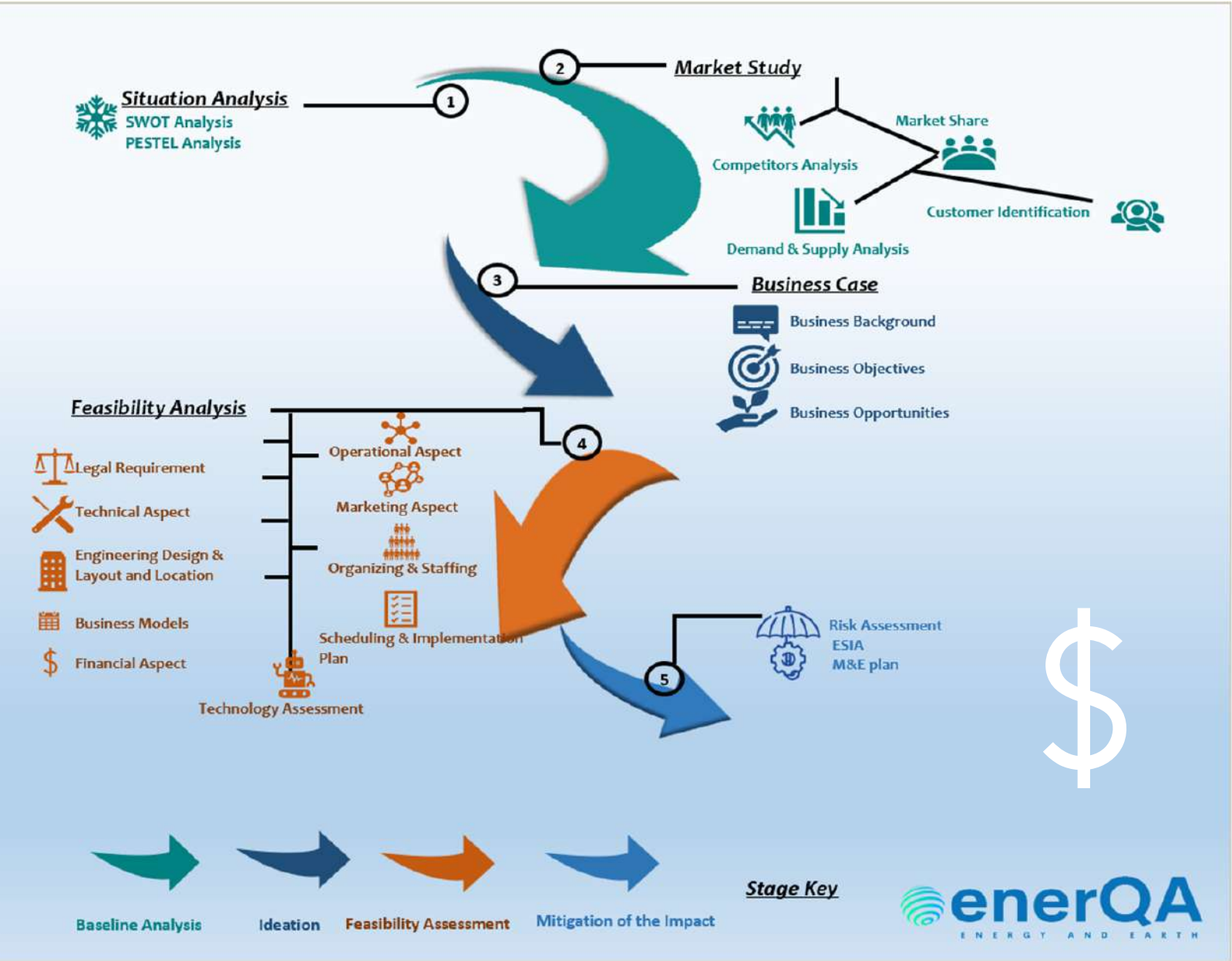
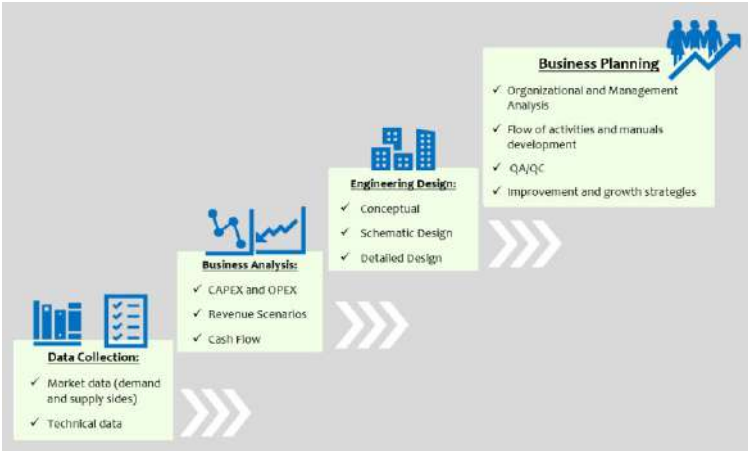
5. Data Sources and Assumptions

- Regional Databases
- Uncertainty and Margins
- Reference Data



Enerqa's business modeling innovative methodology

Enerqa leverages data analytics, AI-driven insights, and sustainability principles to craft robust business plans and feasibility studies. Our approach integrates ESG factors, energy efficiency, and regulatory compliance for long-term viability. We conduct market-driven assessments, scenario modeling, and risk analysis to optimize decision-making. Each strategy is tailored to industry-specific challenges and growth opportunities. With a focus on financial, technical, and environmental feasibility, we ensure sustainable and profitable outcomes. Enerqa transforms business planning into a strategic advantage.



3. Enerqa's Publications

Introduction

Enerqa's Publications: Sharing Knowledge, Driving Impact

At Enerqa Consultancy, we believe in the power of knowledge sharing to accelerate climate action. Our commitment to research and thought leadership is reflected in a diverse portfolio of publications, showcasing our analytical capabilities, data visualization expertise, and ability to translate complex information into actionable insights. This includes contributions to key national reports, alongside a range of other publications.

Demonstrating Our Expertise:

Our publications highlight Enerqa's deep understanding of climate change challenges and solutions, demonstrating our:

- **Research Capabilities:** Rigorous analysis and evidence-based insights drive our publications.
- **Content Creation Capabilities:** We communicate complex information clearly and concisely, making it accessible to a broad audience.
- **Analytical Capabilities:** Our publications showcase our ability to analyse data, identify trends, and develop practical recommendations.
- **Data Visualization Capabilities:** We utilize compelling visuals, including infographics, to effectively communicate key findings and facilitate understanding.
- **Infographic Creation Capabilities:** Our visually engaging infographics simplify complex data and enhance the impact of our research.

A Variety of Formats for Maximum Reach:

Enerqa's publications encompass a range of formats to maximize their reach and impact:

- **National Reports (NIRs, NCs, etc.):** Supporting countries in fulfilling their UNFCCC reporting obligations, including National Inventory Reports and National Communications.
- **Articles:** Sharing insights on current climate change topics and trends.
- **Scientific Papers:** Contributing to the body of knowledge on climate science and solutions.
- **Contributions to International Reports:** Collaborating with leading organizations to inform global policy and action.

The scope of Enerqa's publications is extensive, covering critical areas such as climate science, greenhouse gas emissions, carbon footprint assessment, ESG reporting, energy transition strategies, and hydrogen technologies. The **2024 Enerqa Publications Archive** reflects this diversity, consolidating over 24 major publications that delve into the evolution of climate change understanding, climate resilience in supply chains, and the environmental and economic implications of energy technologies. The company has also explored specialized topics such as the role of artificial intelligence in environmental sustainability, supercritical water technology for waste management, and the impacts of war on climate and resource security.

Enerqa's publications stand as a pillar of knowledge leadership, demonstrating scientific integrity, technical excellence, and real-world impact. Through a combination of rigorous research, industry engagement, and public outreach, the company continues to shape the discourse on climate resilience, energy transition, and sustainable finance. Its work not only informs decision-makers and industry leaders but also empowers individuals and organizations to take meaningful steps toward a sustainable future. As Enerqa expands its knowledge-sharing initiatives, it is well-positioned to drive innovation, foster collaboration, and create lasting change in the global sustainability landscape.

Articles, scientific papers and contributions

Rethinking Tomorrow

Insights from the United Nations Global Environment Outlook (GEO-7)

Avoiding Emissions: A Sustainable Solution

Scope 4 Emissions: The Concept of Avoided Emissions

The Hidden Costs of Your Burger and Pizza: What's Really at Stake?

Explore the World Sustainably

EMPOWER COMMUNITIES

Empowering Communities Through Social Sustainability: A Call to Action

I-RECs: A Catalyst for Renewable Energy Investment in Qatar

Smoking and Climate Change

The current issue and full text archive of this journal is available on Emerald insight at: <https://www.emerald.com/insight/1097-0959.htm>

A review of electricity consumption and CO₂ emissions in Gulf Cooperation Council households and proposed scenarios for its reduction

Tareq Zerehabden Younis Ahmed
Cambridge University, Cambridge, UK and
University of Surrey, Stag Hill Campus, Guildford, UK
Marcello Elayeb Ahmed
Institute of Engineering Research and Material Technology,
National Center for Research, Khartoum, Sudan, and
Qassey A. Ahmed and Asya Adnan Mohamed
University of Khartoum, Khartoum, Sudan

Electricity consumption and CO₂ emissions

Received 22 September 2022
Revised 14 November 2022
Accepted 26 November 2022

HONORARY CREDENTIAL 荣誉证书

This is to certify that
Mr./Ms. Quosay A. Ahmed, Amr Nasreddin H. Abdulhadi, Yasar Fadel 先生/女士
感谢参加“2022国际石油石化技术会议暨新能源及节能技术国际会议”并分享报告。
报告题目:《Projections of the Energy Mix in Sudan》
at the joint International Petroleum and Petrochemical Technology Conference 2022 and International New Energy and Energy Conversion Technology Conference 2022
IPPTC Organizing Committee
INETC Organizing Committee
October, 2022

GEO-7
GLOBAL ENVIRONMENT OUTLOOK

HEALTHY PLANET
HEALTHY PEOPLE

UN
environment
United Nations
Environment Programme

CAMBRIDGE

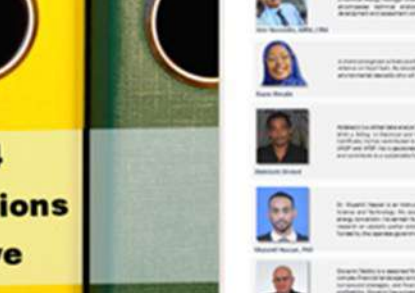
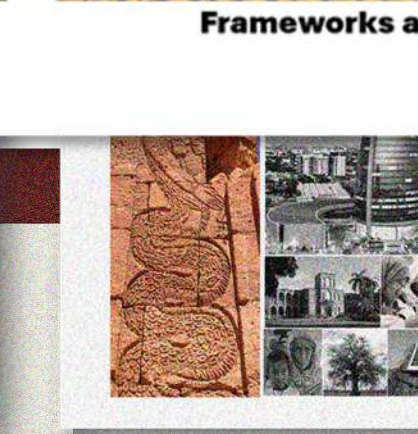
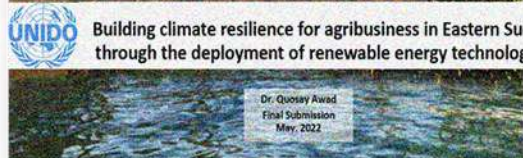
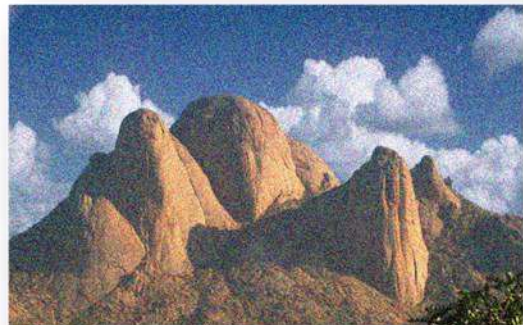
SYNTHESIS OF SUBMISSIONS ON THE 2023 FOCUS AREA OF THE PARIS COMMITTEE ON CAPACITY-BUILDING:

"CAPACITY-BUILDING SUPPORT FOR ADAPTATION, WITH A FOCUS ON ADDRESSING GAPS AND NEEDS RELATED TO FORMULATING AND IMPLEMENTING NATIONAL ADAPTATION PLANS"

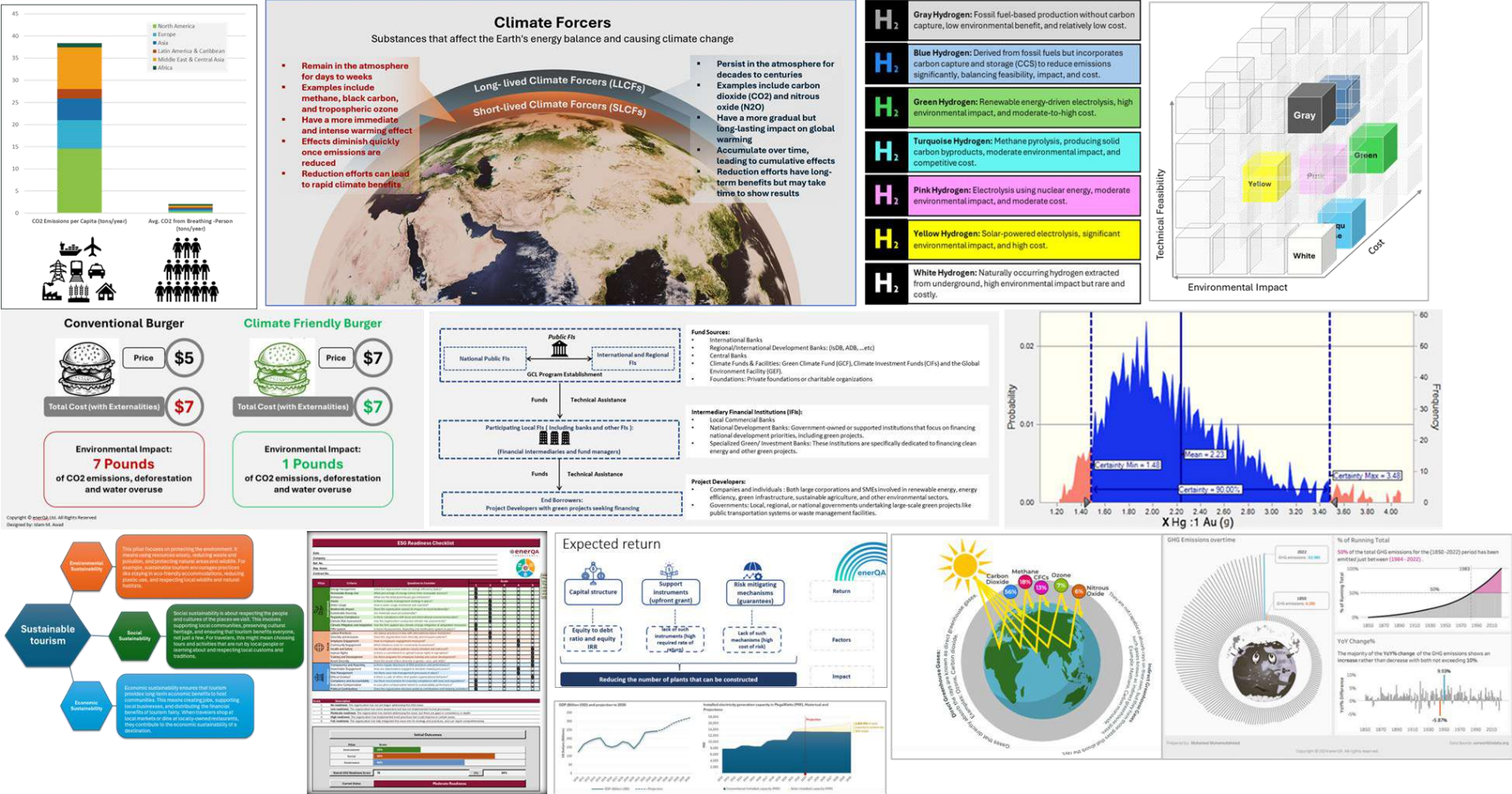
MAY 2023

United Nations Climate Change
Paris Committee on Capacity-building

Articles, scientific papers and contributions



Articles, scientific papers and contributions – Data visualisation



Quosay A. Ahmed, Tarig Z. Ahmed2, Mawahib E. Ahmed and Asia Adlan Mohamed "The Role of GCC Higher Education Systems in Supporting the Process of Nationally Determined Contributions (NDCs) Towards the Achievement of the SDG 13 Climate Action" GRM conference 2023, University of Cambridge, UK (2023)

Quosay A. Ahmed, Amr Nasradin, Yassir Fadul, "Projection of Sudan Energy Mix" International New Energy and Energy Conservation Technology Conference- China, (2022)

Quosay A. Ahmed and ChabilalDhital, "Decision Making Strategies for Engineering Design, Energy Consumption and Emission Mitigation –Hybrid Vehicle Versus Ice Vehicles" presented in the 26th Scientific Conference on Trends in the Development of Heavy-Duty Machines, January 27-31. 2013, Zakopane, Poland

Islam M. and Quosay A. Ahmed "Green Credit Lines in the Gulf Cooperation Council: Evaluating Opportunities, Overcoming Challenges, and Assessing the Impact on Energy Transition in Qatar" GRM 14 Conference, University of Cambridge, United Kingdom, 2024.

Abubaker Younis, Amro Khalafalla, Ismail Elgizouli, Ratiba Abdelgadir, Ai Omer, Mahmoud Onsa, Quosay Ahmed "Economic Analysis and Policy Related Recommendations to Promote Distributed Solar Photovoltaic Systems in Sudan" UofKEJ Vol.12 Issu 1pp 17-22 (Feb. 2022)

Tarig Z. Y. Ahmed, Mawahib E. Ahmed, Quosay A. Ahmed and Asia A. Mohamed "A Review of Electricity Consumption and CO₂ Emissions in Gulf Cooperation Council Households and Proposed Scenarios For its Reduction" Arab Gulf Journal of Scientific Research, ISSN: 1985-9899, Feb. 2024

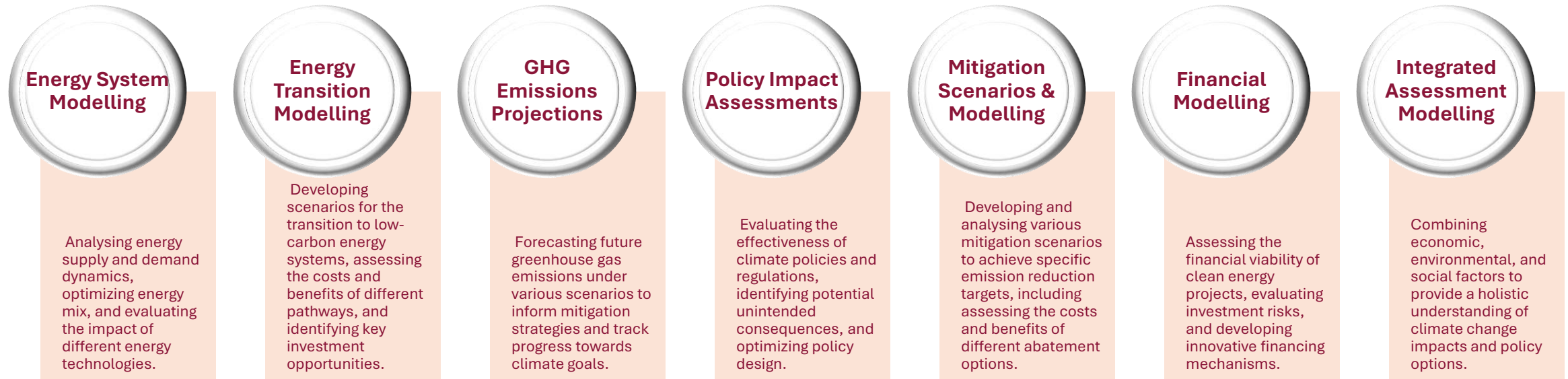
4. Modelling and Simulation

Introduction

At Enerqa Consultancy, we leverage the power of advanced modelling and simulation to inform strategic decision-making and drive effective climate action. Our expertise in this area allows us to analyse complex energy systems, assess policy impacts, project future scenarios, and evaluate mitigation options with a high degree of accuracy. We utilize cutting-edge tools and methodologies to provide clients with actionable insights for a sustainable future.

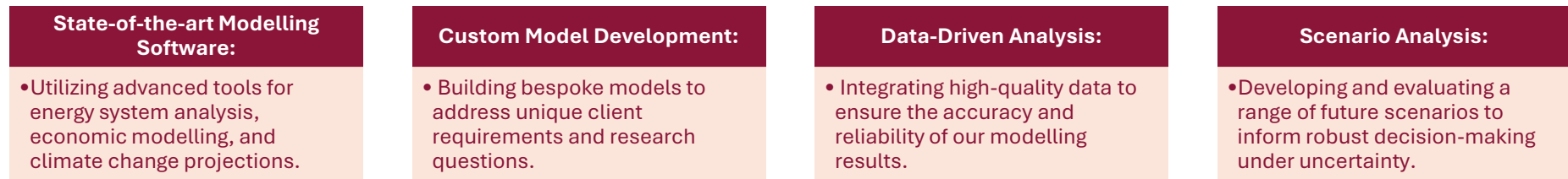
Our Modelling & Simulation Capabilities:

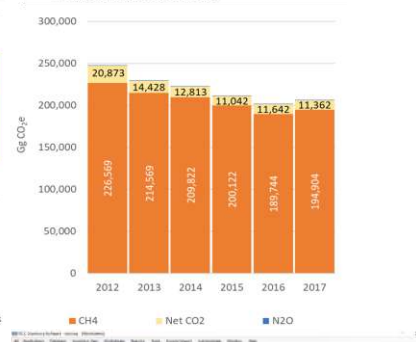
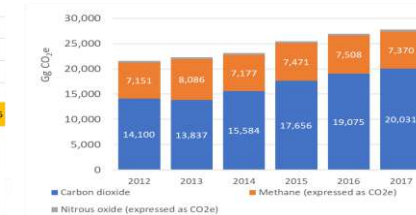
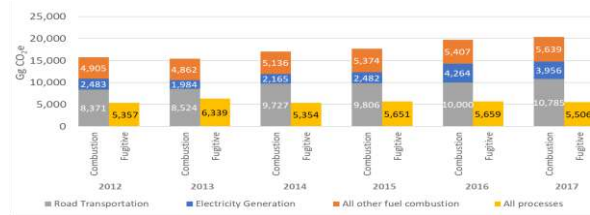
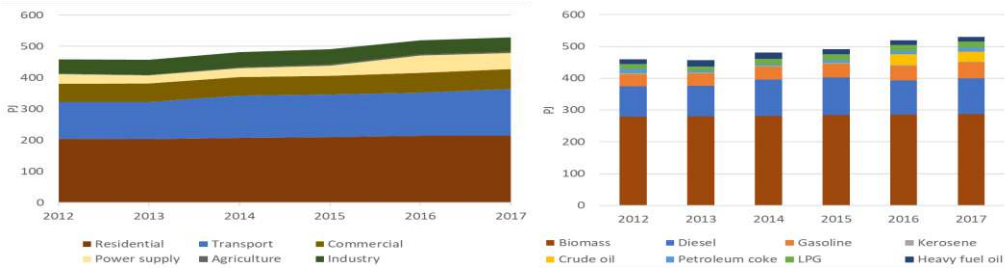
Enerqa's modelling and simulation services encompass a wide range of applications, including:



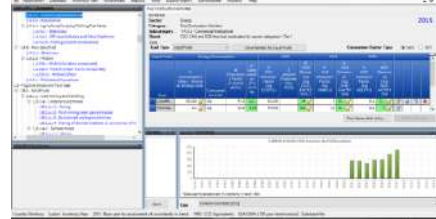
Tools & Methodologies:

We employ a variety of industry-leading software and methodologies, tailored to the specific needs of each project. Our approach combines:

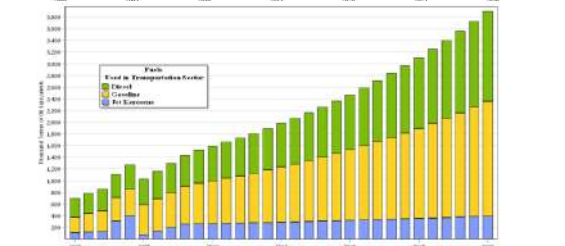
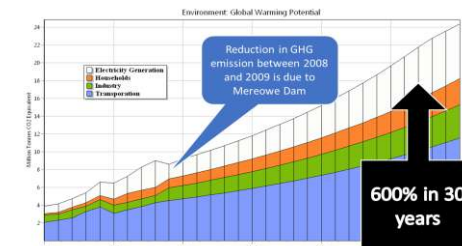
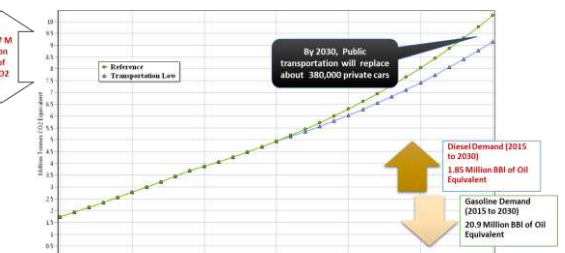
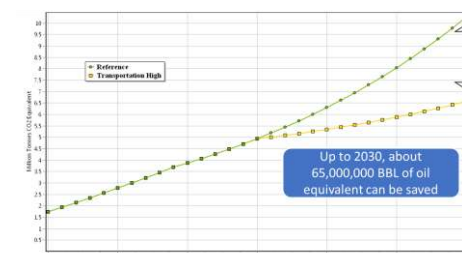
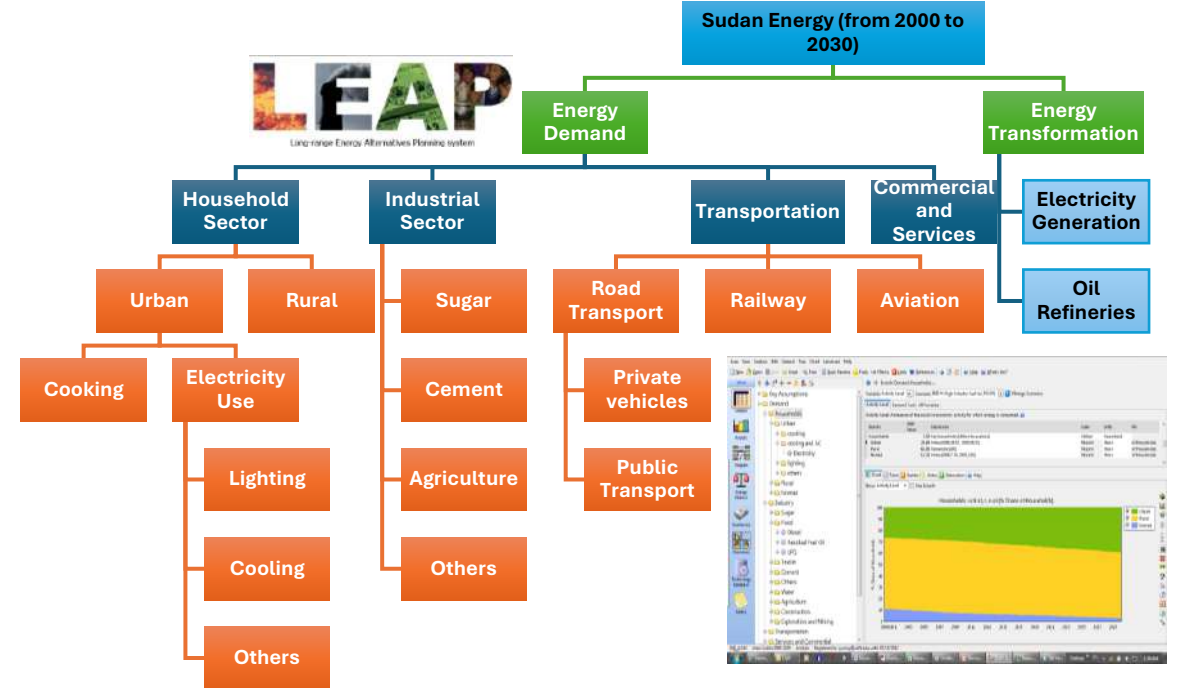




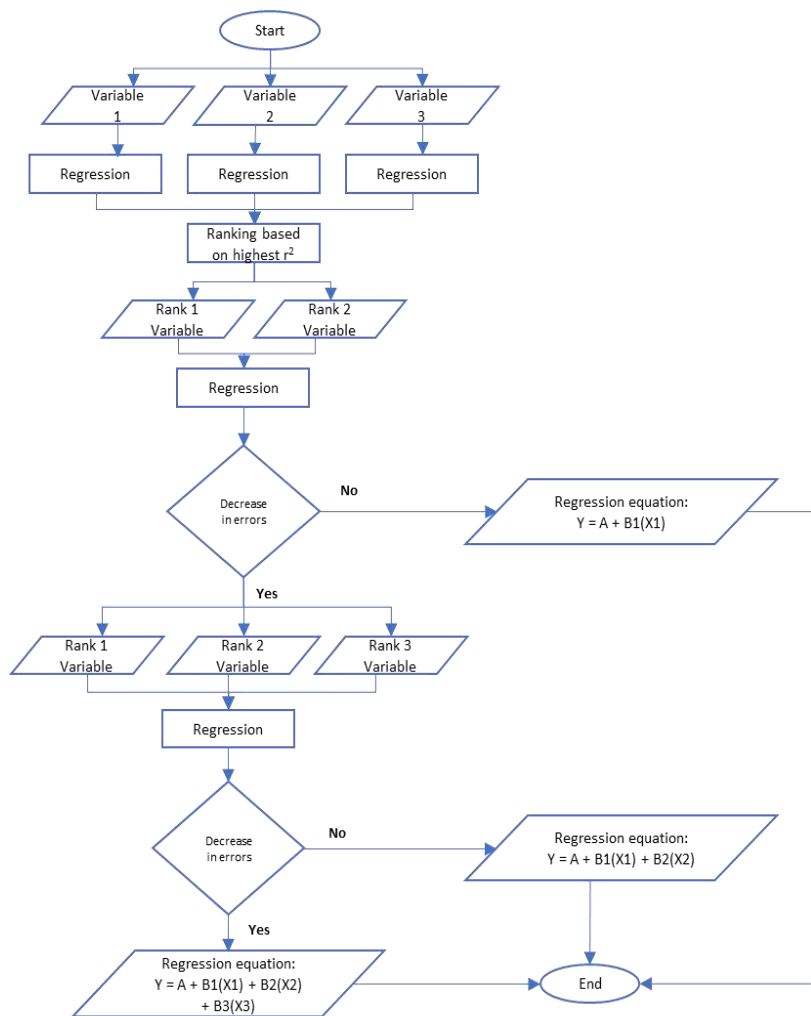
Categories	Total Area (ha)	Net CO2 emissions (Gg CO2)
3.B - Land	18647590	151148.901
3.B.1 - Forest land	18632230.8	18406.1846
3.B.1.a - Forest land Remaining Forest land	11262946	38183.6434
3.B.1.b - Land Converted to Forest land	7369284.8	-19777.4587
3.B.2 - Cropland	23472153.8	95403.5625
3.B.2.a - Cropland Remaining Cropland	13219949	0
3.B.2.b - Land Converted to Cropland	10252204.8	95403.5625
3.B.3 - Grassland	48363147	519.689115
3.B.3.a - Grassland Remaining Grassland	48214047	0
3.B.3.b - Land Converted to Grassland	149100	519.689115



(2012 to 2017) GHG emissions inventory model using IPCC 2006 Inventory Software

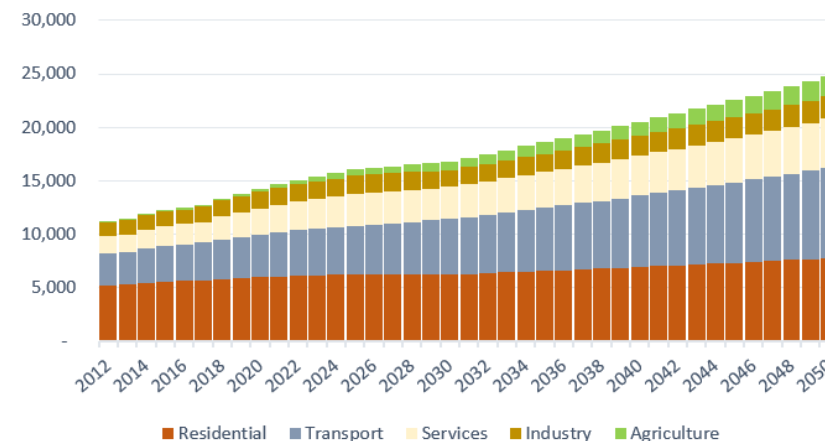


Modelling of GHG emissions mitigation scenarios for Energy sector (between 2015 to 2030) using LEAP software



A forward regression model built to project Final Energy Consumption in each demand sector.

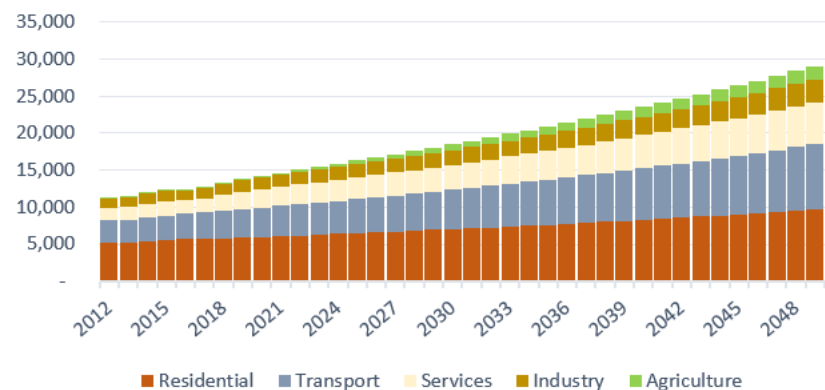
TFEC projections by sector to 2050, ET scenario



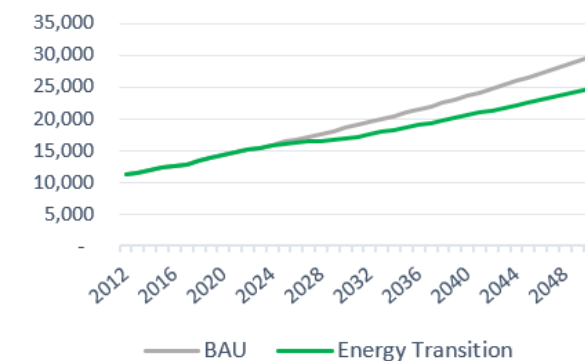
The projections were mainly developed using regression analyses under two scenarios: the Business as Usual (BAU) scenario and the Energy Transition (ET) scenario.

The BAU scenario was developed using Forward Regression, and the ET scenario was developed by applying adjustments to the BAU scenario, from selected most-likely applicable interventions proposed in the country's NDC.

TFEC projections by sector to 2050, BAU scenario



TFEC projections under the BAU and ET to 2050, in kTOE



5. Framework setup and Institutional Arrangements

Introduction

1. Institutional Arrangements & Framework Development

Enerqa has played a pivotal role in establishing institutional frameworks for climate transparency and MRV systems in Sudan. This includes developing a **National Transparency Committee (NTC)** to oversee framework implementation. The company has also designed and proposed institutional arrangements for the transport and energy sectors, ensuring alignment with Sudan's Nationally Determined Contributions (NDCs). Strengthening legal and procedural frameworks has been a key focus, integrating climate reporting into national policies.

2. Capacity Building & Knowledge Transfer

Enerqa has led various training and capacity-building programs for government agencies, private sector stakeholders, and NGOs. These efforts include conducting a capacity needs assessment to identify training requirements in GHG inventory methodologies, data collection, and mitigation strategies. The company has developed and delivered tailored training programs to enhance the technical capacity of national institutions. Additionally, it has facilitated stakeholder engagement and knowledge transfer through workshops, online platforms, and expert secondments.

3. MRV System Strengthening & NDC Tracking

Enerqa has worked extensively on improving MRV systems for tracking emissions and mitigation actions. A key contribution has been refining and implementing an NDC tracking tool to monitor climate commitments. The company has integrated MRV tools with existing data collection systems to enhance accuracy and efficiency. Additionally, it has developed baseline scenarios and projections to assess future emissions trends.

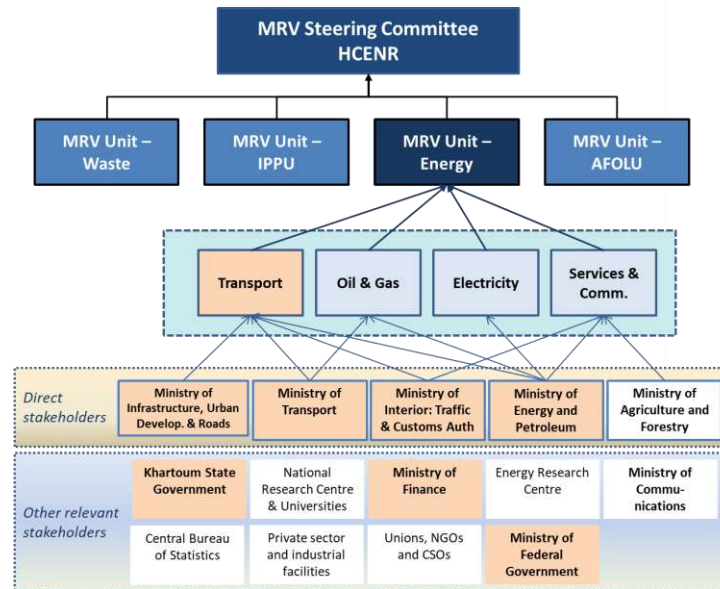
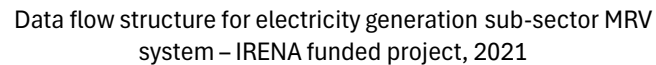
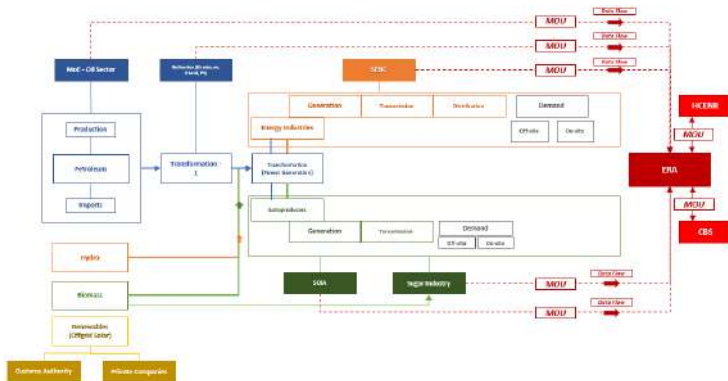
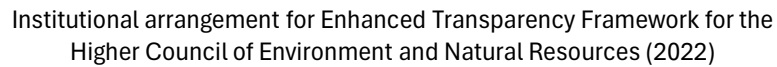
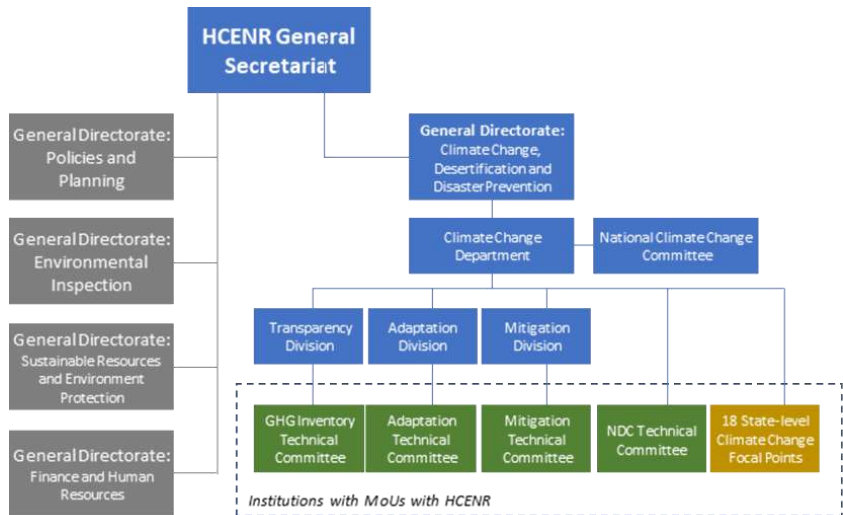


Supporting Your Net Zero Journey with Precision, Insight, and Finance

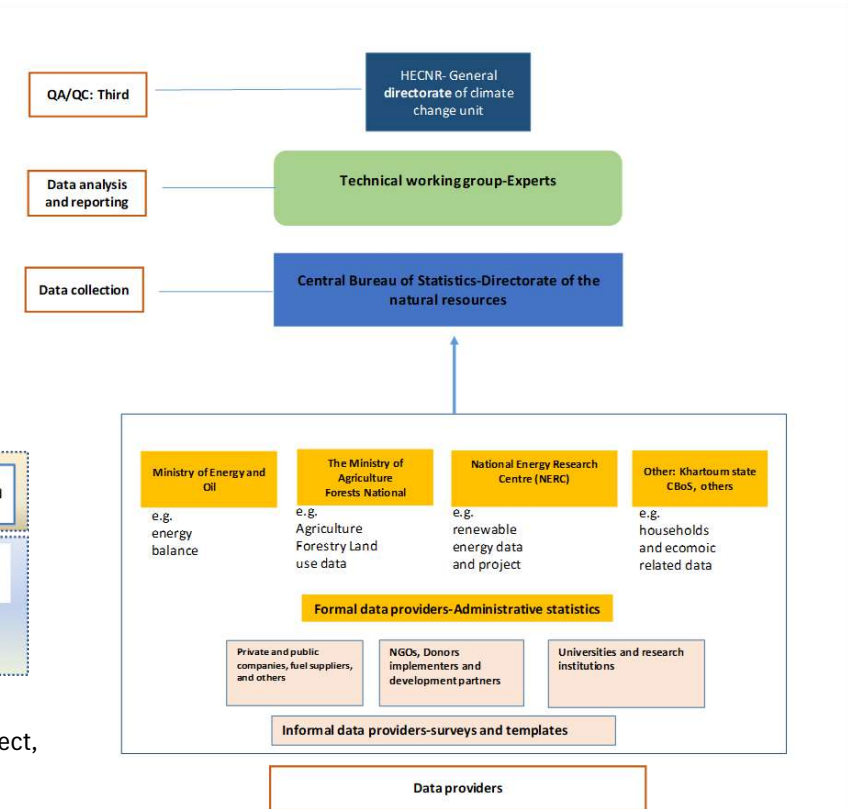
At **Enerqa**, we help businesses and public sector organizations achieve their greenhouse gas (GHG) emissions targets through a comprehensive, data-driven, and finance-aligned approach. Whether you're just beginning your climate strategy or advancing toward net-zero, our end-to-end support ensures measurable impact and actionable outcomes.

ENERQA’s Approach to Emissions Reduction

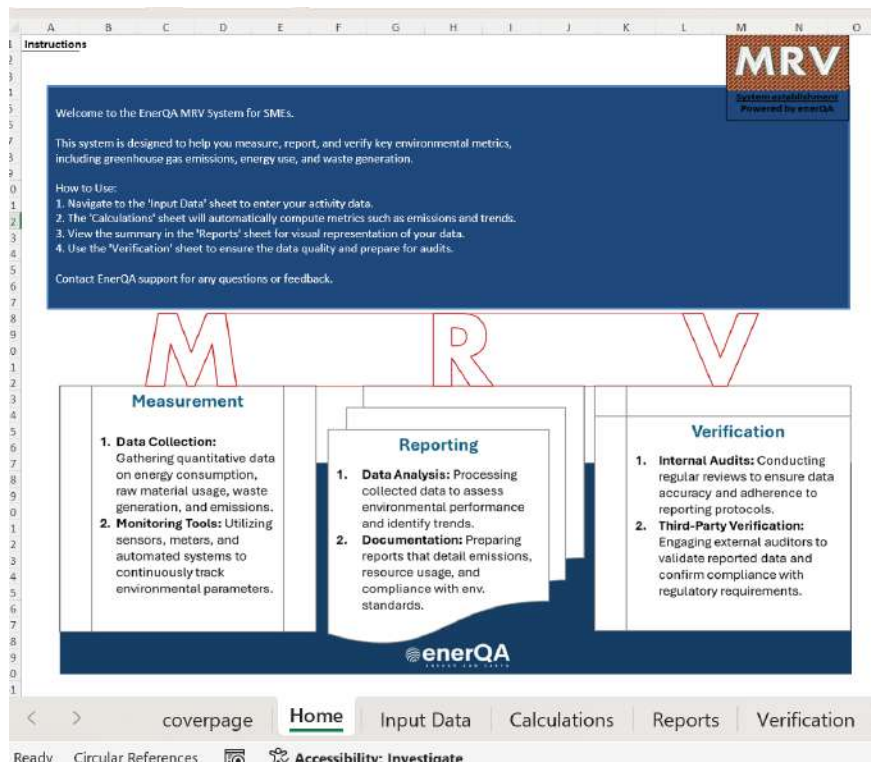




MRV system setup for Transport Sector in Sudan – ICAT funded project,
2024



Data flow framework for energy sector GHG related information, (Capacity Building for Transparency – CBIT project, 2023)



enerQA's MRV system creator for facility scale

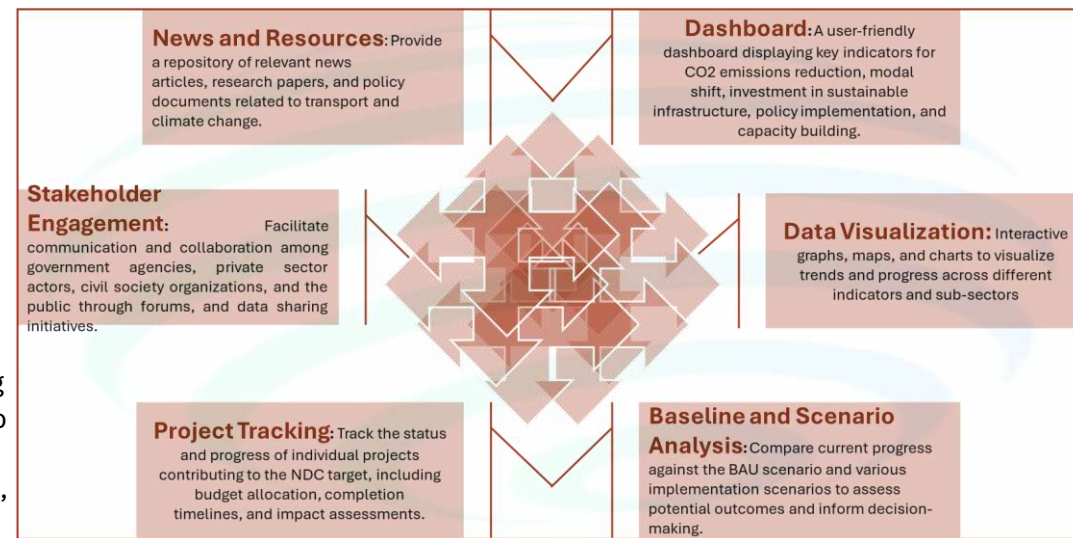


As the world accelerates toward net-zero goals, access to climate finance and carbon credit markets is becoming a crucial advantage for businesses, governments, and project developers. However, navigating these opportunities requires robust data, transparent reporting, and verifiable impact metrics—and this is where Enerqa's MRV (Monitoring, Reporting, and Verification) Tool is revolutionizing the space.

Enerqa's MRV Tool serves as the backbone for organizations seeking to participate in carbon markets, access sustainability-linked finance, and enhance climate accountability. By providing a streamlined, automated, and transparent platform, the tool ensures that emissions reductions are accurately measured, consistently reported, and independently verified—the three critical pillars required for carbon credit certification and climate finance eligibility.

For businesses and industries, the MRV Tool simplifies carbon accounting and ensures compliance with international standards such as the GHG Protocol, ISO 14064, and Science-Based Targets (SBTi). This helps corporations build credible sustainability reports that attract investors and qualify for green finance mechanisms. For governments and regulatory bodies, it strengthens national climate policies, enhances transparency in emissions tracking, and supports the formulation of carbon pricing mechanisms. For project developers and NGOs, the tool provides an efficient way to quantify, report, and verify emission reductions—allowing them to generate high-quality carbon credits and access funding from climate finance institutions, development banks, and sustainability-driven investors.

Unlike traditional MRV systems, Enerqa's tool is fully customizable, integrating advanced data analytics, remote sensing, and automation to reduce reporting errors and enhance credibility. It also ensures seamless integration with other sustainability tools, making it easier for organizations to connect their emissions tracking with ESG reporting, risk assessment, and investment frameworks.



Tracking tool/dashboard for GHG mitigation projects

6. EnerQA Services

OUR SERVICES



C. Climate Change

- C.1. Climate Finance
- C.2. Climate Action (Mitigation and Adaptation)
- C.3. Climate Change Reporting



B. Business Solutions

- B.1. Feasibility Studies and Business Plans
- B.2. Supporting Studies (all services)



E. Energy

- E.1. Energy Management
- E.2. Renewable Energy Projects Development.



ENV. Environment

- S.1. Strategic
- S.2. Operational
- S.3. ESG



Business Solutions

Environment

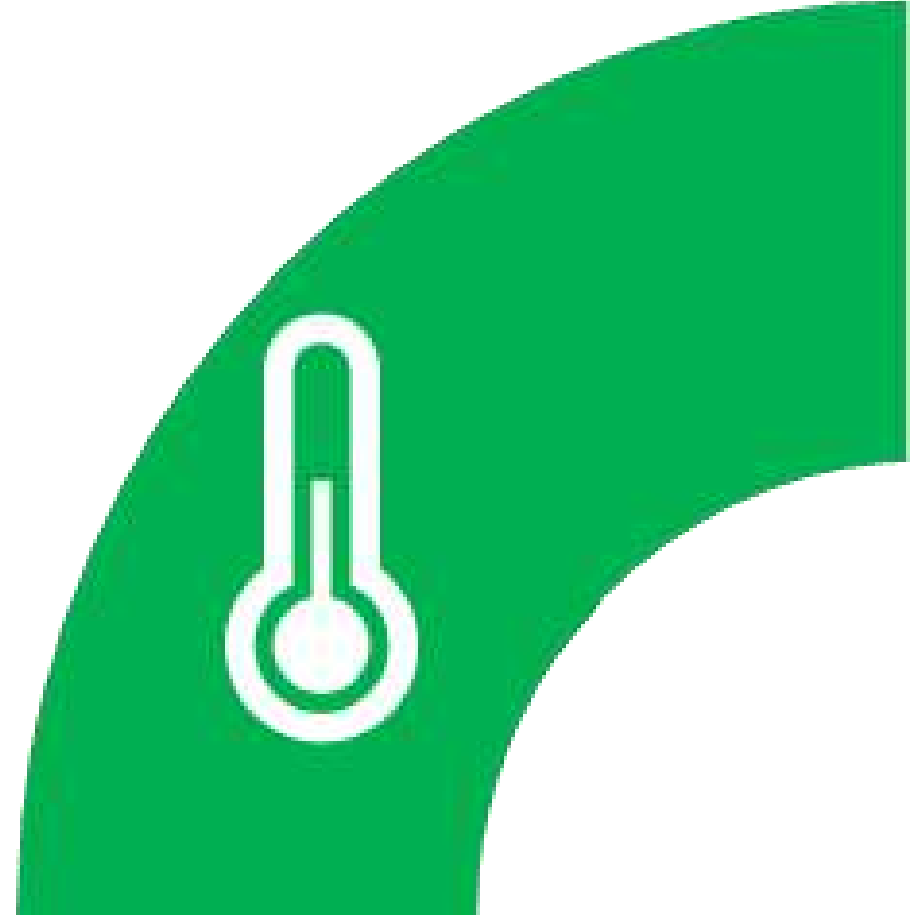
Energy

Climate Change

C.1. Climate Finance

C.2. Climate Action (Mitigation and Adaptation)

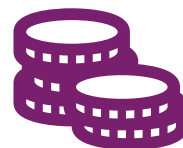
C.3. Climate Change Reporting



C.1.1. Carbon credits and climate finance

Carbon credit:

- Validation of GHG baseline and mitigation impacts
- MRV system update
- Standardization
- Verification
- Issuance of carbon credits



Source of
Revenue

Climate Finance:

- Reviewing mitigation strategies
- Tuning and development of future projects to meet the climate finance requirements
- Building green business models
- Detailed feasibility study considering other sustainability dimensions
- Support the client to access the climate finance



Entitlement to
fund

Projects pipeline

Ongoing

Now

New / Planned

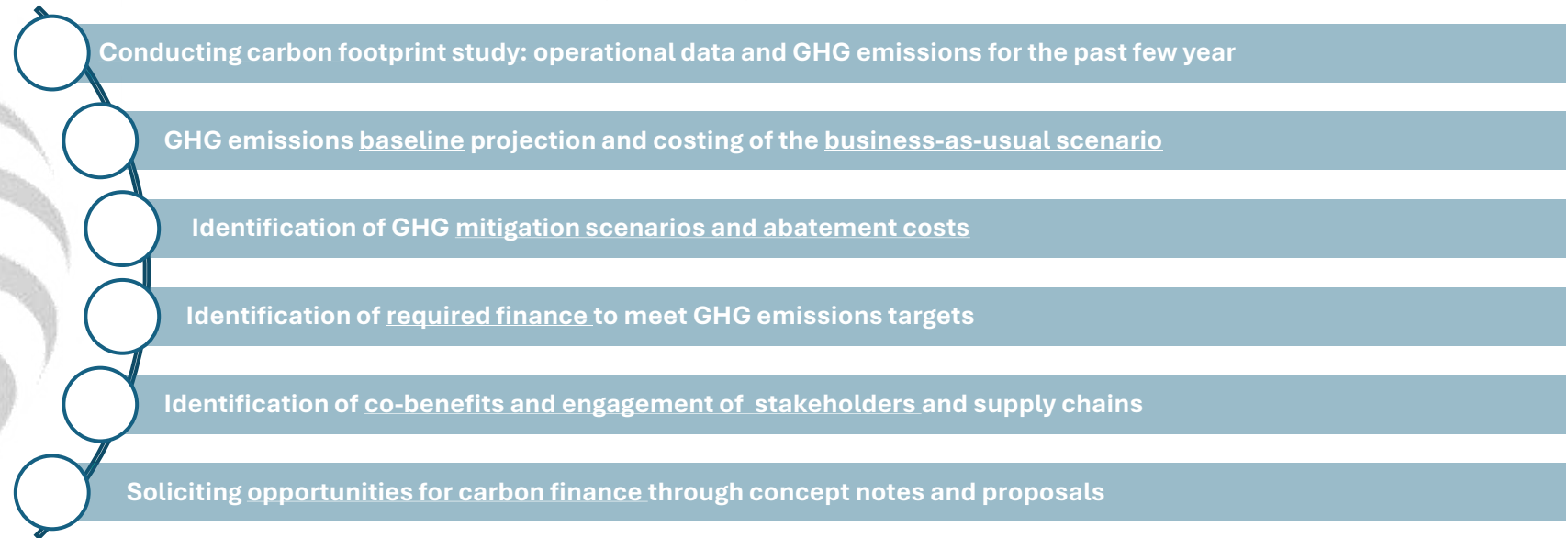
C.2.1. Development of projects ideas, concept notes, proposals, and TORs



C.2.2. Decarbonization and mitigation related studies

At Enerqa, we are at the forefront of supporting SMEs, corporations, and governmental entities in their transition toward low-carbon and sustainable operations. Our expertise in GHG emissions reduction and mitigation planning enables organizations to navigate climate commitments, regulatory requirements, and carbon finance opportunities effectively. Through our comprehensive consultancy services, we help clients assess, plan, and implement mitigation strategies that align with both business objectives and global climate goals. From analyzing carbon footprints and establishing baseline emissions projections to identifying cost-effective mitigation pathways and financing options, we provide data-driven, actionable solutions that drive real impact. Enerqa's approach ensures that organizations not only reduce emissions efficiently but also maximize co-benefits, engage stakeholders, and explore carbon finance mechanisms to support their sustainability journey. With a strong track record in climate advisory and finance, we empower clients to turn climate ambition into measurable action

enerQA's De-carbonization pathway



ENERQA'S CARBON FOOTPRINT CERTIFICATION: LEADING THE WAY TO EMISSIONS REDUCTION

At **Enerqa**, we empower organizations with the tools and methodologies needed to measure, manage, and reduce greenhouse gas (GHG) emissions, aligning their operations with global climate targets and sustainability best practices. Our Carbon Footprint Certification Program follows internationally recognized standards, including the GHG Protocol and ISO 14064/14067, ensuring accuracy, transparency, and accountability in emissions tracking and reduction efforts.

Through a structured, phased approach, Enerqa's certification framework helps businesses and institutions at different stages of their sustainability journey:

- **Baseline Certification:** Establishes a solid foundation for emissions measurement using global standards, providing organizations with a clear picture of their **carbon footprint and environmental impact**.
- **On-Progress Certification:** Implements a transparent and replicable methodology for tracking, reporting, and verifying emissions across Scope 1, 2, and 3—covering direct, indirect, and supply chain emissions.
- **Mitigation Certification:** Provides sector-specific guidance to tailor emissions reduction strategies, helping organizations identify and implement cost-effective and impactful mitigation measures.

By achieving Enerqa's Carbon Footprint Certification, organizations gain a credible pathway to carbon neutrality, enhance their sustainability credentials, and unlock opportunities for carbon finance and green investments. This certification is not just a recognition of commitment—it is a practical roadmap to long-term emissions reduction and climate leadership.



Baseline Certification



On-Progress Certificate



Mitigation Certificate

Methodological Framework

GHG Protocol & ISO Standards

- Use frameworks like the GHG Protocol (corporate, product, and project levels) and ISO 14064, 14067 standards as foundation elements.

Data Collection & Emissions Tracking

- Implement a transparent, replicable methodology for emissions data collection, calculation, and reporting across Scopes 1, 2, and 3 (direct, indirect, and supply chain emissions).

Sector-Specific Guidance

- Tailor methods for each sector, as emission sources and reduction potential vary widely across industries.

Business Solutions

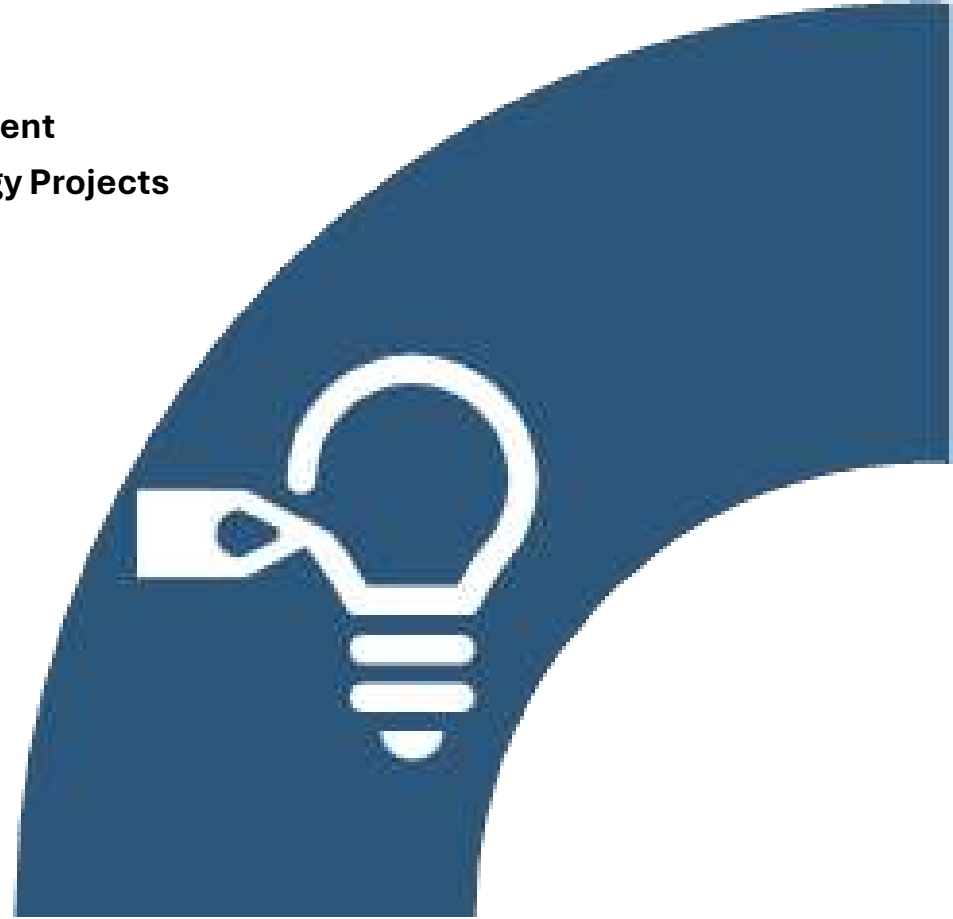
Environment

Energy

Climate Change

E.1. Energy Management

**E.2. Renewable Energy Projects
Development**



Enerqa's Energy Consultancy Services: Driving Efficiency & Renewable Solutions

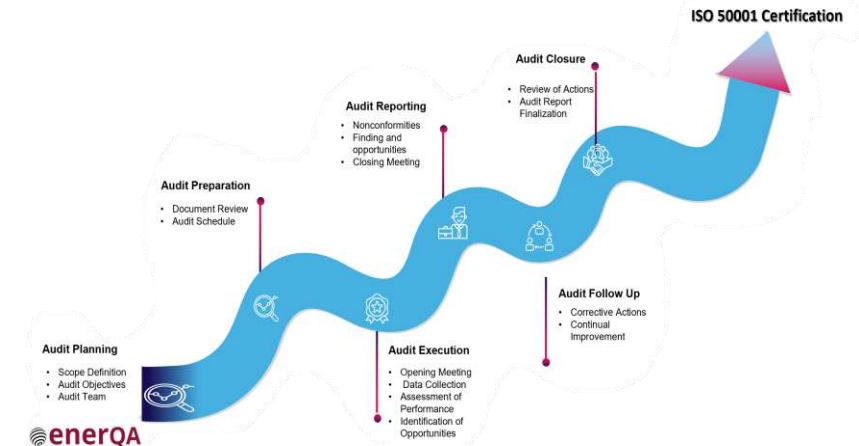
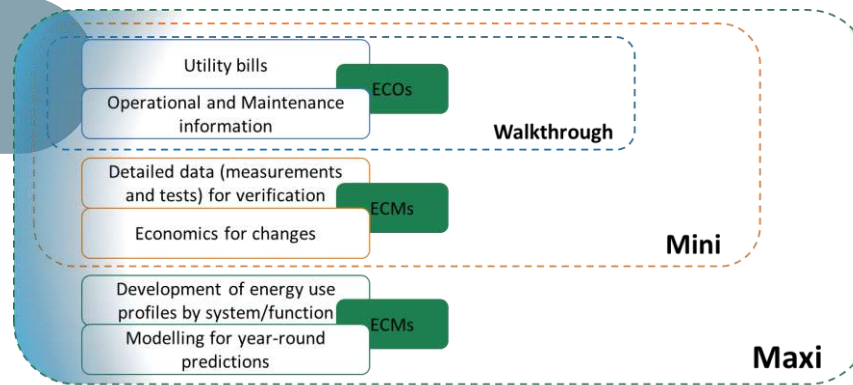
Enerqa's energy consultancy services provide tailored solutions to support the oil and gas sector, electricity systems, and renewable energy deployment, helping clients across public, private, and non-governmental organizations achieve efficiency, sustainability, and regulatory compliance. In the oil and gas industry, we assist companies in optimizing energy use, reducing operational emissions, and implementing energy efficiency measures, ensuring alignment with global decarbonization targets and ESG commitments. Our expertise in flaring reduction, carbon capture utilization and storage (CCUS), and process optimization enables oil and gas firms to enhance profitability while mitigating environmental impact. In the electricity sector, we provide strategic support across the entire value chain—from generation and grid integration to energy storage, transmission, and consumption optimization. Our services include grid stability assessments, energy storage feasibility studies, load forecasting, and demand-side management, helping utilities and energy providers transition toward resilient, low-carbon energy systems. We also support renewable energy deployment by offering technical, economic, and policy advisory on solar, wind, hydro, and bioenergy projects, assisting governments, private companies, and NGOs in developing bankable projects, securing financing, and integrating renewables into existing grids. Whether supporting public agencies in achieving national energy targets, assisting corporations in optimizing energy efficiency, or guiding NGOs in promoting sustainable energy access, Enerqa delivers data-driven, high-impact solutions that drive cost savings, energy security, and environmental responsibility.



E.1. Energy Management

Enerqa supports organizations in assessing, optimizing, and managing their energy consumption to improve efficiency and reduce operational costs. Our services include:

1. **Energy audits** to identify inefficiencies and areas for improvement.
2. **Energy monitoring and management solutions** for better resource allocation.
3. **Policy development and compliance support** to align with sustainability regulations.
4. **Optimization of energy use in industrial, commercial, and public sector operations** to achieve long-term savings and environmental benefits.



Enerqa's energy audits and performance assessments provide a comprehensive evaluation of energy consumption, identifying inefficiencies and uncovering opportunities for cost-effective energy savings. By analyzing utility data, equipment efficiency, operational workflows, and industry benchmarks, we help businesses and organizations pinpoint areas of excessive energy use and recommend tailored solutions to enhance performance. To complement this, our energy monitoring and smart management systems integrate IoT-based tracking, real-time data analytics, and AI-driven insights to provide continuous visibility and control over energy usage. These digital solutions enable automated adjustments, predictive maintenance, and peak load optimization, ensuring maximum efficiency and reducing operational costs. Additionally, we support clients in achieving policy and regulatory compliance by aligning their energy strategies with national and international efficiency standards, such as ISO 50001, LEED, and local energy regulations. Our expertise ensures that businesses meet legal requirements while improving sustainability performance and accessing potential green incentives or energy subsidies. Furthermore, operational optimization and process improvements are at the core of our approach, helping industries, commercial buildings, and public infrastructure enhance energy utilization through equipment upgrades, process streamlining, heat recovery solutions, and system retrofits. Whether it's an industrial facility optimizing its production energy footprint, a commercial complex reducing HVAC inefficiencies, or a municipality improving public infrastructure energy performance, Enerqa's solutions drive tangible cost savings, enhanced reliability, and long-term sustainability for its clients.

E.2. Renewable Energy Projects Development

Accelerating the Transition to Renewable Energy

Enerqa is committed to facilitating the **global shift toward clean energy** by supporting organizations in **developing, financing, and deploying renewable energy projects**. Our expertise covers:

1. **Feasibility Studies & Resource Assessments** – Evaluating the viability of solar, wind, hydro, and bioenergy projects based on location, resources, and financial feasibility.
2. **Project Design, Engineering, and Development** – Providing technical, economic, and environmental assessments to ensure bankability and successful project execution.
3. **Renewable Energy Policy & Investment Advisory** – Helping clients navigate regulations, access incentives, and secure financing for clean energy initiatives.
4. **Grid Integration & Storage Solutions** – Ensuring seamless incorporation of renewable energy into existing infrastructure, minimizing intermittency challenges.

Our renewable energy consultancy services empower businesses, utilities, and government agencies to reduce reliance on fossil fuels, lower emissions, and unlock new opportunities for carbon finance and sustainability leadership.

Achieving a Low-Carbon, Resilient Future

At Enerqa, we recognize that energy efficiency and renewable energy are critical to achieving climate resilience, economic stability, and long-term competitiveness. Through our integrated consultancy approach, we help clients:

- ✓ Cut costs through smarter energy use and renewable adoption
- ✓ Enhance sustainability and corporate responsibility
- ✓ Meet regulatory and ESG (Environmental, Social, and Governance) requirements
- ✓ Position themselves as leaders in the clean energy transition

Whether you're looking to improve energy efficiency, develop a renewable energy strategy, or implement innovative clean energy projects, Enerqa is your trusted partner in driving sustainable energy solutions that deliver financial, environmental, and social benefits.

Energy



Business Solutions

Environment

- S.1. Strategic
- S.2. Operational
- S.3. Environmental, Social and Governance (ESG)



S.1. Strategic Environmental Solutions

Enerqa provides **comprehensive environmental consultancy services**, categorized into **Strategic, Operational, and Environmental, Social, and Governance (ESG) solutions**. Our expertise helps businesses, governments, and organizations integrate **sustainability principles**, comply with **environmental regulations**, and adopt **innovative, science-based approaches** for **climate resilience and resource management**.

Enerqa supports organizations in **long-term sustainability planning** by providing data-driven insights and policy frameworks that enhance environmental governance, ecosystem protection, and sustainable land-use practices. Our strategic services include:

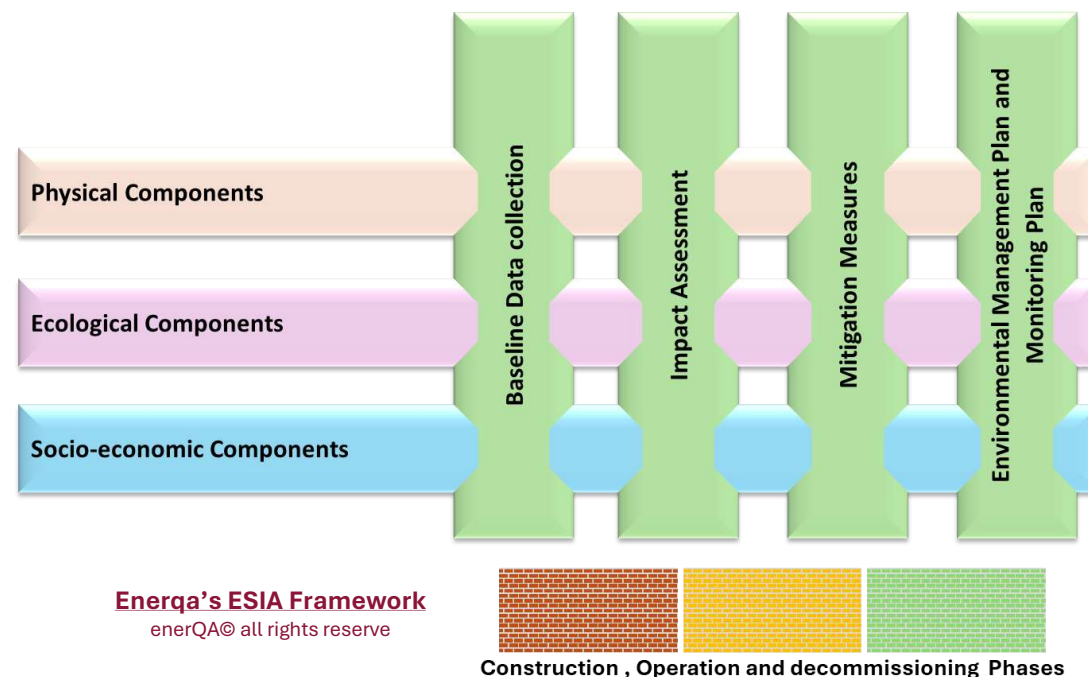
1. **Environmental Governance** – Developing policies, institutional frameworks, and regulatory strategies to improve environmental management.
2. **Environmental-Economic Accounting (SEEA)** – Integrating environmental data into economic decision-making to assess the true cost of natural resource consumption.
3. **Environmental and Social Assessment (SESA)** – Evaluating environmental and social risks for large-scale projects, ensuring sustainability in development planning.
4. **Nature-Based Solutions (Mangroves Restoration & Ecosystem Management)** – Implementing ecosystem-based approaches for climate resilience, carbon sequestration, and biodiversity conservation.
5. **Bioremediation Solutions** – Leveraging biological processes to clean up contaminated environments, including soil and water remediation.
6. **Geo-Database Development** – Establishing spatial data platforms for better environmental monitoring and decision-making.
7. **Integrated Coastal Zone Management** – Supporting sustainable coastal development, protecting marine ecosystems, and mitigating coastal erosion.
8. **Land Use Planning (Food Security)** – Designing sustainable land-use frameworks that balance agricultural productivity, conservation, and urban expansion.



S.2. Operational Environmental Services

Enerqa provides **practical, impact-driven solutions** to help industries, governments, and businesses comply with environmental regulations, minimize risks, and enhance sustainability performance. Our operational services include:

1. **Environmental and Social Impact Assessment (ESIA)** – Conducting in-depth evaluations of potential environmental and social impacts for new and existing projects.
2. **Environmental Due Diligence & Exit Audits** – Ensuring regulatory compliance and risk assessment during business transactions, acquisitions, and project closures.
3. **Health, Safety & Environment (HSE) Audits & Inspections** – Evaluating workplace safety, environmental risks, and compliance with occupational health and safety standards.
4. **Waste Management Solutions** – Designing and implementing sustainable waste management strategies, including circular economy approaches, waste-to-energy solutions, and landfill optimization.



1. Circular Economy Approaches

Enerqa promotes a circular economy by designing waste reduction, recycling, and material recovery strategies to extend product lifecycles and minimize landfill use. We conduct waste stream analysis, optimize material recovery processes, and facilitate industrial symbiosis, where waste by-products from one industry serve as raw materials for another. Additionally, we support eco-design and sustainable packaging initiatives, helping businesses develop materials that are recyclable, biodegradable, or reusable.

2. Waste-to-Energy (WtE) Solutions

We help clients convert waste into valuable energy sources by implementing biogas, biomass, and incineration with energy recovery solutions. Our services include feasibility studies, technology selection, and project integration into national sustainability goals. By leveraging anaerobic digestion, gasification, and pyrolysis technologies, we support industries and municipalities in harnessing waste for electricity generation, heating, and biofuel production, reducing reliance on fossil fuels while managing waste effectively.

3. Landfill Optimization and Waste Disposal Management

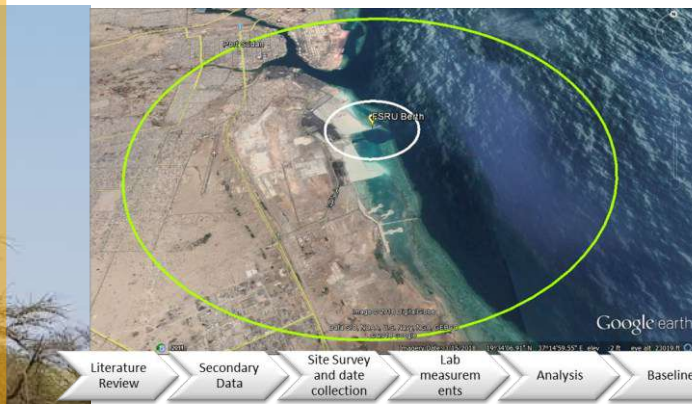
For non-recyclable and residual waste, Enerqa provides landfill optimization strategies to enhance site efficiency, regulatory compliance, and environmental safety. We specialize in landfill site selection, leachate management, and methane gas capture, turning landfill emissions into energy sources. Additionally, we promote landfill diversion strategies such as composting, advanced waste sorting, and recycling initiatives, significantly reducing the volume of waste sent to landfills.

S.3. Environmental, Social, and Governance (ESG) Solutions

Environment

Enerqa's **ESG consulting services** help organizations integrate sustainability and responsible business practices into their operations, ensuring compliance with global ESG frameworks. Our services include:

1. **GIS and Remote Sensing-Based Environmental Mapping** – Conducting spatial analysis to support environmental planning and impact assessment, including:
 - **Environmental Sensitivity Mapping** – Identifying ecologically vulnerable areas to guide sustainable development.
 - **Natural Resources Mapping (Non-Renewable)** – Assessing the distribution of finite resources to support sustainable extraction and conservation policies.
 - **Biodiversity Mapping** – Monitoring species distribution and habitat quality to enhance conservation efforts.
 - **Land Degradation Mapping** – Evaluating soil erosion, deforestation, and desertification risks to guide land restoration strategies.
2. **ESG Compliance & Reporting** – Assisting organizations in aligning with international sustainability reporting standards such as GRI, SASB, TCFD, and CDP.
3. **Sustainability & Corporate Responsibility Strategies** – Helping businesses integrate sustainability into their corporate strategy and achieve measurable ESG goals.
4. **Climate Risk & Resilience Planning** – Identifying climate-related risks and developing adaptation strategies to enhance business and community resilience.



Business Solutions

Environment

Energy

- B.1. Feasibility Studies and Business Plans**
- B.2. Business Modelling and Supporting Studies**



B.1. Feasibility Studies and Business Plans

Enerqa specializes in conducting feasibility studies and developing business plans to help organizations assess project viability, mitigate risks, and build robust financial and operational strategies. These studies provide a data-driven foundation for investment decisions, strategic planning, and market expansion.

Feasibility Studies

A feasibility study is a critical component in evaluating a project's technical, financial, operational, and environmental feasibility. Enerqa ensures that every feasibility study includes:

1. **Risk Assessment** – Identifying and mitigating financial, technical, and operational risks.
2. **Financial Viability Analysis** – Evaluating funding structures, investment requirements, and profitability projections.
3. **Market Analysis** – Conducting situation analysis, industry trends assessment, and competitor benchmarking.
4. **Technical and Operational Feasibility** – Ensuring that projects align with technical best practices, infrastructure readiness, and sustainability considerations.
5. **Environmental and Social Impact Assessment (ESIA) and M&E Planning** – Ensuring compliance with regulatory, environmental, and social governance frameworks.



Through comprehensive feasibility analysis, Enerqa provides strategic recommendations on whether to proceed, modify, or restructure projects, ensuring long-term success.

Business Plan Development

Enerqa helps organizations develop structured, goal-oriented business plans that serve as dynamic roadmaps for achieving business growth and securing financing. These plans include:

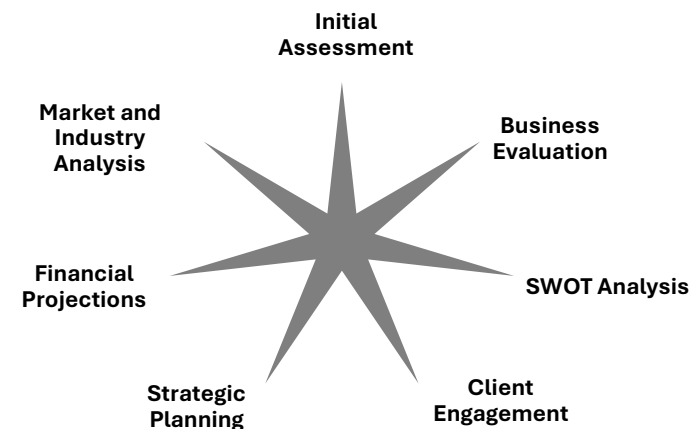
1. **Strategic Goal Setting** – Defining a clear vision, mission, and objectives for business development.
2. **Operational Optimization** – Identifying efficiency improvements across processes, HR structures, and resource utilization.
3. **Performance Improvement & Risk Mitigation** – Integrating financial projections, SWOT analysis, and competitive strategies to enhance market positioning.
4. **Market Expansion & Access to Finance** – Helping businesses identify funding opportunities, build investment cases, and secure financial support from institutions.

Enerqa's business plans align with market realities, regulatory frameworks, and financial sustainability principles, empowering businesses to thrive in competitive markets.

B.2. Business Modelling and Supporting Studies

Tailored Business Models for Complex Projects: Ensuring a Win-Win for All Stakeholders

At Enerqa, we develop customized business models that align financial viability, stakeholder interests, and sustainability goals, ensuring a win-win outcome for all parties involved. Our expertise lies in structuring investment-ready, ESG-compliant, and risk-optimized frameworks that drive profitability while meeting green finance requirements and regulatory standards.



1. Stakeholder-Centered Business Model Development

Enerqa's approach ensures that investors, businesses, governments, and communities benefit equitably. We design models that maximize value creation, promote multi-stakeholder collaboration, and incorporate fair risk-sharing mechanisms, ensuring long-term project success and sustainability.

2. ESG and Green Finance Compliance Integration

We embed ESG principles and green finance frameworks into our business models to secure funding, ensure regulatory alignment, and enhance reputational value. Our models comply with global sustainability standards and facilitate access to climate finance, green bonds, and impact investment opportunities.

3. Financial Structuring and Risk Management for Long-Term Success

Enerqa designs resilient financial models that integrate diversified revenue streams, blended finance strategies, and tailored risk mitigation plans. This approach enhances project bankability, minimizes financial uncertainty, and ensures sustainable growth.

4. Circular and Inclusive Business Models for Sustainability

We incorporate circular economy principles and social impact strategies to ensure that business models reduce waste, optimize resources, and drive inclusive economic development. Our models support job creation, local engagement, and environmental responsibility while ensuring long-term profitability.

5. Enabling Scalability and Adaptability

Enerqa develops scalable and adaptable business models that evolve with market trends, technological advancements, and regulatory changes. Through data-driven decision-making and performance monitoring, we help businesses future-proof their strategies and ensure long-term impact.

Green Credit Line Development

Enerqa supports financial institutions and businesses in establishing Green Credit Lines, enabling them to finance sustainable, renewable energy, and energy efficiency projects. Our services include:

- 1. Needs Assessment & Strategy Development** – Identifying funding gaps, investment opportunities, and risk mitigation strategies.
- 2. Regulatory Compliance & Approval** – Ensuring that financial products align with green finance regulations, ESG frameworks, and banking standards.
- 3. Product Customization** – Structuring innovative financial products tailored to green investments, impact-driven funding, and sustainable infrastructure.
- 4. Risk Assessment & Mitigation** – Managing financial and investment risks while enhancing business resilience and sustainability performance.
- By establishing **Green Credit Lines**, Enerqa helps financial institutions expand their portfolios, enhance sustainability reputations, and gain competitive advantages in climate finance.

Region	Number of Incubators	Common Types	Success Stories	Economic Impact
North America	~1,500+	University, Corporate, Private	Stripe, Dropbox, Airbnb, Uber, Lyft, Tesla, SpaceX, etc.	Creates ~900,000 jobs annually (NBIA); high survival rates for incubated startups
Europe	~2,000+	Public, University, Private	TransferWise, Spotify, etc.	Significant job creation; EU funding drives innovation and regional development
Asia	~2,500+	Government-led, Corporate, Private	Alibaba Group, Tencent, Baidu, etc.	Rapid growth in tech hubs (e.g., China, India); contributes to GDP growth
Latin America	~500+	Public, NGO-led, Private	Rapido, etc.	Boosts local economies; reduces poverty through entrepreneurship
Africa	~400+	NGO-led, Public, University	Andela, JUMIA, etc.	Drives job creation and innovation; supports UN Sustainable Development Goals (SDGs)
Middle East	~300+	Government-led, Corporate	Careem, etc.	Diversifies economies away from oil; fosters innovation in tech and smart cities
Oceania	~200+	University, Private, Government-led	Green, afterpay, etc.	Supports small economies; promotes innovation in niche industries

A green credit line is a dedicated **financial mechanism** that enables the financial institutions (FIs) to support and finance environmentally responsible projects particularly RE/EE projects and initiatives. This consultancy service focus on providing comprehensive guidance to financial institutions aiming to establish green credit lines, allowing them to play a pivotal role in advancing sustainability while maintaining financial viability.

Sustainable Growth

Enhanced Reputation

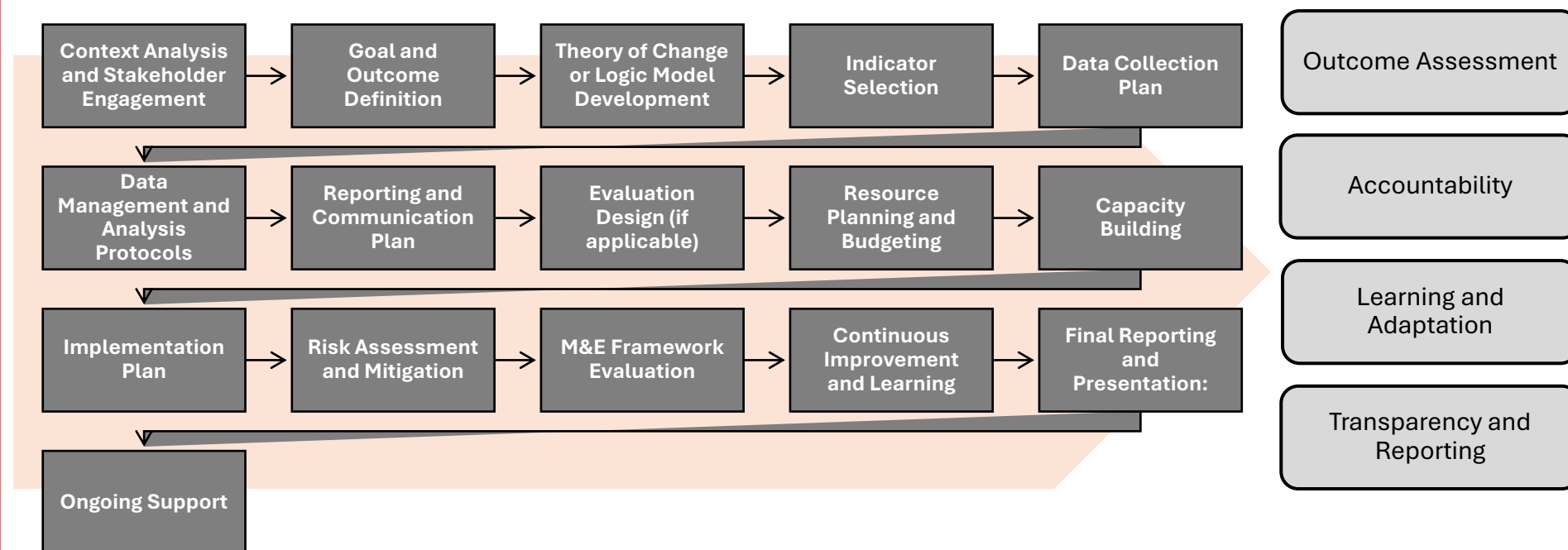
Competitive Advantage

Risk Mitigation

Compliance Assurance

Development of M&E Framework

enerQA team helps in Developing a Monitoring and Evaluation (M&E) framework as a consulting service which involves creating a structured plan for assessing and improving the effectiveness and impact of programs, projects, or initiatives as well as M&E costed plan.



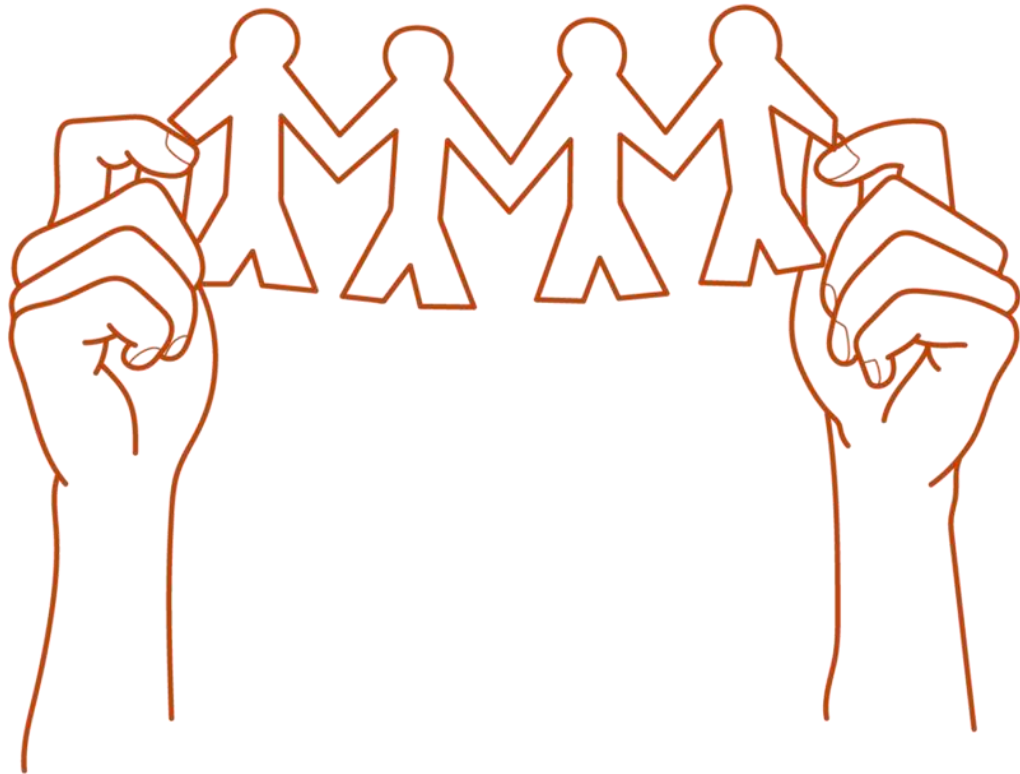
Why Choose Enerqa for M&E Consultancy?

- ✓ Bespoke, impact-oriented M&E frameworks tailored to project needs.
- ✓ Integration of ESG & sustainability metrics for added value.
- ✓ Cutting-edge technology & real-time analytics for smarter decision-making.
- ✓ A learning-driven, participatory approach that ensures M&E enhances performance, not just compliance.
- ✓ Capacity-building & long-term institutional strengthening for sustained success.

With Enerqa's M&E services, organizations move beyond compliance and embrace strategic, data-driven decision-making that maximizes impact, accountability, and sustainability.

Capacity Building

Unleash peoples' full potential



Capacity Building and training



At Enerqa, we believe that knowledge and expertise are the cornerstones of sustainable progress. Our capacity-building services are designed to equip individuals, organizations, and institutions with the skills, tools, and insights needed to drive impactful change in climate action, energy transition, environmental sustainability, and business development.

1. Training Programs

We develop customized training programs and materials, tailored to the specific needs of organizations. Our approach leverages diverse learning platforms, including interactive sessions, online modules, and hands-on training, ensuring effective knowledge transfer and engagement.

2. Workshops and Events

Enerqa organizes awareness workshops aimed at educating stakeholders on emerging trends, best practices, and strategic approaches in sustainability and climate action. Our tailored, hands-on workshops provide practical insights, ensuring that participants gain the necessary skills to apply knowledge effectively.

3. In-House Capacity Building and Evaluation

We offer technical assistance, mentoring, and evaluation services to strengthen the institutional and technical capabilities of organizations. Additionally, we provide impact assessment reports to measure the effectiveness and long-term benefits of capacity-building efforts.

4. Knowledge Transfer and Best Practices

Enerqa facilitates knowledge exchange through expert-led discussions and case studies, ensuring that organizations stay ahead of industry challenges by leveraging proven solutions and real-world insights.

5. Continuous Learning and Updates

We support ongoing learning and development through regular updates, training refreshers, and dedicated support services, helping organizations adapt to evolving challenges and emerging opportunities.

With a structured, strategic, and results-driven approach, Enerqa's capacity-building programs empower stakeholders across industries to develop resilient, innovative, and sustainable solutions for the future.

7. Enerqa's projects

Please Note: Certain projects referenced in this section were undertaken by key personnel in their individual capacities prior to the formal establishment of Enerqa Ltd. in 2022. These experiences have been fully integrated into the company's collective expertise, as the involved individuals now serve as founders and shareholders, ensuring continuity and enhanced institutional capacity for current and future engagements.

Assignment	Client
Net Zero Strategy Development	Qatar Free Zone
Green Climate Fund (GCF) - Concept Note for “Community based waste management system”	SEPS
PIF for “Building institutional and technical capacity of Sudan’s for transparency related activities under the UNFCCC and the Paris Agreement”	Higher Council for Environment & Natural Resources
Carbon footprint study and GHG emission baseline development for Al-Waha Farm	DAL Group
Project proposal for NAMA facilities “Building climate resilience for agribusiness in Eastern Sudan through the deployment of renewable energy applications”	UNIDO
MRV System for Energy and Transport Sectors in Sudan- Initiative Climate Change Transparency	UNEP-CCC
Participating in PP of the Sudan’s Capacity Building Initiative for Transparency project	UNDP/GEF
3rd Sudan National Greenhouse Gases (GHG) Inventory Report	HCENR / UNDP
GHG emissions baseline from road transport sector in Sudan – estimation of country specific emission factor	HCENR
Development of feed in tariff NAMA policy for a Utility Scale Power Generation from Wind Energy	Ministry of Water Resources, Irrigation and Electricity
Capacity Building component in the CBIT Sudan project	HCENR / UNDP
Modelling with IPCC Inventory Software 2006: “Sudan’s GHG inventory for the third national communication – Energy Sector”	HCENR
Promoting Low Carbon Development Project - Nationally Appropriate Mitigation Actions (Energy Sector)	HCENR
Impacts of climate change on Sudanese Oil and Gas industry	Ministry of Petroleum
Institutional Arrangements for the Ministry of Finance with regard to Climate Change impacts and Nationally Determined Contributions	Ministry of Finance and National Planning - Sudan
Modelling of climate change impacts on Sudan’s economy	World Research Institute (WRI)
Developing Standardized Baseline for The Project “Promoting the Use of Electric Water Pumps for Irrigation in Sudan	Ministry of Energy and Petroleum
National circumstances and institutional arrangements for the third national communication and biennial update reports of Sudan.	HCENR/UNDP
Emissions estimation from venting and flaring: Energy Auditing and Benchmarking Study for Khartoum Refinery	Ministry of Petroleum and Gas
Gap assessment in institutional arrangement, capacity building and formulation of MRV system in Sudan	HCENR/ UNDP



#	Assignment	Client
1	Environmental Impact Assessment for Copper Recycling Factory in Qatar	Seashore Group.
2	Environmental Impact Assessment study for road construction and drainage system establishment	Mazzraty Co.
3	Environmental compliance study for Aluminum Recycling Facility in Doha	Al-Assryia Factory
4	Environmental permit and compliance for mix gases industrial facility	National Industrial Gas Plants
5	Participating in the Global Environment Outlook (GEO)-7	UNEP
6	Environmental and Social Impact Assessment (ESIA) for Alternative fuel source (SRF)	AL Shamal Cement Factory
7	Environmental Exit Assessment for petroleum concessions	Greater Nile Petroleum
8	Environmental and Social Impact Assessment for block 6 and pipelines	Petro-Energy E&P
9	ESIA "Red Sea LNG Floating Solution Facilities and Port Sudan NG Pipeline"	SUNAGAS
10	Seepage study- evaporation ponds in Central Processing Facilities and number of Field Processing Facilities	Petro-Energy E&P
11	ESIA for waste management facilities - Gadarif State	Rawaa Co. for Gold Extraction
12	ESIA for feeder road between Id Al-Fursan –Kubum road ; South Darfur Livelihood and Agricultural Infrastructure Project	World vision Germany/Sudan
13	Participated in developing Strategic Environmental and Social Assessment (SESA) & ESM- Sudan REDD+ Program	World Bank
14	ESIA Study for Lau Tau Agribusiness Project, at Taraba State, Nigeria	Dangote Sugar Refinery
15	Participatory Land Use Planning (PLUP) and ESIA for five localities in Red Sea State	Welthunger hlfe Organization



#	Assignment	Client
1	Design of PV solar system (10+10 MW) for two farms in Qatar	Almana Group
2	Design of Solar energy system to Sulteen Fram project	SAIC Group
3	Modelling of Sudan's energy situation and forecast using LEAP software, Sudan's Second National Communication Project.	HCENR
4	Terminal Evaluation of the Solar Water Pump Project – Northern State - Sudan	UNDP
5	Renewable energy deployment within the " Resilience of agropastoral populations in eastern Sudan" project	UNIDO
6	Preparation of Joint Program Document for "Solar Revolution for Transforming Lives in the two states of Sudan through a National Solar Fund Scheme"	SDG Fund- Joint Project UNDP, UNHCR and UNICEF
7	GLB mega solar project using innovative gravitational power storage system	GLB Investment
8	Integrated Management Approach for Fuel Shortage in Sudan (Transport Sector)- research project	Ministry of Higher Education
9	Fugitives Emissions in Energy Auditing and Benchmarking Study for El-obayid Refinery	Ministry of Petroleum
10	Design of a Measurement, Reporting and Verification system for NDC enhancement the power sector	IRENA
11	Energy analysis, GHG emissions analysis and Levelized Cost of Energy for the current diesel water pumping systems and proposed solar PV pumping systems-	Renewable Energy Director
12	Design and formalization of institutional arrangements for energy statistics in the power sector	IRENA

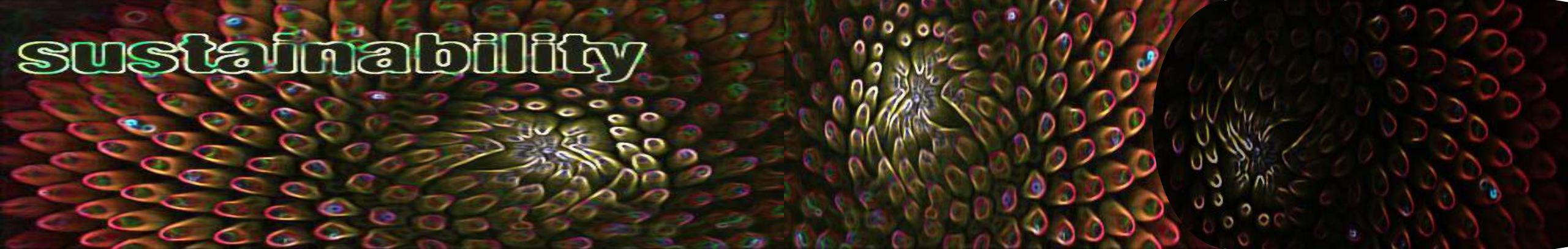


JOINT SDG FUND



#	Assignment	Client
1	Bioplastic facility in Qatar	Arab Qatari Company for Dairy Production W.L.L
2	Bankable feasibility study for aluminum factory – Qatar Development Bank	Alsuwadi – Assryah
3	Business plan for Sudanese Microfinance for Development Facility	Islamic Development Bank
4	Monitoring and Evaluation (M&E) Plan for Sudanese Microfinance for Development Facility	Islamic Development Bank
5	PPP contracts formulation – Alawqaf Islamic Foundation	Ministry of Al-Awgaf -
6	Feasibility Study for meat processing facilities, agricultural products and others	University of Khartoum
7	Plant/ Facility layout, design and technical specifications of machinery and equipment for different manufacturing facilities	African Development Bank
8	Technical and Economic Feasibility for the Replacement of Diesel Water Pumps with Solar Water Pumps, South Kordofan and Gedaref States	SEHC- UNDP funded Project
9	Assets revaluation for petroleum upstream facilities	Sharif Petroleum Co.
10	Review of the pre-feasibility study of the Saudi-Polychem Project	Esnad Co.
11	Khartoum Refinery's Asset Management	KRC
12	Business plan for multi-functional facilities in 3 cities	UofK
13	Feasibility and technical study of deployment of Diesel Fuel Unit	2BOPCO Operating Co.
14	Business Plan for Khairat Organic Fertilizer factory	SOCORO - ESTIDAMA
15	Investment map for the Republic of South Sudan	Sustainable Development Enterprise (SDE)





Our Sustainability Checklist is a structured tool used to outline the specific roles and responsibilities of the consultancy services for each project. Depending on the project requirements, **ticks will be applied to relevant items**, indicating which aspects of information provision are applicable. Below is an explanation of how each item on the checklist is used:

Sustainability Checklist for Consultancy Project

- ☐ Climate Data Aggregation
- ☐ Stakeholder Engagement
- ☐ Capacity Building
- ☐ Policy Support
- ☐ Tools and Platform Development
- ☐ Sustainability Reporting
- ☐ Awareness
- ☐ Data Visualization
- ☐ Compliance and Reporting
- ☐ Research and Innovation
- ☐ Content Creation
- ☐ Modeling
- ☐ Data Collection

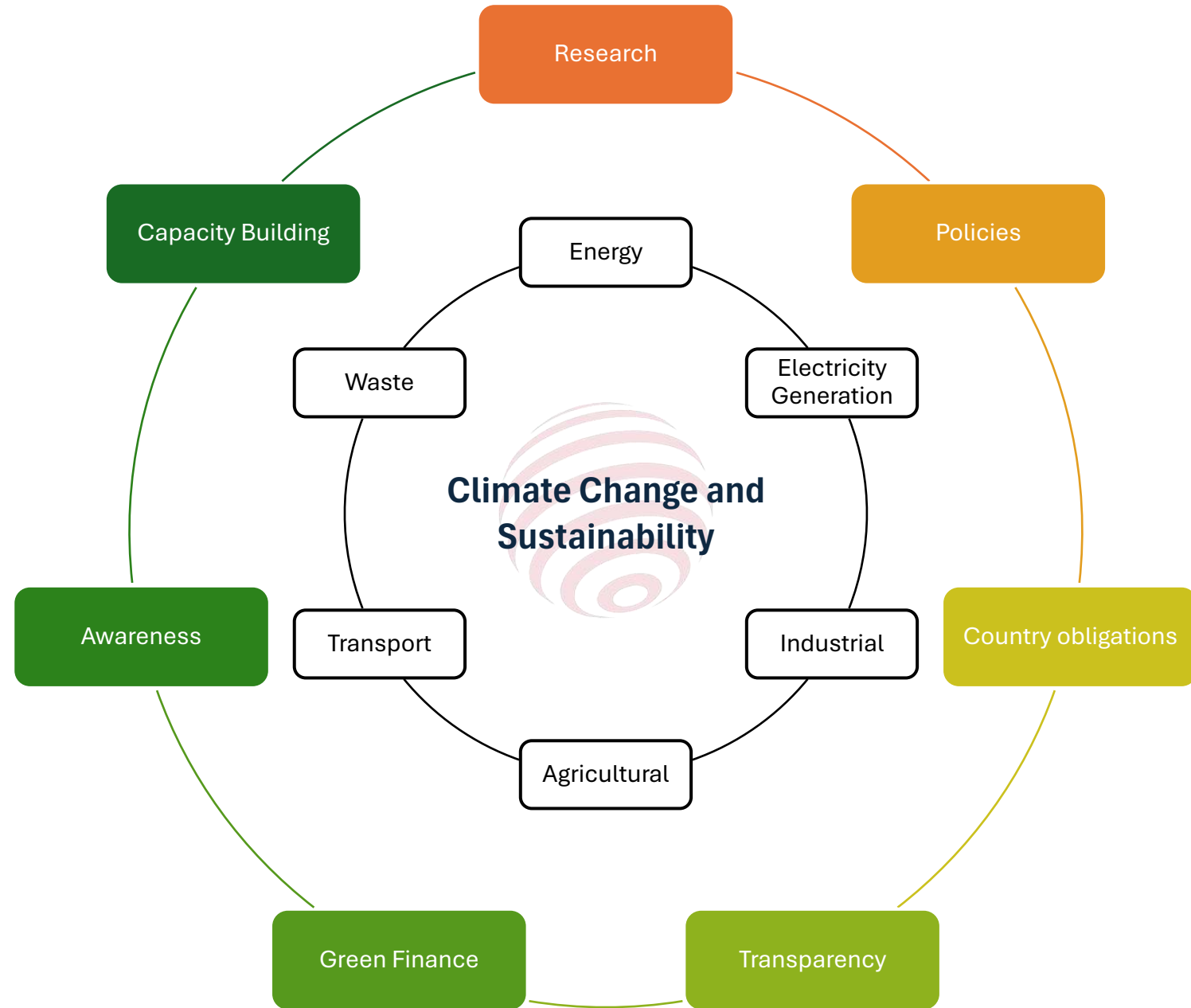
- 1.Data Collection and Compilation** – This item is checked when the project requires gathering raw data from various sources, such as surveys, databases, reports, or monitoring systems.
- 2.Data Verification and Validation** – A tick on this item indicates that the information provider is responsible for ensuring the accuracy, consistency, and reliability of collected data before it is used for reporting or analysis.
- 3.Regulatory Compliance and Reporting** – This applies to projects that require adherence to national or international regulations, standards, or frameworks, ensuring proper documentation and compliance.
- 4.Stakeholder Coordination** – If checked, this means the information provider is responsible for engaging with relevant stakeholders, including government agencies, private sector entities, and NGOs, to gather insights and ensure smooth communication.
- 5.ESG and Sustainability Metrics** – A tick on this item signifies that the project involves environmental, social, and governance (ESG) factors, requiring assessment and reporting on sustainability performance.
- 6.Technical Studies and Research** – When selected, this item highlights the need for specialized research, feasibility studies, and technical assessments related to climate change, energy, or other sustainability topics.
- 7.Gap Analysis and Benchmarking** – If this is checked, the information provider must conduct a comparative analysis to identify gaps in policies, practices, or performance against industry best practices or regulatory requirements.
- 8.Risk Assessment and Mitigation** – This item applies when the project requires identifying potential risks (financial, environmental, operational) and developing strategies to mitigate them.
- 9.Audit and Quality Assurance** – A tick here indicates that the information provider must conduct audits or quality checks to ensure the integrity of data, reports, and compliance measures.
- 10.Capacity Building and Training** – If this item is marked, the project involves training personnel, conducting workshops, or developing knowledge-sharing programs to enhance institutional capacity.

Sectors and Scopes

Enerqa has completed almost 50 projects in the areas of climate change, energy, environment, and green business models.

Our work has been diverse; working with public sector, UN organizations and NGOs, and private sector clients, from facility level to state and country levels, across a wide range of industries, under different scopes and objectives.

At Enerqa, we are committed to delivering comprehensive, impactful, and sustainable solutions for climate, energy, environment, and business development. With a global network of experts and strategic partners, we ensure that every project benefits from cutting-edge expertise, local insights, and innovative approaches. Through our structured methodologies, data-driven strategies, and commitment to excellence, we empower governments, businesses, and organizations to navigate complex challenges, drive sustainability, and achieve long-term success. As we move forward, Enerqa remains a trusted partner, bridging technical expertise with actionable solutions to shape a more resilient, low-carbon, and prosperous future.



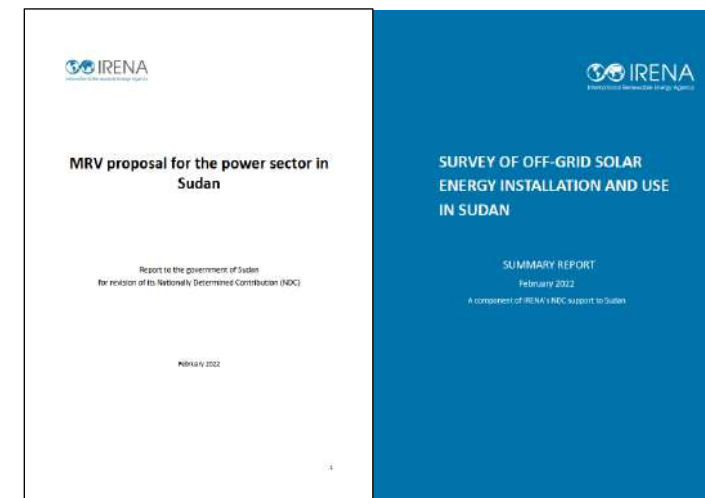
Technical Assistance and Capacity Building for the NDC Enhancement International Renewable Energy Agency (IRENA) Funded Project

Project activities

- Energy data audit and gap analysis to identify existing data, data sources and gaps
- Data collection and fill of identified data gaps
- GHG emissions calculation and provision of structured datasets
- Development of a Measurement, Reporting and Verification (MRV) system for electricity generation in Sudan, to be part of the National MRV system
- Capacity building

The project also included two surveys:

- Off Grid Solar PV Installation and Use
- Bagasse use in electricity generation and associated GHG emissions in the sugar production industry



Information Provider Scope Checklist

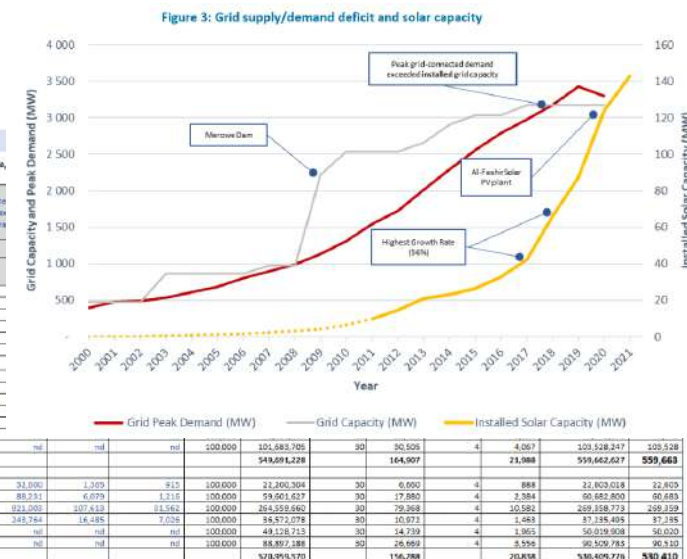
- ☒ Climate Data Aggregation
- ☒ Stakeholder Engagement
- ☒ Capacity Building
- ☒ Policy Support
- ☒ Tools and Platform Development
- ☐ Sustainability Reporting
- ☒ Awareness
- ☒ Data Visualization
- ☒ Compliance and Reporting
- ☒ Research and Innovation
- ☒ Content Creation
- ☒ Modeling
- ☒ Data Collection

Annex G: Summary sheet for bagasse consumption in sugar factories and GHG emissions

Table G.1: Summary of worked data,

Year	Factory	Gen From National Grid (MWh)	Total Bagasse Consumption (tons)	Total Steam Produced (tons)	Bagasse Consumption for Electricity Generation (tons)	NCV (MJ/kg)	Energy Input to Electricity Generation (MWh)	Energy Input to Electricity Generation (TJ)	Sto. Emission
					A	B	C	D	
							A x (B x 1000)	C / 10 ⁶	
2018	Assiout	11,310	91,620	104,400	45,072	7.51028	338,505,040	338.51	
	New Halfa	5,561	244,180	279,035	97,864	7.51028	733,483,840	733.48	
	Esnein	754,417	839,049	3,185,291	364,271	7.1118	2,595,585,531	2,595.58	
	White Nile	nd	202,128	329,476	45,288	7.51028	339,753,124	339.75	
	Sennar	nd	nd	nd	183,838	7.60776	1,054,881,289	1,054.88	
	Gumel	nd	nd	nd	154,300	7.51028	1,158,618,692	1,158.64	
	Total				839,880		6,215,943,733	6,215.94	
2019	Assiout	19,080	94,160	109,650	24,478	7.51028	200,343,485	200.36	
	New Halfa	6,808	251,016	257,024	62,409	7.51028	694,017,178	694.02	
	Esnein	245,756	295,179	3,140,380	325,167	7.1118	2,312,844,955	2,312.84	
	White Nile	nd	89,328	324,473	63,195	7.51028	489,635,434	489.64	
	Sennar	nd	nd	nd	99,044	7.60776	783,116,161	783.12	
	Gumel	nd	nd	nd	135,893	7.51028	1,016,837,050	1,016.84	
	Total				743,886		5,496,942,283	5,496.93	
2020	Assiout	13,817	55,520	91,320	29,560	7.51028	222,003,043	222.00	
	New Halfa	6,624	198,187	220,574	79,360	7.51028	596,016,266	596.02	
	Esnein	245,000	393,022	3,188,801	371,449	7.1118	2,645,595,603	2,645.60	
	White Nile	nd	88,378	418,853	48,866	7.51028	365,720,777	365.72	
	Sennar	nd	nd	nd	62,127	7.60776	481,287,127	481.29	
	Gumel	nd	nd	nd	118,567	7.51028	886,971,879	886.97	
	Total				710,059		5,205,595,605	5,205.60	

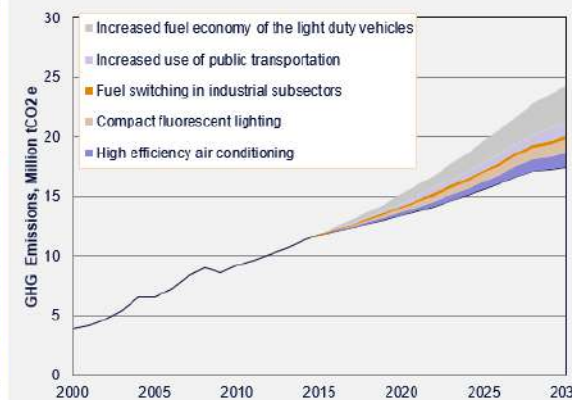
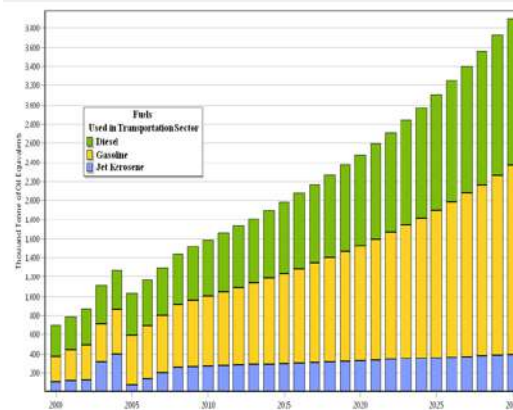
Note: Figures in blue are data received from the factories as-is; the rest of the figures either estimated or calculated.



Project activities

- # Technical Support and Capacity Building for Applying ICAT Guidance in Sudan's Transport Sector

- ☒ Climate Data Aggregation
- ☒ Stakeholder Engagement
- ☒ Capacity Building
- ☒ Policy Support
- ☒ Tools and Platform Development
- ☐ Sustainability Reporting
- ☒ Awareness
- ☒ Data Visualization
- ☒ Compliance and Reporting
- ☐ Research and Innovation
- ☒ Content Creation
- ☒ Modeling
- ☒ Data Collection



1. Vehicle Population and Type:			
DATA CATEGORY	DATA ELEMENT	DESCRIPTION	VERIFICATION METHOD
VEHICLE POPULATION	Total Number of Vehicles	Overall number of vehicles registered in the country, categorized by type (e.g., cars, motorcycles, trucks, buses).	Government vehicle registration records, surveys, vehicle inspections.
VEHICLE TYPE	Specific Vehicle Category	Detailed breakdown of vehicle population by subcategories (e.g., passenger cars, SUVs, light trucks, heavy trucks, motorcycles).	Same as above, with additional vehicle classification if needed.
VEHICLE AGE	Age Distribution of Vehicles	Number of vehicles in each age category (e.g., 0-5 years, 6-10 years, etc.).	Registration records, vehicle inspections, surveys.
TECHNOLOGY LEVEL	Emission Standards	Classification of vehicles based on their emission standards (e.g., Euro 4, Euro 5, etc.).	Registration records, vehicle inspections, manufacturer data.

2. Fuel Consumption:			
DATA CATEGORY	DATA ELEMENT	DESCRIPTION	VERIFICATION METHOD
FUEL SALES DATA	Sales per Fuel Type	Total volume of each fuel type (e.g., gasoline, diesel, LPG) sold within the country.	Fuel retail outlets, wholesale distributors, tax records.
VEHICLE SPECIFIC FUEL CONSUMPTION	Average Fuel Consumption by Vehicle Type	Fuel consumption per kilometer traveled for different vehicle categories, often expressed as liters per 100 kilometers (L/100km) or miles per gallon (MPG).	Manufacturer data, fuel economy tests, in-country studies.
ACTIVITY FACTORS	Driving Patterns and Fuel Use	Adjustments for specific driving conditions (e.g., urban vs. rural, cold starts) and non-road mobile machinery fuel use.	Surveys, traffic models, fuel consumption studies.

3. Average Annual Traveled Kilometers:			
DATA CATEGORY	DATA ELEMENT	DESCRIPTION	VERIFICATION METHOD
VEHICLE TYPE SPECIFIC VKT	Average Annual Distance Traveled per Vehicle Type	Estimated average kilometers traveled per year for each vehicle category.	Traffic surveys, GPS tracking, distance-based tolls, household travel surveys.
ROAD NETWORK	Characteristics of Road Infrastructure	Information on road types (highways, rural roads), traffic volumes, and average speeds.	Traffic surveys, infrastructure databases, satellite imagery.
REGIONAL VARIATIONS	VKT Differences by Region	Adjustments for regional variations in travel patterns and infrastructure.	Same as above, with focus on regional discrepancies.

Model development for LCOE, solar pump size and cost savings for irrigation

UNDP-GEF Funded Project

Project activities

- Stakeholder identification
- Desk review and secondary data collection
- Surveys designs and questionnaires preparations
- Awareness sessions design and awareness material preparation
- Primary data collection
- Awareness sessions and material dissemination
- Data analysis and preparation of structured datasets
- Validation workshop
- Final report and datasets

3.4 Inputs			
3.4.1 Technical			
Diesel pump driver rated power (S.Kordofan),		18	
Diesel pump driver rated power (Gedaref), HP		20	
Operating time (days)		270	
Solar pump efficiency		100%	
Diesel pump driver efficiency		18%	
$E_d = (E_{11} + R) / (G_d > G_{threshold}) P_{G1} P_{G2} P_{G3}$			
R (Losses due to friction in irrigation system)		10%	
Gd (Fraction of the day during which solar radiation is above the threshold at which the pump starts to work)		0.95	
μG (Yield of the photovoltaic generator)		0.85	
μI (Yield of AC/DC converter)		0.9	
μMB (Yield of the pump)		0.43	
Denominator of the equation		31.25%	
Power losses due to high temperature		10%	
3.4.2 Financial			
OPEX of Solar Pumps (% of initial investment)		2%	
Lifetime t, Solar & Diesel (years)		20	
Cost per Watt (Solar pump set), U.S. Dollars			
1.2			

Gedaref		
Average flow rate (m3/hr)	25.075	
Average daily irrigation	10.13	
Average daily flow rate (m3/day)	254.00975	
Solar irradiance (kWh/m2); sunlight hours		
	6.2	
Vertical Head (m)		
5		
10		
15		
20		

South Kordofan		
Average flow rate (m3/hr)	25.075	
Average daily irrigation	9.39	
Average daily flow rate	235.45425	
Solar irradiance (kWh/m2); sunlight hours		
	6	
Vertical Head (m)		
10		
15		
20		
30		

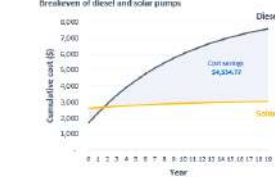
Information Provider Scope Checklist

- ☒ Climate Data Aggregation
- ☒ Stakeholder Engagement
- ☒ Capacity Building
- ☒ Policy Support
- ☒ Tools and Platform Development
- ☐ Sustainability Reporting
- ☒ Awareness
- ☒ Data Visualization
- ☐ Compliance and Reporting
- ☒ Research and Innovation
- ☒ Content Creation
- ☒ Modeling
- ☒ Data Collection

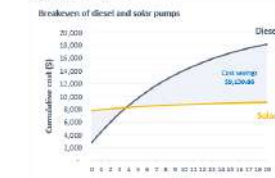
LCOE (solar pumping system) 5\$/kWh 0.33

2.3 Breakdown of solar and diesel cumulative costs

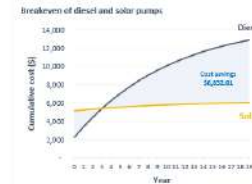
Head (m) 5



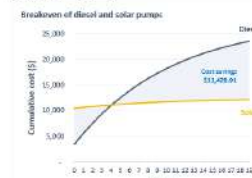
Head (m) 15



Head (m) 10



Head (m) 20



Locality	Integrated Land (hectares)	Fuel_Cycle1 (kg)	Fuel_Cycle2 (kg)	Fuel_Cycle3 (kg)	Energy Produced Cycle 1 (kWh)	Energy Produced Cycle 2 (kWh)	CO2 Emission	CH4 Emission	N2O Emission	CO2e (kg)	Fuel_Cycle1 (kg)	Fuel_Cycle2 (kg)	Fuel_Cycle3 (kg)	Energy Produced Cycle 1 (kWh)	Energy Produced Cycle 2 (kWh)	CO2 Emission	CH4 Emission	N2O Emission	CO2e (kg)	
Wadigra	4	795	675.75	25,097.150.00	0.02	2,351.14	0.28	0.02	2,351.14	0.02	456.00	456.00	25,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	1,722.51
Wadigra	4	636	540.45	23,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	456.00	456.00	23,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	1,722.51
East Courtyrads	2	318	270.30	11,622.900.00	0.01	961.26	0.12	0.01	961.26	0.01	228.00	228.00	11,622.900.00	0.01	961.26	0.12	0.01	961.26	0.01	961.26
East Courtyrads	2	277	235.45	17,484.950.00	0.01	1,391.86	0.17	0.01	1,391.86	0.01	228.00	228.00	17,484.950.00	0.01	1,391.86	0.17	0.01	1,391.86	0.01	1,391.86
East Courtyrads	2	277	235.45	17,484.950.00	0.01	1,391.86	0.17	0.01	1,391.86	0.01	228.00	228.00	17,484.950.00	0.01	1,391.86	0.17	0.01	1,391.86	0.01	1,391.86
Al-Gadala	3	795	675.75	25,097.150.00	0.02	2,351.14	0.28	0.02	2,351.14	0.02	456.00	456.00	25,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	1,722.51
Al-Gadala	3	636	540.45	23,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	456.00	456.00	23,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	1,722.51
Tharad	2	318	270.30	11,622.900.00	0.01	961.26	0.12	0.01	961.26	0.01	228.00	228.00	11,622.900.00	0.01	961.26	0.12	0.01	961.26	0.01	961.26
Tharad	2	277	235.45	17,484.950.00	0.01	1,391.86	0.17	0.01	1,391.86	0.01	228.00	228.00	17,484.950.00	0.01	1,391.86	0.17	0.01	1,391.86	0.01	1,391.86
Tharad	2	277	235.45	17,484.950.00	0.01	1,391.86	0.17	0.01	1,391.86	0.01	228.00	228.00	17,484.950.00	0.01	1,391.86	0.17	0.01	1,391.86	0.01	1,391.86
Al-Gadala	3	795	675.75	25,097.150.00	0.02	2,351.14	0.28	0.02	2,351.14	0.02	456.00	456.00	25,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	1,722.51
Al-Gadala	3	636	540.45	23,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	456.00	456.00	23,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	1,722.51
Al-Gadala	3	599	515.15	20,111.450.00	0.02	1,391.86	0.17	0.02	1,391.86	0.02	456.00	456.00	20,111.450.00	0.02	1,391.86	0.17	0.02	1,391.86	0.02	1,391.86
Al-Gadala	3	795	675.75	25,097.150.00	0.02	2,351.14	0.28	0.02	2,351.14	0.02	456.00	456.00	25,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	1,722.51
Al-Gadala	3	636	540.45	23,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	456.00	456.00	23,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	1,722.51
Al-Gadala	3	599	515.15	20,111.450.00	0.02	1,391.86	0.17	0.02	1,391.86	0.02	456.00	456.00	20,111.450.00	0.02	1,391.86	0.17	0.02	1,391.86	0.02	1,391.86
Al-Gadala	3	795	675.75	25,097.150.00	0.02	2,351.14	0.28	0.02	2,351.14	0.02	456.00	456.00	25,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	1,722.51
Al-Gadala	3	636	540.45	23,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	456.00	456.00	23,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	1,722.51
Al-Gadala	3	599	515.15	20,111.450.00	0.02	1,391.86	0.17	0.02	1,391.86	0.02	456.00	456.00	20,111.450.00	0.02	1,391.86	0.17	0.02	1,391.86	0.02	1,391.86
Al-Gadala	3	795	675.75	25,097.150.00	0.02	2,351.14	0.28	0.02	2,351.14	0.02	456.00	456.00	25,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	1,722.51
Al-Gadala	3	636	540.45	23,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	456.00	456.00	23,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	1,722.51
Al-Gadala	3	599	515.15	20,111.450.00	0.02	1,391.86	0.17	0.02	1,391.86	0.02	456.00	456.00	20,111.450.00	0.02	1,391.86	0.17	0.02	1,391.86	0.02	1,391.86
Al-Gadala	3	795	675.75	25,097.150.00	0.02	2,351.14	0.28	0.02	2,351.14	0.02	456.00	456.00	25,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	1,722.51
Al-Gadala	3	636	540.45	23,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	456.00	456.00	23,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	1,722.51
Al-Gadala	3	599	515.15	20,111.450.00	0.02	1,391.86	0.17	0.02	1,391.86	0.02	456.00	456.00	20,111.450.00	0.02	1,391.86	0.17	0.02	1,391.86	0.02	1,391.86
Al-Gadala	3	795	675.75	25,097.150.00	0.02	2,351.14	0.28	0.02	2,351.14	0.02	456.00	456.00	25,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	1,722.51
Al-Gadala	3	636	540.45	23,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	456.00	456.00	23,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	1,722.51
Al-Gadala	3	599	515.15	20,111.450.00	0.02	1,391.86	0.17	0.02	1,391.86	0.02	456.00	456.00	20,111.450.00	0.02	1,391.86	0.17	0.02	1,391.86	0.02	1,391.86
Al-Gadala	3	795	675.75	25,097.150.00	0.02	2,351.14	0.28	0.02	2,351.14	0.02	456.00	456.00	25,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	1,722.51
Al-Gadala	3	636	540.45	23,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	456.00	456.00	23,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	1,722.51
Al-Gadala	3	599	515.15	20,111.450.00	0.02	1,391.86	0.17	0.02	1,391.86	0.02	456.00	456.00	20,111.450.00	0.02	1,391.86	0.17	0.02	1,391.86	0.02	1,391.86
Al-Gadala	3	795	675.75	25,097.150.00	0.02	2,351.14	0.28	0.02	2,351.14	0.02	456.00	456.00	25,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	1,722.51
Al-Gadala	3	636	540.45	23,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	456.00	456.00	23,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	1,722.51
Al-Gadala	3	599	515.15	20,111.450.00	0.02	1,391.86	0.17	0.02	1,391.86	0.02	456.00	456.00	20,111.450.00	0.02	1,391.86	0.17	0.02	1,391.86	0.02	1,391.86
Al-Gadala	3	795	675.75	25,097.150.00	0.02	2,351.14	0.28	0.02	2,351.14	0.02	456.00	456.00	25,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	1,722.51
Al-Gadala	3	636	540.45	23,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	456.00	456.00	23,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	1,722.51
Al-Gadala	3	599	515.15	20,111.450.00	0.02	1,391.86	0.17	0.02	1,391.86	0.02	456.00	456.00	20,111.450.00	0.02	1,391.86	0.17	0.02	1,391.86	0.02	1,391.86
Al-Gadala	3	795	675.75	25,097.150.00	0.02	2,351.14	0.28	0.02	2,351.14	0.02	456.00	456.00	25,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	1,722.51
Al-Gadala	3	636	540.45	23,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	456.00	456.00	23,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	1,722.51
Al-Gadala	3	599	515.15	20,111.450.00	0.02	1,391.86	0.17	0.02	1,391.86	0.02	456.00	456.00	20,111.450.00	0.02	1,391.86	0.17	0.02	1,391.86	0.02	1,391.86
Al-Gadala	3	795	675.75	25,097.150.00	0.02	2,351.14	0.28	0.02	2,351.14	0.02	456.00	456.00	25,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	1,722.51
Al-Gadala	3	636	540.45	23,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	456.00	456.00	23,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	1,722.51
Al-Gadala	3	599	515.15	20,111.450.00	0.02	1,391.86	0.17	0.02	1,391.86	0.02	456.00	456.00	20,111.450.00	0.02	1,391.86	0.17	0.02	1,391.86	0.02	1,391.86
Al-Gadala	3	795	675.75	25,097.150.00	0.02	2,351.14	0.28	0.02	2,351.14	0.02	456.00	456.00	25,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	1,722.51
Al-Gadala	3	636	540.45	23,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	456.00	456.00	23,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	1,722.51
Al-Gadala	3	599	515.15	20,111.450.00	0.02	1,391.86	0.17	0.02	1,391.86	0.02	456.00	456.00	20,111.450.00	0.02	1,391.86	0.17	0.02	1,391.86	0.02	1,391.86
Al-Gadala	3	795	675.75	25,097.150.00	0.02	2,351.14	0.28	0.02	2,351.14	0.02	456.00	456.00	25,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	1,722.51
Al-Gadala	3	636	540.45	23,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	456.00	456.00	23,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	1,722.51
Al-Gadala	3	599	515.15	20,111.450.00	0.02	1,391.86	0.17	0.02	1,391.86	0.02	456.00	456.00	20,111.450.00	0.02	1,391.86	0.17	0.02	1,391.86	0.02	1,391.86
Al-Gadala	3	795	675.75	25,097.150.00	0.02	2,351.14	0.28	0.02	2,351.14	0.02	456.00	456.00	25,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	1,722.51
Al-Gadala	3	636	540.45	23,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	456.00	456.00	23,145.800.00	0.02	1,722.51	0.28	0.02	1,722.51	0.02	1,722.51
Al-Gadala	3	599	515.15	20,111.450.00	0.02	1,391.86	0.17	0.02	1,391.86	0.02	456.00	456.00	20,111.450.00	0.02	1,391.86	0.17	0.02	1,391.86	0.02	1,391.86
Al-Gadala	3	795	675.75	25,097.150.00	0.02	2,351.1														

Waste recycling assessment for export potential and green finance in Qatar

Project activities

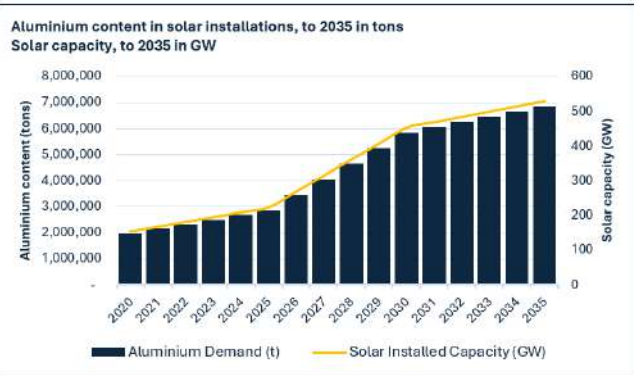
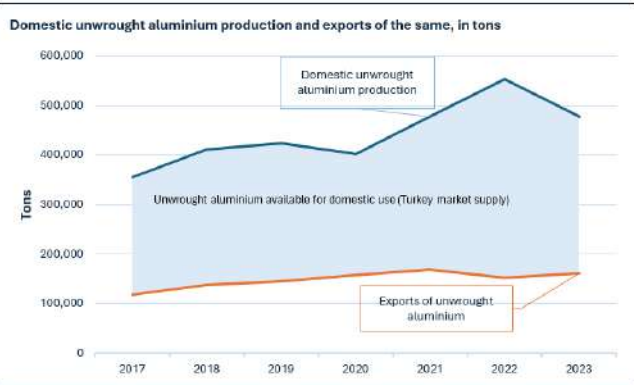
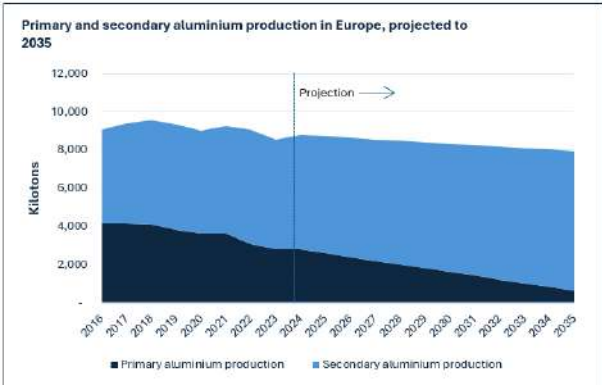
- Stakeholder identification
- Desk review and secondary data collection
- Interviews designs
- Primary data collection
- Analysis and assessment: Market, Technical and Financial
- Validation meetings
- Final report and datasets

Information Provider Scope Checklist

- ☒ Climate Data Aggregation
- ☒ Stakeholder Engagement
- ☐ Capacity Building
- ☐ Policy Support
- ☒ Tools and Platform Development
- ☐ Sustainability Reporting
- ☐ Awareness
- ☒ Data Visualization
- ☐ Compliance and Reporting
- ☒ Research and Innovation
- ☒ Content Creation
- ☒ Modeling
- ☒ Data Collection

QATAR		Phase 1						Phase 2					
		2025			2026			2027			2028		
Descriptions	Quantity	Unit Cost	Total Cost	Quantity	Unit Cost	Total Cost	Quantity	Unit Cost	Total Cost	Quantity	Unit Cost	Total Cost	
Labor Salary	27	7,000	189,000	27	7,000	189,000	51	7,000	357,000	51	7,000	357,000	
Admin Salary	8	25,000	200,000	8	25,000	200,000	8	25,000	200,000	8	25,000	200,000	
Procurement Salary	3	15,000	45,000	3	15,000	45,000	3	15,000	45,000	3	15,000	45,000	
Subcontractor Salary	3	15,000	45,000	3	15,000	45,000	3	15,000	45,000	3	15,000	45,000	
Subcontractor Salary	3	10,000	30,000	3	10,000	30,000	3	10,000	30,000	3	10,000	30,000	
Managing Partner Salary	8	80,000	640,000	8	80,000	640,000	8	80,000	640,000	8	80,000	640,000	
Electricity	380,000	0.08	30,800	481,250	0.08	38,500	500,135	500,135	500,135	500,135	500,135	500,135	
Water	185	3.04	562	209	3.04	636	338	338	338	338	338	338	
Rg	323,000	0.47	152,791	393,750	0.47	185,063	284,869	284,869	284,869	284,869	284,869	284,869	
Fuel	3	2,500	7,500	3	2,500	7,500	3	2,500	7,500	3	2,500	7,500	
Transportation	1,800	400	720,000	1,875	400	750,000	2,400	400	960,000	2,400	400	960,000	
Customer Service	6,800,000	8%	544,000	8,099,611	8%	647,969	10,738,455	8%	859,076	13,270,000	8%	647,969	
Insurance	3	104,896	314,688	3	104,896	314,688	3	104,896	314,688	3	104,896	314,688	
Land Rent	3	10,475	31,425	3	10,475	31,425	3	10,475	31,425	3	10,475	31,425	
Laboratory	3	10,000	30,000	3	10,000	30,000	3	10,000	30,000	3	10,000	30,000	
Landfill	3	11,888	35,664	3	11,888	35,664	3	11,888	35,664	3	11,888	35,664	
Total Operational Cost per Month		5,445,584		6,468,294		8,408,276		9,888,393		11,888,393		13,888,393	
Total Operational Cost per Year		65,347,008		77,619,528		100,900,352		118,660,716		142,660,716		166,660,716	

UNITED		Phase 1						Phase 2					
		2025			2026			2027			2028		
Descriptions	Quantity	Unit Cost	Total Cost	Quantity	Unit Cost	Total Cost	Quantity	Unit Cost	Total Cost	Quantity	Unit Cost	Total Cost	
Labor Salary	27	3,000	81,000	27	3,000	81,000	51	3,000	153,000	51	3,000	153,000	
Admin Salary	8	4,130	33,040	8	4,130	33,040	8	4,130	33,040	8	4,130	33,040	
Procurement Salary	3	4,130	12,390	3	4,130	12,390	3	4,130	12,390	3	4,130	12,390	
Subcontractor Salary	3	4,130	12,390	3	4,130	12,390	3	4,130	12,390	3	4,130	12,390	
Subcontractor Salary	3	2,740	8,220	3	2,740	8,220	3	2,740	8,220	3	2,740	8,220	
Managing Partner Salary	8	10,000	80,000	8	10,000	80,000	8	10,000	80,000	8	10,000	80,000	
Electricity	380,000	0.08	30,400	481,250	0.08	38,500	500,135	500,135	500,135	500,135	500,135	500,135	
Water	185	3.04	562	209	3.04	636	338	338	338	338	338	338	
Rg	323,000	0.15	48,450	393,750	0.15	59,063	59,069	59,069	59,069	59,069	59,069	59,069	
Fuel	3	2,500	7,500	3	2,500	7,500	3	2,500	7,500	3	2,500	7,500	
Transportation	1,800	330	594,000	1,875	330	620,625	2,400	330	792,000	2,400	330	792,000	
Customer Service	6,800,000	0	0	8,099,611	0	0	10,738,455	0	0	13,270,000	0	0	
Insurance	3	28,884	86,652	3	28,884	86,652	3	28,884	86,652	3	28,884	86,652	
Land Rent	3	2,800	8,400	3	2,800	8,400	3	2,800	8,400	3	2,800	8,400	
Laboratory	3	3,530	10,590	3	3,530	10,590	3	3,530	10,590	3	3,530	10,590	
Landfill	3	3,183	9,549	3	3,183	9,549	3	3,183	9,549	3	3,183	9,549	
Total Operational Cost per Month		395,677		448,793		5,808,576		6,888,576		8,108,576		9,188,576	
Total Operational Cost per Year		4,748,124		5,385,516		69,702,792		82,662,912		97,302,912		110,262,912	



Risk Category	Risk	Probability	Impact	Total Score	Impact on the Project	Priority
1. Market Risks	1.1 Price Volatility	5	5	25	Severe financial loss, inability to forecast revenue streams, and pressure on pricing strategy.	Critical
	1.2 Demand Variability	4	4	16	Revenue shortfalls due to reduced demand from key industries like construction and	Critical
	Competition	4	3	12	Loss of market share to major players like Qatium and EGA, reducing profitability.	High
2. Operational Risks	2.1 Supply Chain Disruptions	4	4	16	Production halts due to lack of raw materials, impacting delivery timelines and revenues.	Critical
	2.2 Equipment Downtime	3	4	12	Production delays, increased repair costs, and reduced operational efficiency.	High
	2.3 Labor Shortages	3	3	9	Difficulty meeting production targets due to insufficient skilled workforce	Moderate
3. Financial Risks	3.1 High Dependency on Imports	4	4	16	Increased project costs and reduced profitability due to high import taxes on materials	High
	3.2 Cost Overruns	5	5	25	Project delays, increased project costs, and risk to financial viability.	Critical
	3.3 Exchange Rate Fluctuations	3	3	9	Higher costs of imported materials and machinery, reducing profit margins.	Moderate
4. Regulatory Risks	4.1 Environmental Compliance	4	3	12	Failure to meet loan repayment schedules could result in penalties and reputational harm.	High
	4.2 Trade Policies	3	3	9	Fines, operational halts, or loss of licenses due to non-compliance with environmental laws.	Moderate
	5.1 Hazardous Waste Management	3	4	12	Higher tariffs or restrictions increasing operational and export costs.	Moderate
5. Environmental Risks	5.2 Emissions and Energy Usage	2	2	4	Environmental damage, fines, and reputational damage if waste is not managed effectively.	High
					Potential penalties, operational disruptions, and reputational harm from non-compliance.	Low

Integrated Community-based Solid Waste Management System

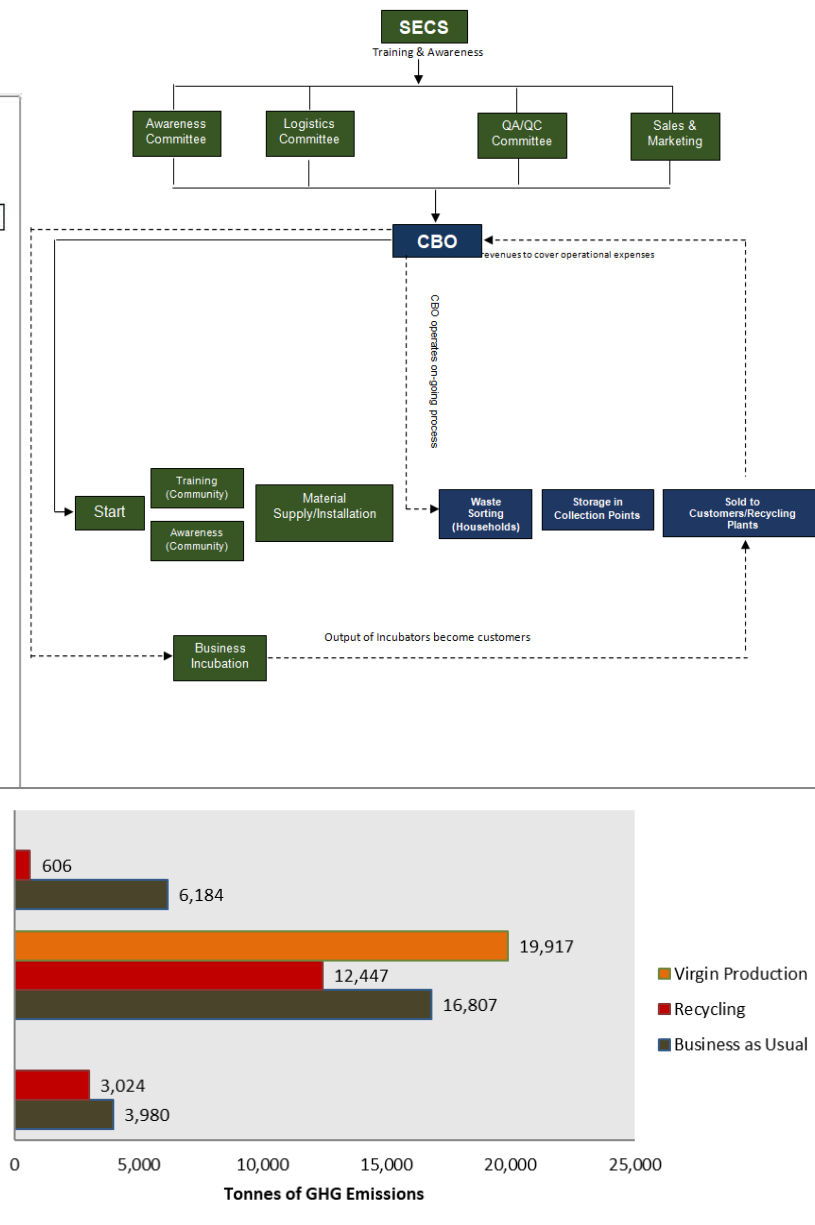
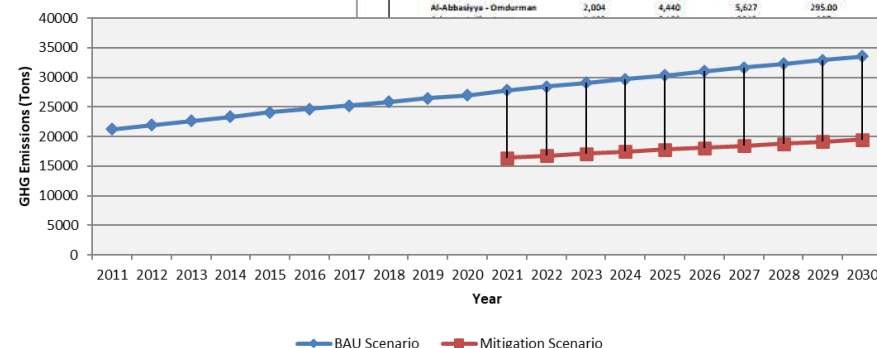
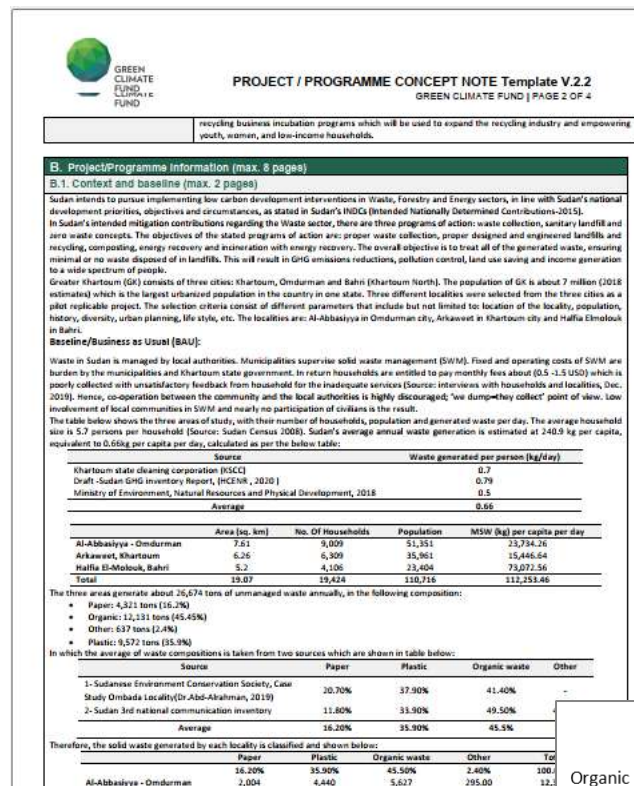
GCF Concept Note

Project activities

- Stakeholder identification
- Desk review and secondary data collection
- Interviews and surveys designs
- Primary data collection
- Analysis
- Business model development
- Validation workshop

Information Provider Scope Checklist

- ☒ Climate Data Aggregation
- ☒ Stakeholder Engagement
- ☒ Capacity Building
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- ☒ Tools and Platform Development
- ☐ Sustainability Reporting
- ☒ Awareness
- ☒ Data Visualization
- ☐ Compliance and Reporting
- ☒ Research and Innovation
- ☒ Content Creation
- ☒ Modeling
- ☒ Data Collection



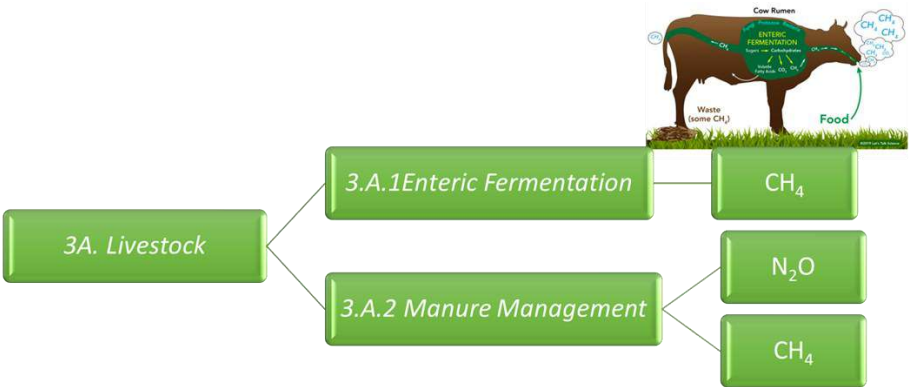
Carbon footprint study for Agricultural mega-project

Project activities

- Data Collection
- Defining Emission Sources
- GHG Emissions Calculations
- Baseline and Trend Analysis
- Emission Categorization
- Mitigation Scenario Development
- Reporting and validation

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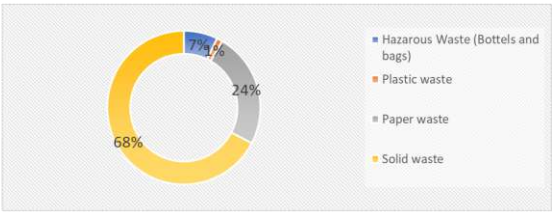
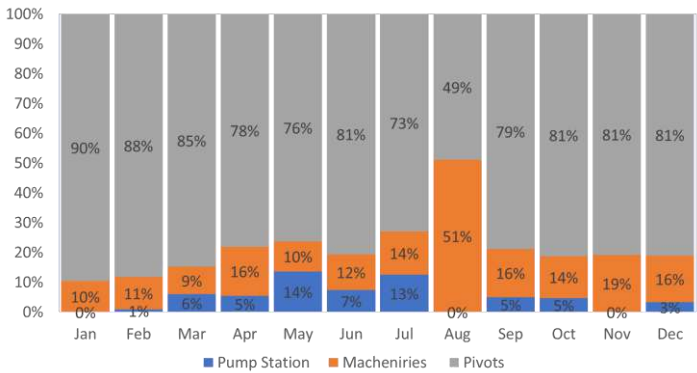


The following information was obtained from the client :

- Established by DAL Agriculture in 2009
- There are 105 operating pivots in the farm
- Crops: Alfaalfa, Rhodes, Maize, silage and citrus trees
- Harvested crops are for export
- The farm has a feedlot for 1,800 heads of cattle

The farm Location and facilities:

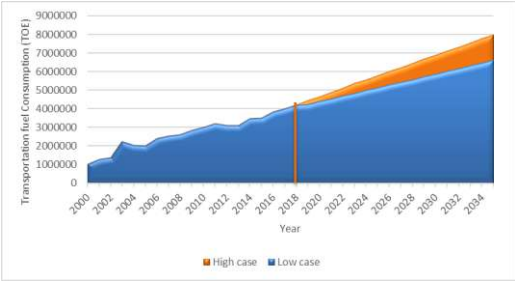
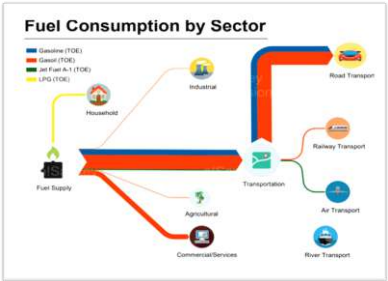
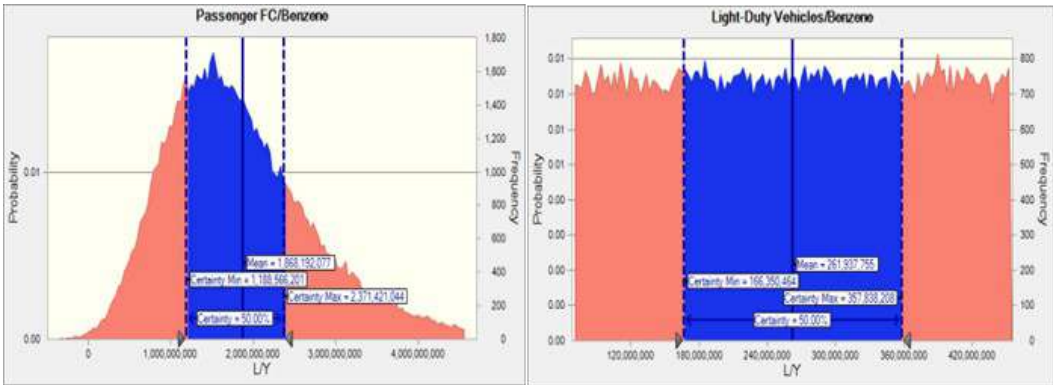
- Location: 50 km (from Soba bridge) Southeast of Khartoum
- Area: 22,000 Feddan (92.4 million sqm), in which ~15,000 Feddan pivot irrigated
- One pump station with 6 pumps (only 2 are operating): pumping water from Blue Nile into two major canals
- Feedlot area : 45 Feddan
- Human resources: 560 persons



Bottom-up approach for estimation of activity data and GHG emissions for road transport in Sudan

Project activities

- 1. **Data Collection:** Gathering vehicle activity data from traffic and customs authorities, including vehicle types, fuel consumption, and registration information.
- 2. **Data Classification and Analysis:** Reclassifying vehicles into IPCC categories (e.g., passenger cars, light-duty vehicles, heavy-duty vehicles, and motorcycles). Analyzing fuel consumption patterns and vehicle usage.
- 3. **Questionnaire Survey:** Conducting surveys to estimate real fuel consumption, vehicle kilometers traveled (VKT), and other indicators.
- 4. **Emission Factor Calculation:** Developing country-specific emission factors for CO2, CH4, and N2O using Sudan-specific fuel properties and vehicle data. Utilizing Monte Carlo simulations and experimental methods for methane and nitrous oxide measurements.
- 5. **Comparative Analysis:** Comparing Sudan's road transport emission factors with IPCC values and other countries to assess accuracy and variations.
- 6. **Forecasting and Scenario Analysis:** Predicting future emissions under low and high-case scenarios based on historical fuel consumption trends and expected growth in vehicle numbers.



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- ☒ Modeling
- ☒ Data Collection

Country	CO ₂ EF (Kg/TJ)	
	Gasoline	Diesel
Sudan	73,117	74,939
Canada	68,980	69,910
France	74,440	74,860
Italy	72,860	73,650
USA	67,620	70,100
Japan	67,900	68,900
Germany	74,030	75,290
Norway	71,300	73,550



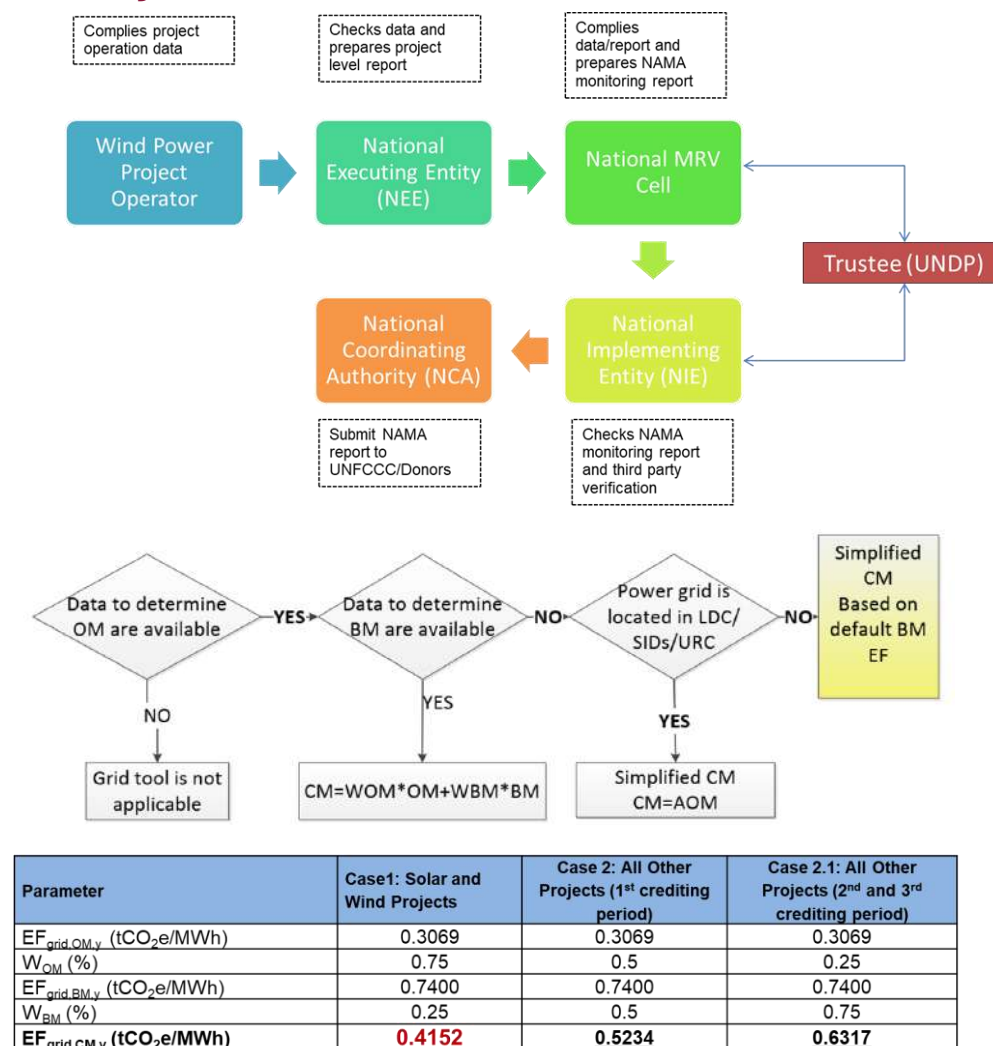
Renewable Energy Feed-in Tariff (RE-Wind-FiT) NAMA Project, Funded by GEF and UNDP

Project activities

- Policy and Regulatory Measures:** Introduction of Feed-in Tariff (FiT) to ensure financial viability and promote private sector participation in renewable energy projects. Establishment of a Partial Risk Guarantee Fund for Renewable Energy (PRGFRE).
- Capacity Building:** Technical training programs for government agencies, private sector employees, and financial institutions, focusing on renewable energy technologies, wind farm assessment, and financial evaluation.
- Institutional Arrangements:** Formation of a NAMA Steering Group, National Coordinating Authority, and National Implementing Entity to oversee and manage project activities. Establishment of a centralized Monitoring, Reporting, and Verification (MRV) system.
- Technology Support and Procurement:** Acquisition of software tools and simulation technologies for wind resource assessment, wind farm design, and operational monitoring.
- Awareness and Stakeholder Engagement:** Conducting campaigns and consultations with key stakeholders, including the government, private sector, and community representatives, to support renewable energy adoption.
- Monitoring, Reporting, and Verification (MRV):** Setting up a comprehensive MRV system to measure GHG emission reductions, track sustainable development impacts, and monitor financial performance.

Information Provider Scope Checklist

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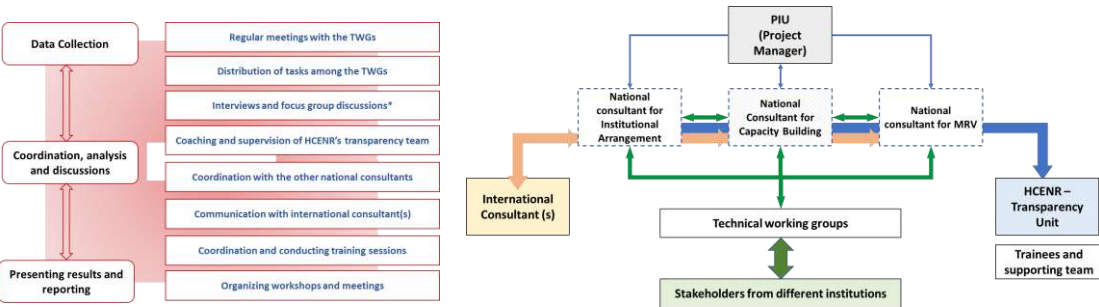
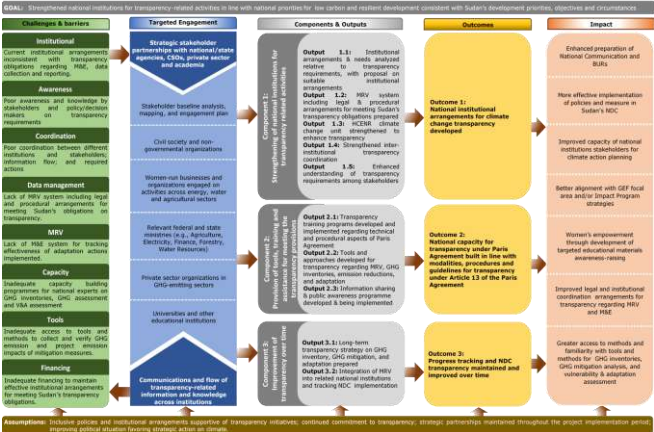
Parameter	Case1: Solar and Wind Projects	Case 2: All Other Projects (1 st crediting period)	Case 2.1: All Other Projects (2 nd and 3 rd crediting period)
EF _{grid,OM,y} (tCO ₂ e/MWh)	0.3069	0.3069	0.3069
W _{OM} (%)	0.75	0.5	0.25
EF _{grid,BM,y} (tCO ₂ e/MWh)	0.7400	0.7400	0.7400
W _{BM} (%)	0.25	0.5	0.75
EF _{grid,CM,y} (tCO ₂ e/MWh)	0.4152	0.5234	0.6317

Capacity Building component in the CBIT Project, funded by GEF and UNDP

Project activities

- Needs Assessment:** Conducting gap and needs assessments on the institutional arrangements and MRV system requirements. Identifying key stakeholders and sectors for transparency-related activities.
- Capacity Building:** Designing and implementing training programs for stakeholders at various levels, including policy makers, technical staff, and local actors. Providing training on GHG inventory preparation, mitigation actions, and adaptation assessments. Developing training-of-trainers (ToT) programs for sustainable capacity building.
- Institutional Strengthening:** Establishing or enhancing inter-institutional coordination mechanisms for transparency, including a transparency unit within HCENR. Developing and approving legal and procedural frameworks for transparency. Forming technical working groups to facilitate MRV implementation.
- Development of MRV Systems:** Designing and establishing MRV systems to measure, report, and verify GHG emissions, mitigation, and adaptation activities. Creating templates, tools, and data-sharing protocols to support MRV operations. Setting up a QA/QC system for GHG inventory data.
- Knowledge Management and Awareness:** Organizing public awareness campaigns and workshops to educate stakeholders on climate change transparency obligations. Developing educational materials and sharing best practices through an online transparency platform. Conducting workshops for gender inclusion and empowering women in climate-related actions.
- Implementation of Transparency Tools:** Creating a national data center and online transparency portal to manage and share MRV-related data and reports. Providing access to relevant tools, templates, and methodologies for data collection and analysis.
- Monitoring, Reporting, and Evaluation (M&E):** Establishing long-term strategies for monitoring and reporting Sudan's progress towards NDCs. Tracking progress in transparency efforts and sharing lessons learned at national and international levels.
- Stakeholder Engagement:** Conducting consultations and meetings with national and state-level stakeholders. Collaborating with government ministries, private sector entities, and academic institutions to enhance transparency coordination.

Code	Entity	Climate Change	Transparency	MRV	Technical and human resources	Policy and legal framework	Financial resources	Information and communication	Stakeholder engagement
1.1	Ministry of Planning	✓	✓	✓	✓	✓	✓	✓	✓
1.2	Ministry of Agriculture	✓	✓	✓	✓	✓	✓	✓	✓
1.3	Ministry of Environment	✓	✓	✓	✓	✓	✓	✓	✓
1.4	Ministry of Health	✓	✓	✓	✓	✓	✓	✓	✓
1.5	Ministry of Education	✓	✓	✓	✓	✓	✓	✓	✓
1.6	Ministry of Industry	✓	✓	✓	✓	✓	✓	✓	✓
1.7	Ministry of Transport	✓	✓	✓	✓	✓	✓	✓	✓
1.8	Ministry of Energy	✓	✓	✓	✓	✓	✓	✓	✓
1.9	Ministry of Water	✓	✓	✓	✓	✓	✓	✓	✓
1.10	Ministry of Land	✓	✓	✓	✓	✓	✓	✓	✓
1.11	Ministry of Urban Planning	✓	✓	✓	✓	✓	✓	✓	✓
1.12	Ministry of Housing	✓	✓	✓	✓	✓	✓	✓	✓
1.13	Ministry of Social Services	✓	✓	✓	✓	✓	✓	✓	✓
1.14	Ministry of Labour	✓	✓	✓	✓	✓	✓	✓	✓
1.15	Ministry of Justice	✓	✓	✓	✓	✓	✓	✓	✓
1.16	Ministry of Religious Affairs	✓	✓	✓	✓	✓	✓	✓	✓
1.17	Ministry of Culture	✓	✓	✓	✓	✓	✓	✓	✓
1.18	Ministry of Tourism	✓	✓	✓	✓	✓	✓	✓	✓
1.19	Ministry of Information	✓	✓	✓	✓	✓	✓	✓	✓
1.20	Ministry of Communications	✓	✓	✓	✓	✓	✓	✓	✓
1.21	Ministry of Transport	✓	✓	✓	✓	✓	✓	✓	✓
1.22	Ministry of Energy	✓	✓	✓	✓	✓	✓	✓	✓
1.23	Ministry of Water	✓	✓	✓	✓	✓	✓	✓	✓
1.24	Ministry of Land	✓	✓	✓	✓	✓	✓	✓	✓
1.25	Ministry of Urban Planning	✓	✓	✓	✓	✓	✓	✓	✓
1.26	Ministry of Housing	✓	✓	✓	✓	✓	✓	✓	✓
1.27	Ministry of Social Services	✓	✓	✓	✓	✓	✓	✓	✓
1.28	Ministry of Labour	✓	✓	✓	✓	✓	✓	✓	✓
1.29	Ministry of Justice	✓	✓	✓	✓	✓	✓	✓	✓
1.30	Ministry of Religious Affairs	✓	✓	✓	✓	✓	✓	✓	✓
1.31	Ministry of Culture	✓	✓	✓	✓	✓	✓	✓	✓
1.32	Ministry of Tourism	✓	✓	✓	✓	✓	✓	✓	✓
1.33	Ministry of Information	✓	✓	✓	✓	✓	✓	✓	✓
1.34	Ministry of Communications	✓	✓	✓	✓	✓	✓	✓	✓
1.35	Ministry of Transport	✓	✓	✓	✓	✓	✓	✓	✓
1.36	Ministry of Energy	✓	✓	✓	✓	✓	✓	✓	✓
1.37	Ministry of Water	✓	✓	✓	✓	✓	✓	✓	✓
1.38	Ministry of Land	✓	✓	✓	✓	✓	✓	✓	✓
1.39	Ministry of Urban Planning	✓	✓	✓	✓	✓	✓	✓	✓
1.40	Ministry of Housing	✓	✓	✓	✓	✓	✓	✓	✓



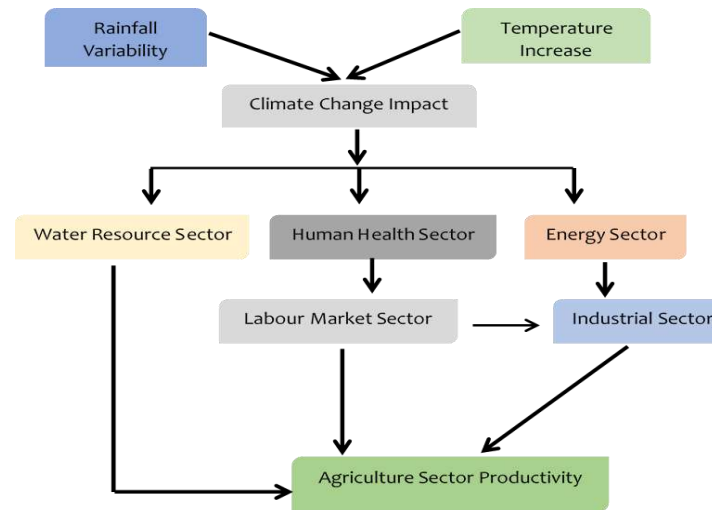
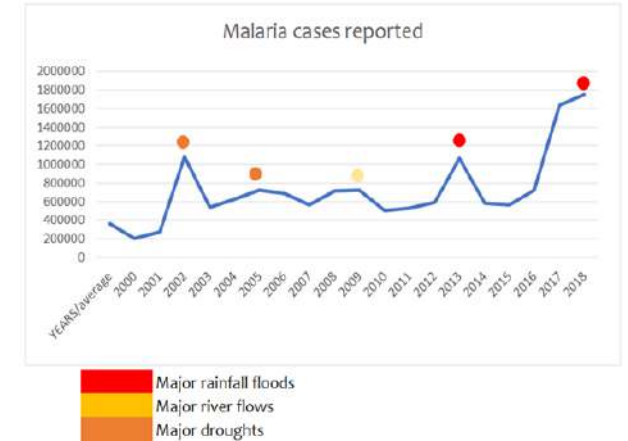
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Building Sudan's Economic Resilience: Climate Change Impact Assessment and Strategic Response (Funded by World Research Institute – WRI)

Project activities

1. **Assessment of Climate Change Impacts on Sudan's Economy:** The report evaluates how climate change affects Sudan's GDP, employment, agriculture, food security, and financial stability. It examines the local and global impacts of rising temperatures, rainfall variability, and extreme weather events such as floods, droughts, and desertification.
2. **Economic Vulnerability and Sectoral Analysis:** A detailed assessment is conducted on agriculture, water resources, energy, infrastructure, and health sectors, identifying vulnerabilities and economic losses associated with climate change. The study uses historical data and climate projections to forecast future trends and risks.
3. **Mitigation and Adaptation Strategies:** The report outlines Sudan's Nationally Determined Contributions (NDCs) and evaluates existing GHG mitigation projects in forestry, agriculture, and energy. It highlights climate adaptation programs in rangeland management, livestock sustainability, and resilient agriculture. The analysis includes Sudan's Climate Change Adaptation Plan (NAP) and other national strategies.
4. **Financial Instruments and Climate Finance:** A review of available climate finance mechanisms, including the Green Climate Fund (GCF), Adaptation Fund (AF), and other international funding sources, is included. The report discusses how Sudan can mobilize financial resources to support climate action initiatives and increase investment in low-carbon development.
5. **Theoretical Modeling for Climate Change Impact:** A Computable General Equilibrium (CGE) model is proposed to quantify economic losses due to climate change and to assess the economic benefits of adaptation and mitigation measures. The model focuses on the impacts of rainfall variability and temperature increase on key sectors such as water, health, and energy.
6. **National Initiatives and Policy Recommendations:** The report provides policy recommendations aligned with Sudan's climate commitments under the Paris Agreement and the Istanbul Plan of Action (2011-2020). It emphasizes poverty reduction, economic resilience, and the integration of climate policies into national development planning.



Information Provider Scope Checklist

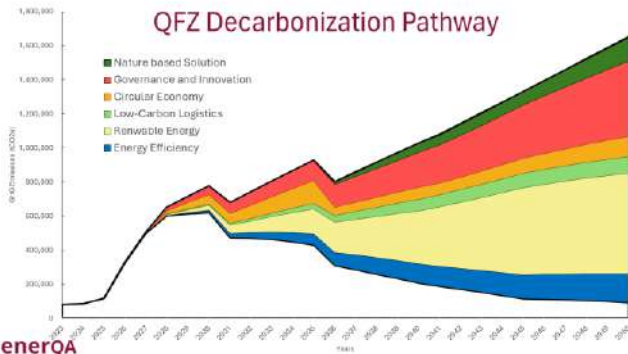
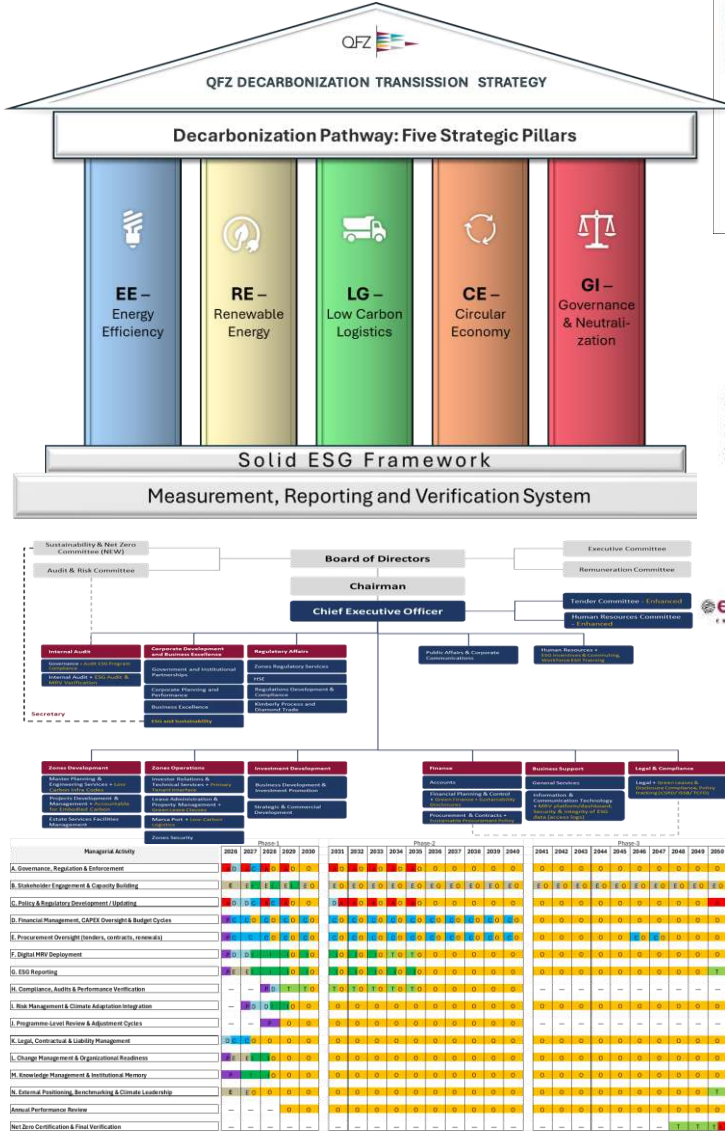
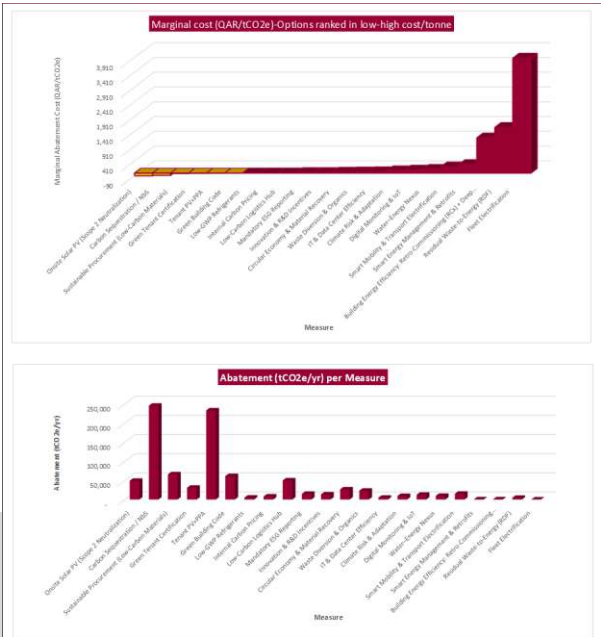
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- ☐ Content Creation
- ☒ Modeling
- ☒ Data Collection

Qatar Free Zone (QFZ) Net Zero Strategy Development Project

Project activities

1. Create GHG inventory and baseline
2. Model different mitigation actions that cover all future expansions and align with SBTi
3. Model MAC curve and calculate ROI for all mitigation measures
4. Establish governance framework and institutional structure for Net Zero implementation.
5. Deploy digital MRV platform and ESG reporting system for Scope 1–3 emissions tracking.
6. Implement energy efficiency programs and operational optimization across QFZ assets.
7. Scale renewable energy deployment (on-site PV and tenant PPAs).
8. Decarbonize logistics, mobility, and industrial operations within the Free Zones.
9. Advance circular economy initiatives, sustainable procurement, and green building standards.
10. Develop long-term carbon neutralization through nature-based solutions and continuous performance monitoring
11. Create a dashboard for the entire strategy
12. Training and capacity building

- ✓ Scope 1,2 and 3
- ✓ >700 Tenants
- ✓ Two zones (24 million SQM)



- ✓ Climate Data Aggregation
- ✓ Stakeholder Engagement
- ✓ Capacity Building
- ✓ Policy Support
- ✓ Tools and Platform Development
- ✓ Sustainability Reporting
- ✓ Awareness
- ✓ Data Visualization
- Compliance and Reporting
- ✓ Research and Innovation
- Content Creation
- ✓ Modeling
- ✓ Data Collection

8. Leading the way

Enerqa: Leading the Way

In an era where sustainability and resilience define success, organizations across industries are facing unprecedented challenges and opportunities. Whether it's achieving net-zero targets, transitioning to clean energy, securing climate finance, integrating ESG, or developing sustainable business models, the need for expert guidance, cutting-edge tools, and strategic foresight has never been greater.

At Enerqa, we are leading the way by providing holistic, data-driven, and impact-oriented consultancy services across six core areas:



1. Sustainability & ESG: Turning Commitments into Measurable Impact

Enerqa helps businesses, governments, and institutions navigate the **complex world of ESG, corporate sustainability, and green finance**. Our services include **sustainability strategy development, ESG readiness assessments, sustainability-linked finance advisory, and impact measurement**. Through our proprietary tools and expert guidance, we enable clients to **comply with global sustainability frameworks, improve transparency, and gain competitive advantages in climate-conscious markets**.



2. Environment: Driving Smart, Science-Based Environmental Solutions

With a strong focus on **circular economy, biodiversity conservation, environmental impact assessment, and resource efficiency**, Enerqa ensures that **projects, policies, and businesses minimize their ecological footprint while maximizing sustainability benefits**. Our approach integrates **data analytics, risk assessment, and best practices** to help organizations comply with environmental regulations and implement sustainable resource management strategies.



3. Climate Change: Unlocking Climate Finance and Decarbonization Strategies

Enerqa plays a **pivotal role** in helping organizations and governments design and implement **climate action plans, GHG mitigation pathways, and carbon credit strategies**. Through our **MRV (Monitoring, Reporting, and Verification) systems, emissions modeling, and climate risk assessments**, we provide **credible, verifiable solutions** that enable access to **carbon markets, climate finance mechanisms, and global decarbonization programs**.

Enerqa: Leading the Way



4. Energy: Powering the Future with Efficiency and Clean Energy Solutions

Enerqa provides comprehensive energy consultancy services, including renewable energy project development, energy audits, grid integration, and energy transition planning. Whether optimizing energy efficiency for industries, supporting governments in designing national energy policies, or helping investors finance clean energy projects, we ensure that organizations meet their sustainability targets while reducing costs and carbon emissions.



BUSINESS MODEL

5. Business Modeling: Creating Sustainable, Profitable, and Scalable Business Models

Enerqa excels in developing tailored business models for complex projects, structuring investment-ready feasibility studies, and supporting organizations in accessing sustainable financing. We help clients align with green investment frameworks, develop circular economy-based revenue models, and integrate ESG compliance into their financial structures—ensuring business growth and sustainability go hand in hand.



6. Capacity Building: Empowering Organizations with Knowledge and Expertise

Sustainability and climate action require more than just policies—they demand knowledge, training, and skills development. Enerqa provides customized training programs, workshops, and in-house capacity-building initiatives to equip teams, businesses, and policymakers with the expertise needed to navigate sustainability challenges, implement best practices, and drive long-term impact.

At **Enerqa**, we pride ourselves on being a leading consultancy firm that transforms knowledge into action across the domains of climate change, energy, environment, and business sustainability. Our extensive experience, global network, and deep technical expertise allow us to deliver innovative, data-driven, and impactful solutions that support both public and private sector organizations in meeting their sustainability and development goals.

With a team of highly skilled professionals, including PhD and MSc holders, and a vast network of experts from over 25 countries, Enerqa offers integrated solutions tailored to the evolving needs of our clients. Our consultancy spans strategic planning, technical assessments, project management, feasibility studies, and implementation support, ensuring that our clients achieve measurable, lasting results.

Global Reach, Local Impact

Enerqa's success is built on strong partnerships and collaborations with leading engineering firms, sustainability consultants, financial institutions, and technology providers worldwide. Our global presence enables us to leverage international best practices, while our regional expertise ensures that our solutions are locally relevant and implementable.

From GHG inventory development and renewable energy transition strategies to circular economy solutions and climate finance advisory, Enerqa delivers holistic, future-ready strategies. Our work aligns with national policies, international climate commitments (such as the Paris Agreement), and global sustainability standards, making us a trusted partner for governments, multilateral organizations, and private sector entities.

Commitment to Excellence and Sustainability

Our structured approach, which includes needs assessments, feasibility studies, monitoring & evaluation, and exit strategies, ensures that our projects are not just well-designed, but also resilient, efficient, and sustainable. We continuously invest in cutting-edge modeling tools, research methodologies, and capacity-building initiatives to support organizations in navigating the complexities of climate action, energy management, and environmental stewardship.

Shaping the Future Together

As we look ahead, Enerqa remains committed to leading the way in sustainable transformation, helping our clients decarbonize, optimize resource efficiency, and build resilient, climate-friendly solutions. We believe in the power of collaboration, innovation, and expertise to create a better, more sustainable future for all.

With over 50 successful projects, a growing roster of specialists, and a proven track record in advisory excellence, Enerqa is your trusted partner in driving impactful change.

Together, we are shaping a greener, more sustainable, and resilient world.



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